Application Guide

Symbio™ 700 Controller

with Voyager 3 Packaged Rooftop Air Conditioners

⚠ SAFETY WARNING

Only qualified personnel should install and service the equipment. The installation, starting up, and servicing of heating, ventilating, and air-conditioning equipment can be hazardous and requires specific knowledge and training. Improperly installed, adjusted or altered equipment by an unqualified person could result in death or serious injury. When working on the equipment, observe all precautions in the literature and on the tags, stickers, and labels that are attached to the equipment.

September 2025

BAS-APG048D-EN

Introduction

Read this manual thoroughly before operating or servicing this unit.

Warnings, Cautions, and Notices

Safety advisories appear throughout this manual as required. Your personal safety and the proper operation of this machine depend upon the strict observance of these precautions.

The three types of advisories are defined as follows:

⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
⚠ CAUTION	Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury. It could also be used to alert against unsafe practices.
NOTICE	Indicates a situation that could result in equipment or property-damage only accidents.

⚠ WARNING

Proper Field Wiring and Grounding Required!

Failure to follow code could result in death or serious injury.

All field wiring MUST be performed by qualified personnel. Improperly installed and grounded field wiring poses FIRE and ELECTROCUTION hazards. To avoid these hazards, you MUST follow requirements for field wiring installation and grounding as described in NEC and your local/state/national electrical codes.

⚠ WARNING

Personal Protective Equipment (PPE) Required!

Failure to wear proper PPE for the job being undertaken could result in death or serious injury.

Technicians, in order to protect themselves from potential electrical, mechanical, and chemical hazards, MUST follow precautions in this manual and on the tags, stickers, and labels, as well as the instructions below:

- **Before installing/servicing this unit, technicians MUST put on all PPE required for the work being undertaken (Examples; cut resistant gloves/sleeves, butyl gloves, safety glasses, hard hat/bump cap, fall protection, electrical PPE and arc flash clothing). ALWAYS refer to appropriate Safety Data Sheets (SDS) and OSHA guidelines for proper PPE.**
- **When working with or around hazardous chemicals, ALWAYS refer to the appropriate SDS and OSHA/GHS (Global Harmonized System of Classification and Labelling of Chemicals) guidelines for information on allowable personal exposure levels, proper respiratory protection and handling instructions.**
- **If there is a risk of energized electrical contact, arc, or flash, technicians MUST put on all PPE in accordance with OSHA, NFPA 70E, or other country-specific requirements for arc flash protection, PRIOR to servicing the unit. NEVER PERFORM ANY SWITCHING, DISCONNECTING, OR VOLTAGE TESTING WITHOUT PROPER ELECTRICAL PPE AND ARC FLASH CLOTHING. ENSURE ELECTRICAL METERS AND EQUIPMENT ARE PROPERLY RATED FOR INTENDED VOLTAGE.**

⚠ WARNING

Follow EHS Policies!

Failure to follow instructions below could result in death or serious injury.

- **All Trane personnel must follow the company's Environmental, Health and Safety (EHS) policies when performing work such as hot work, electrical, fall protection, lockout/tagout, refrigerant handling, etc. Where local regulations are more stringent than these policies, those regulations supersede these policies.**
- **Non-Trane personnel should always follow local regulations.**

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Revision History

- Updated cooling only tables in Conventional Thermostat Sequence of Operation chapter.
- Updated Discharge Air Reset section in Variable Volume Discharge Air Control Sequence of Operation chapter.
- Updated Dehumidification - Hot Gas Reheat chapter.
- Updated Troubleshooting chapter.
- Updated Appendix A Compressor Staging.
- Updated Appendix B Voyager 3 Configuration.

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Introduction

The Symbio™ 700 installed on Voyager™ 3 rooftop units is a factory installed and programmed controller, providing digital control and protection of the equipment.

The Symbio 700 has two model options:

- **Standard controller** — provides advanced troubleshooting and access to the Symbio Service and Installation mobile app.
- **Advanced controller** — Building Automation System interface via BACnet® (ANSI/ASHRAE Standard 135-2016), Modbus™, Air-Fi®, or LonTalk®. Also allows TPG2 and up to eight additional I/O via XM30 or XM32 expansion modules.

The Symbio 700 offers multiple equipment configuration options regardless of controller model. The Voyager rooftop unit can be configured as the following system types:

- Conventional Thermostat Control (T-Stat)
- Constant Volume Zone Temperature Control (CVZT)
- Variable Volume Zone Temperature Control (VVZT)
- Variable Volume Discharge Air (VVDA)

These configurations can be used with standard cooling systems.

This guide provides information about the configuration, control capabilities and troubleshooting of the Voyager system with Symbio 700 controller.

Additional Documentation

- *Symbio™ Service and Installation App Quick Start Guide* (BAS-SVN043*-EN)
- *Symbio™ 700 Controller User Guide* (BAS-SVU054*-EN)
- *BACnet® and Modbus™ Integration to Voyager 3 Packaged Rooftop Air-Conditioners with Symbio™ 700 Controls Integration Guide* (BAS-SVP066*-EN)
- *LonTalk® Integration to Voyager 3™ Packaged Rooftop Air-Conditioners with Symbio™ 700 Controls Integration Guide* (BAS-SVP067*-EN)
- *Voyager™ 3 Packaged Rooftop Air Conditioners with Symbio™ 700 Controls 27.5 to 50 Tons- 60 Hz 22.9 to 41.7 Tons (82-142 kW)- 50 Hz Installation Operation and Maintenance* (RT-SVX34*-EN)

Symbio 700 Overview

Field Connection

The Symbio™ 700 controller optimizes inputs and outputs (I/O) for multiple applications. For initial installation of a Voyager 3 with Symbio 700, the field landed inputs are outlined below.

Figure 1. Symbio 700 field connections

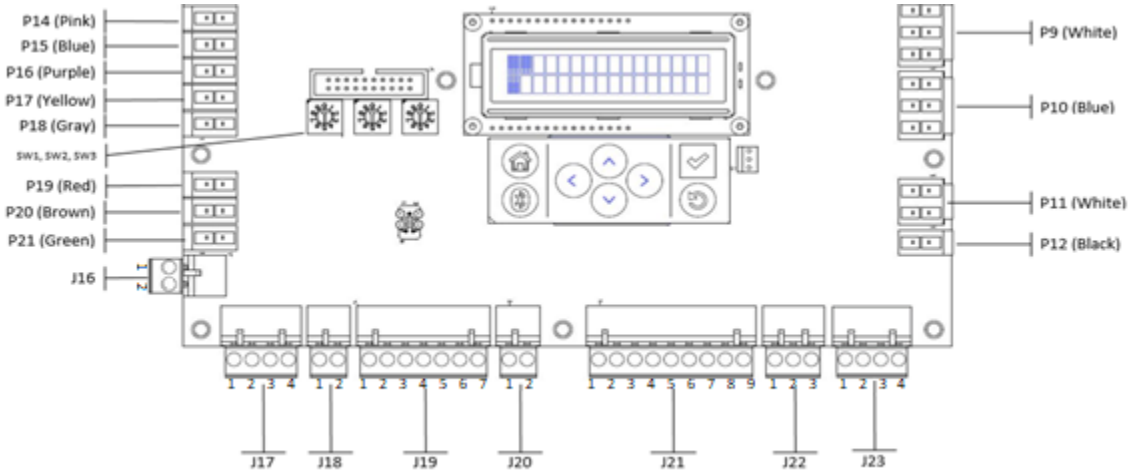


Table 1. Field Connections

Connector	Function	Pin #	Signal
J16	Demand Shed/Demand Limit Connection	1	24VAC
		2	Demand Shed/Demand Limit Input
J17	BACnet Communication Connections	1	BACnet +
		2	BACnet -
		3	BACnet +
		4	BACnet -
J18	Equipment Shutdown Input Connections	1	24VAC
		2	Equipment Shutdown Input
J19	Zone Sensor Connections	1	Zone Temperature
		2	GND
		3	Cool Setpoint
		4	Mode
		5	Heat Setpoint
		6	GND
		7	24VAC
J20	Occupancy Connections	1	24VAC
		2	Occupancy Switch

Table 1. Field Connections (continued)

Connector	Function	Pin #	Signal
J21	Thermostat Connections	1	24VAC
		2	Y1
		3	W1/O
		4	G
		5	W2
		6	Y2
		7	X2
		8	1.5K Ohms Pull-down
		9	GND
J22	Space CO_2	1	24VDC
		2	0-10 VDC input
		3	GND
J23	Space Relative Humidity	1	24VDC
		2	4-20 mA input
		3	GND

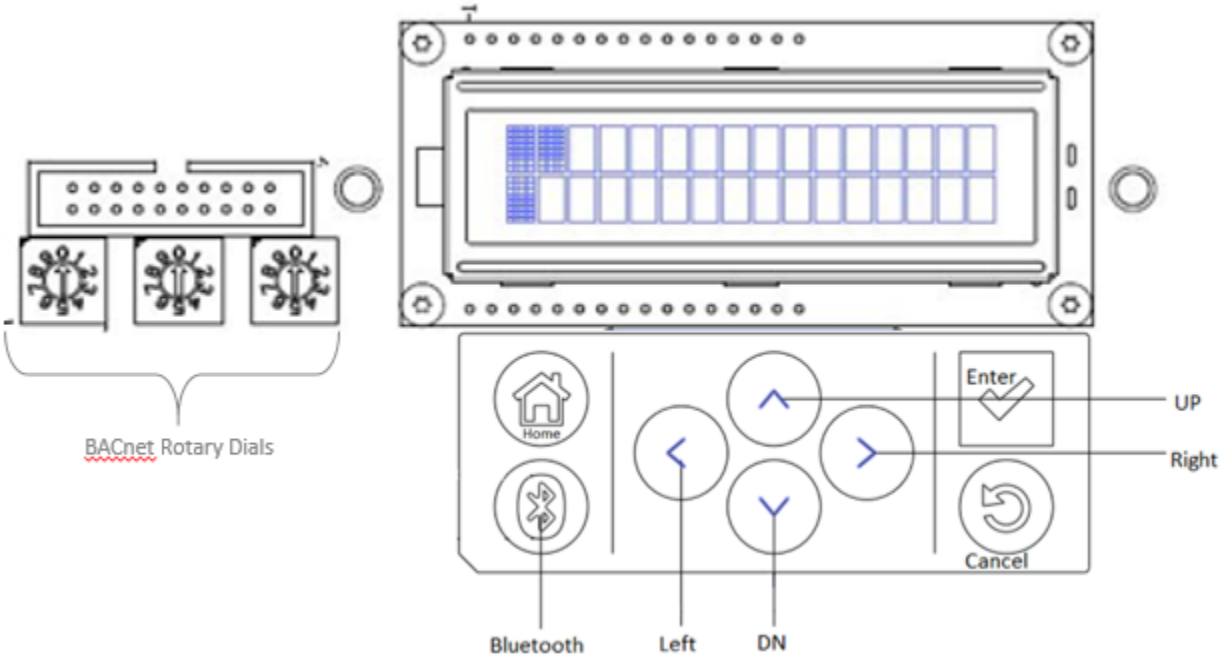
Unit Configuration

The Voyager 3 system can be configured via an onboard user interface or via the Symbio Service and Installation mobile app, see “Voyager 3 Configuration,” p. 90 for configuration options and descriptions.

Onboard User Interface

The onboard user interface provides a 2 × 16 Backlit LCD display and navigational buttons. This allows the user to view status, configure, and troubleshoot the unit without additional tools.

Figure 2. Symbio 700 onboard user interface



The interface provides an intuitive menu structure: alarms, status, service, settings, and utilities. Configuration of the unit is accomplished under the utilities menu item. A complete list of functions is outlined in *Symbio™ 700 Controller, User Guide* (BAS-SVU054*-EN).

To configure the unit, navigate to the utilities menu and press **Enter**. Once in the utilities menu the user has additional submenu options. This allows the user to navigate and configure the appropriate setting quickly and easily.

Mobile App

The Trane Symbio™ Service and Installation mobile app is required to setup, edit, and confirm the communication protocol and associated settings.

The free download of Trane Symbio Service and Installation mobile app is available on the App Store® for iOS, and on Google Play® for Android™.

Figure 3. Trane Symbio service and installation mobile app



Bluetooth Pairing

Quick Connection Instructions

Follow these instructions to quickly connect the mobile app to the Symbio™ 700 controller:

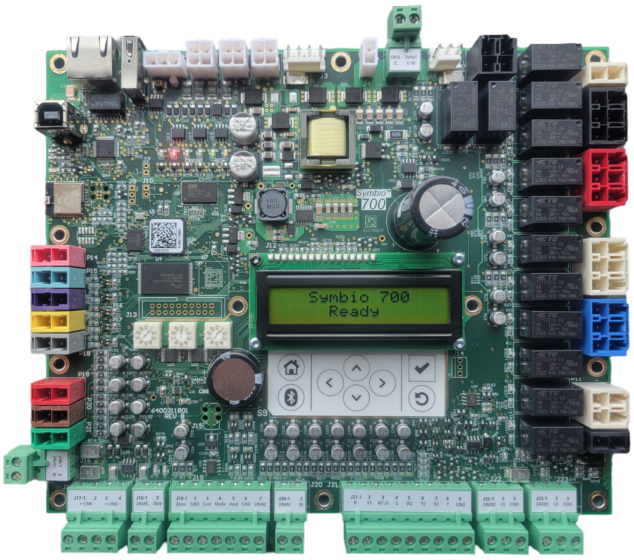
1. Turn on Bluetooth®[1].
2. Tap  .
3. Start the app. Tap **View Available Devices**.
4. Select the controller.
5. Tap **OK** to pair.
6. Tap  .

[1] *The Bluetooth® word mark and logos are registered trademarks owned by Bluetooth SIG, Inc. and any use of such marks by the company is under license.*

Connecting to the Symbio™ 700 controller

1. Enable Bluetooth®[2] on your smart device.
2. Access the Symbio 700 controller in the low voltage portion of the equipment.

Figure 4. Symbio 700 controller



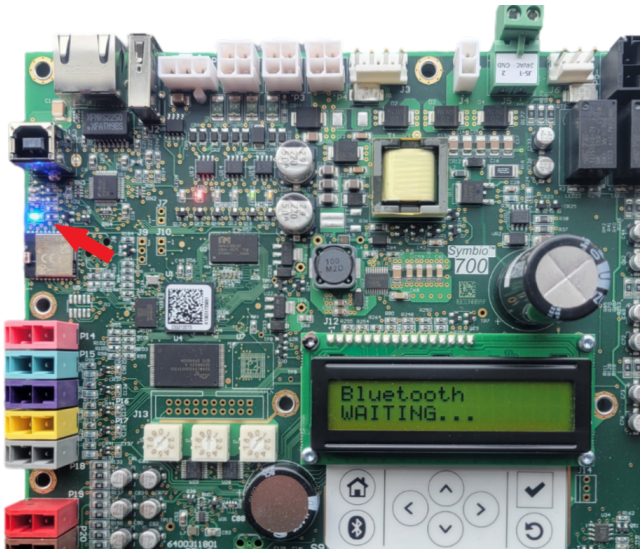
3. Tap  on Symbio 700 keyboard/display to turn on Bluetooth.
4. Confirm the status of Bluetooth communications. A solid blue LED indicates a successful pairing.

Table 2. Bluetooth communication status

	Blue LED	Display	Description
Tap for On/Off	Off	NOT CONNECTED	Bluetooth Off
	Blinking	WAITING...	Bluetooth On — Not Paired
	On Solid	CONNECTED	Bluetooth On — Connected/Paired

2. *The Bluetooth® word mark and logos are registered trademarks owned by Bluetooth SIG, Inc. and any use of such marks by the company is under license.*

Figure 5. Symbio 700 bluetooth status



5. Start the mobile app on your smart device.

Figure 6. Login screen

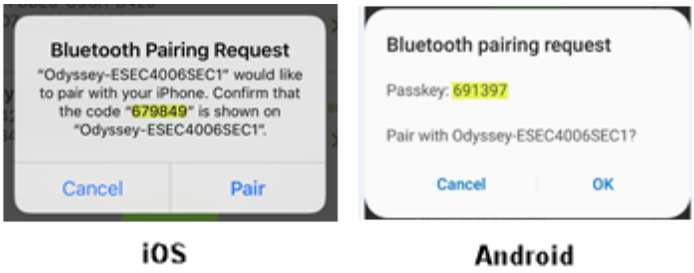


6. On the login screen, tap **View Available Devices** in the lower section of the screen. Or Trane personnel can login using their Trane Connect username and password.
7. On the Unit List page, select the Symbio 700 controller to which you want to pair. If the controller is not listed, tap the refresh arrow in the upper right-hand corner of the screen.

Note: *If a Symbio 700 is not the original Symbio controller as shipped with the equipment, the Bluetooth equipment list will list the controller serial number, instead of the equipment serial number.*

8. When prompted, pair the app to the Symbio 700 controller. A popup message displays a 6-digit random number. The same number is shown on the display of the Symbio™ 700 controller until the pairing is complete, allowing the user to confirm connection to the intended controller.

Figure 7. Bluetooth pairing



9. Tap ✔ on the Symbio 700 on-board keyboard/display to complete the connection.

When the LED light is a solid blue and the display reads Bluetooth Connected, the Bluetooth pairing and connection is complete.

Important: *To keep the list of previously-connected devices manageable, the Bluetooth smart devices list is limited to 10 devices. When 10 or more Bluetooth devices are defined on the smart device, connection to the Symbio 700 controller is not allowed.*

- ***iOS devices*** *- delete any unused devices until there are less than 10 items.*
- ***Android devices*** *- the devices list is automatically limited to 10 items.*

The Symbio Installation and Service tool is required to view, manage, and configure the following:

- Building Automation System configuration (Advanced Controller Configuration)
  - BACnet® over Zigbee® (Air-Fi® Wireless)
  - BACnet IP (Internet Protocol)
  - BACnet MS/TP
  - LonTalk®
  - Modbus RTU
  - Modbus TCP
- Historical Alarms
- Firmware Update (includes both the Symbio 700 Module and the Options Modules)
- Backing up and Restoring the database
- Transfer Settings from one controller to another
- Return the Symbio 700 to its Factory Default configuration by using the Factory Default Database (if available)
- Update the license file

For more detailed information on the Symbio Service and Installation Mobile Application, see *Symbio™ Service and Installation App, Quick Start Guide* (BAS-SVN043*-EN).

Options Module

The Symbio 700 extends its control capabilities through the use of additional hardware modules. These modules are installed, wired, and tested in the unit when ordered from the factory. The modules can also be field installed. The following table summarizes the Symbio 700 functions that require an additional hardware module.

Table 3. Symbio 700 options

Symbio 700 Function	Customer Options Module	Indoor Options Module	Fresh Air Options Module	Stepper Motor Module
Humidistat	X			
External Auto/Stop	X			
Ventilation Override	X			
Alarm Output	X			
Remote Minimum Position	X			
Condensate Overflow		X		
Electric Heat		X		
VFD Bypass		X		
Duct Static Pressure Control		X		
0 to 100% Economizer			X	
0 to 100% Econoimizer w/Traq			X	
StatiTrac – Building Pressure Control			X	
Relief Fan			X	
Air Filter Status			X	
Dehumidification – Hot Gas Reheat	X	X		X

Start-Up Sequence

Under normal conditions, the Symbio™ 700 will start-up in approximately 60 seconds once power is applied to the system. During this process, the controller checks that a valid system configuration is present and proceeds to normal control operation. After start-up, the system will begin to respond to operational requests.

Conventional Thermostat Sequence of Operation

When the Voyager system is configured to operate with a conventional thermostat, the controller provides protection for the system (see General Support Sequence section) and continues to provide insight to operating conditions. A conventional thermostat can be applied with CVZT and single zone VAV configured systems. While not recommended, a conventional thermostat can be applied to single zone variable volume configured systems, but the system is limited to staged fan control instead of a fully variable sequence.

When under conventional thermostat control, the equipment responds directly to operating requests from the thermostat device. Each thermostat input corresponds to a specific unit function, as described in the following tables. Equipment protection functions and compressor minimum on/off timers remain in-control, even when under conventional thermostat control.

Table 4. Cooling only, without outside air

Inputs						Outputs				
X	**Y1**	**Y2**	**W1/O**	**W2**	**G**	**Supply Fan On/Off Request**	**Supply Fan Speed Request [a]**	**Compressor Cool Stage Request [a]**	**Auxiliary Heat Stage Request [a]**	**Heat Cool Mode Status**
—	OPEN	OPEN	OPEN	OPEN	CLOSED	ON	Min	None	None	Fan Only
—	OPEN	OPEN	CLOSED	OPEN	X	ON	Min	None	Stage 1[b]	Heat
—	OPEN	OPEN	X	CLOSED	X	ON	Min	None	Full Stage[c]	Heat
—	CLOSED	OPEN	OPEN	OPEN	X	ON	Min	Stage 1	None	Cool
—	OPEN	CLOSED	OPEN	OPEN	X	ON	Min	Stage 1	None	Cool
—	CLOSED	CLOSED	OPEN	OPEN	X	ON	Max	Stage 5	None	Cool
—	X	X	X	X	X	OFF	0	None	None	OFF

[a] See Appendix A, compressor staging information for details and additional information.

[b] Staged electric heat and staged gas heat operate at stage 1. Modulating gas heat modulates to the Discharge Air Heating Setpoint BAS - Active and supply fan operates at the minimum allowable speed.

[c] Staged electric heat and staged gas heat operate at full capacity. Modulating gas heat modulates to the Discharge Air Heating Setpoint BAS - Active and supply fan operates a maximum speed.

Table 5. Cooling only, with outside air

Inputs								Outputs					
Economizer System Status	**X**	**Y1**	**Y2**	**W1/O**	**W2**	**G**	**DCV Fan ON Request**	**Supply Fan On/Off Request**	**Supply Fan Speed Request [a]**	**OA Damper Position Request**	**Compressor Cool Stage Request [a]**	**Auxiliary Heat Stage Request**	**Heat Cool Mode Status**
X	—	OPEN	OPEN	OPEN	OPEN	CLOSED	X	ON	Min	Min	None	None	Fan Only
X	—	OPEN	OPEN	CLOSED	OPEN	X	X	ON	Min	Min	None	Stage 1[b]	Heat
X	—	OPEN	OPEN	X	CLOSED	X	X	ON	Min	Min	None	Full Stage[c]	Heat
Enabled[d]	—	CLOSED	OPEN	OPEN	OPEN	X	X	ON	Min/Max	Econ	0	None	Cool
Enabled[d]	—	OPEN	CLOSED	OPEN	OPEN	X	X	ON	Min/Max	Econ	0	None	Cool
Enabled[e]	—	CLOSED	CLOSED	OPEN	OPEN	X	X	ON	Max	Full	Stage 1	None	Cool
Disabled	—	CLOSED	OPEN	OPEN	OPEN	X	X	ON	Min	Min	Stage 1	None	Cool
Disabled	—	OPEN	CLOSED	OPEN	OPEN	X	X	ON	Min	Min	Stage 1	None	Cool
Disabled	—	CLOSED	CLOSED	OPEN	OPEN	X	X	ON	Max	Min	Stage 5	None	Cool
X	—	X	X	X	X	X	ON	ON	Min	Min	None	None	Fan Only
X	—	X	X	X	X	X	X	OFF	0	0	None	None	OFF

[a] See Appendix A for supply fan and compressor staging information details and additional information.
[b] Staged electric heat and staged gas heat operate at stage 1. Modulating gas heat modulates to the Discharge Air Heating Setpoint BAS - Active and supply fan operates at the minimum allowable speed.
[c] Staged electric heat and staged gas heat operate at full capacity. Modulating gas heat modulates to the Discharge Air Heating Setpoint BAS - Active and supply fan operates a maximum speed.
[d] When economizer is requested, the outdoor air damper will modulate above minimum position to control discharge air temperature to Economizer Discharge Air Setpoint - Active (default 55°F).
[e] Economizer requested and Y1 + Y2 are closed, the outdoor air damper will open to 100% and the supply fan will increase to maximum speed, before a compressor stage is added.

Modulating Gas Heat - Thermostat Control

When the Symbio 700, Space Controller is configured for Conventional TStat and modulating gas heat is installed, a discharge air temperature sensor is required for modulating control. On a W1, W2 or W1+W2 call, gas heat is modulated to deliver discharge air at the Discharge Air Heating Setpoint BAS setpoint (default 100°F).

If the discharge air temperature is invalid, the control will revert to staged control with W1 calling 50% modulating heat capacity and W2 or W1+W2 calling 100% capacity.

Space Temperature Control

System Types of VVZT and CVZT operate to provide space comfort heating and cooling. A system mode wired input or Heat Cool Mode Request input determines the heating or cooling mode of operation. If a heat cool system mode input is not provided, the Symbio™ 700 operates per Heat Cool Mode Request default value, Auto is the default setting. In Auto, the controller will automatically determine it is appropriate to heat or cool based on space temperature, setpoints, and heating/cooling configured for the unit. The space temperature and space temperature setpoints determine a space heating or cooling demand. If space temperature is above the cooling setpoint, this represents a space cooling demand.

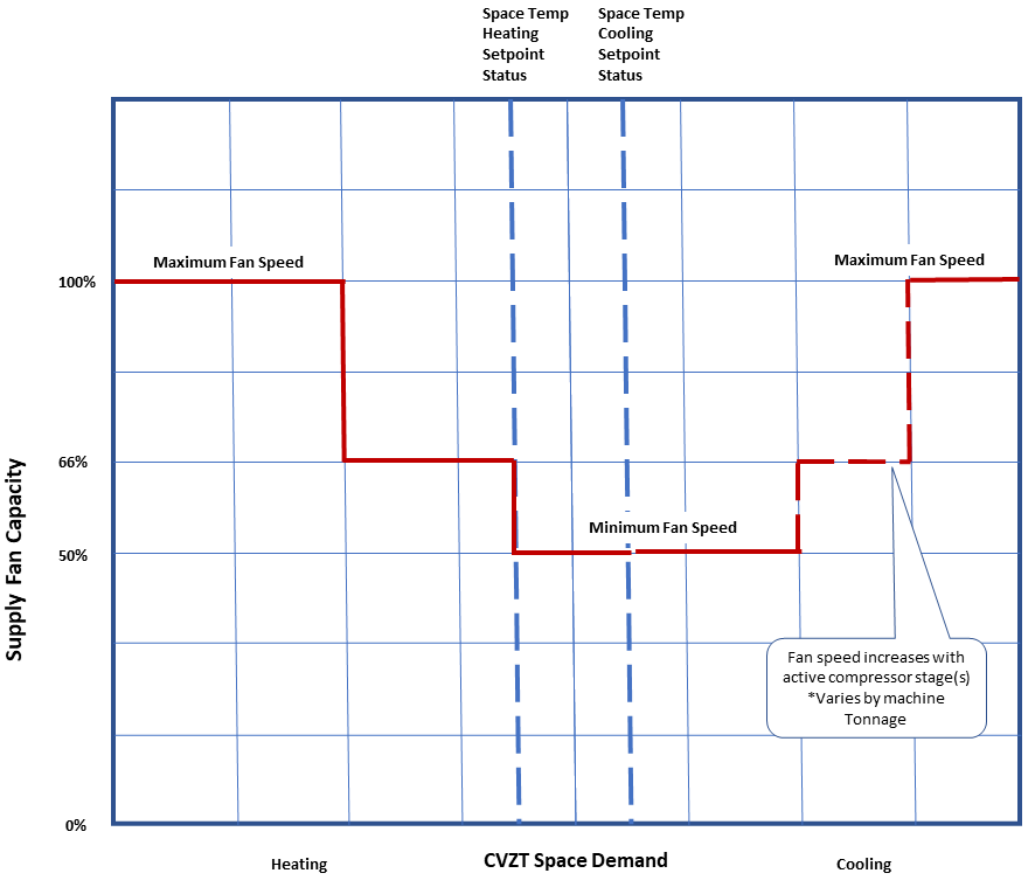
Symbio 700 supports two types of space temperature control: Single Loop Space Temperature Control and Single Zone Variable Air Volume.

Single Loop Space Temperature Control

Configured VVZT and CVZT System Types operate in Single Loop Space Temperature Control when only a space temperature input is provided the Symbio™ 700 control (no discharge air temperature sensor installed). In heating modes, with staged gas heat or staged electric heat installed, the supply fan will operate at maximum speed. In cooling modes, the supply fan will operate at the lowest speed allowed for the stage of cooling capacity and 100% when all compressor stages are on. When space temperature is between the heating and cooling setpoint, the supply fan operates at a minimum speed. Alternately, the supply fan can be setup to cycle off when there is no demand for heating or cooling via Supply Fan Configuration Command setting.

Cooling capacity increases as space temperature increases above the cooling setpoint. Heat capacity increases when space temperature decrease below the heating setpoint. Capacity decreases has space temperature approaches the desired space setpoint.

Figure 8. Multi-speed fan sequence of operation



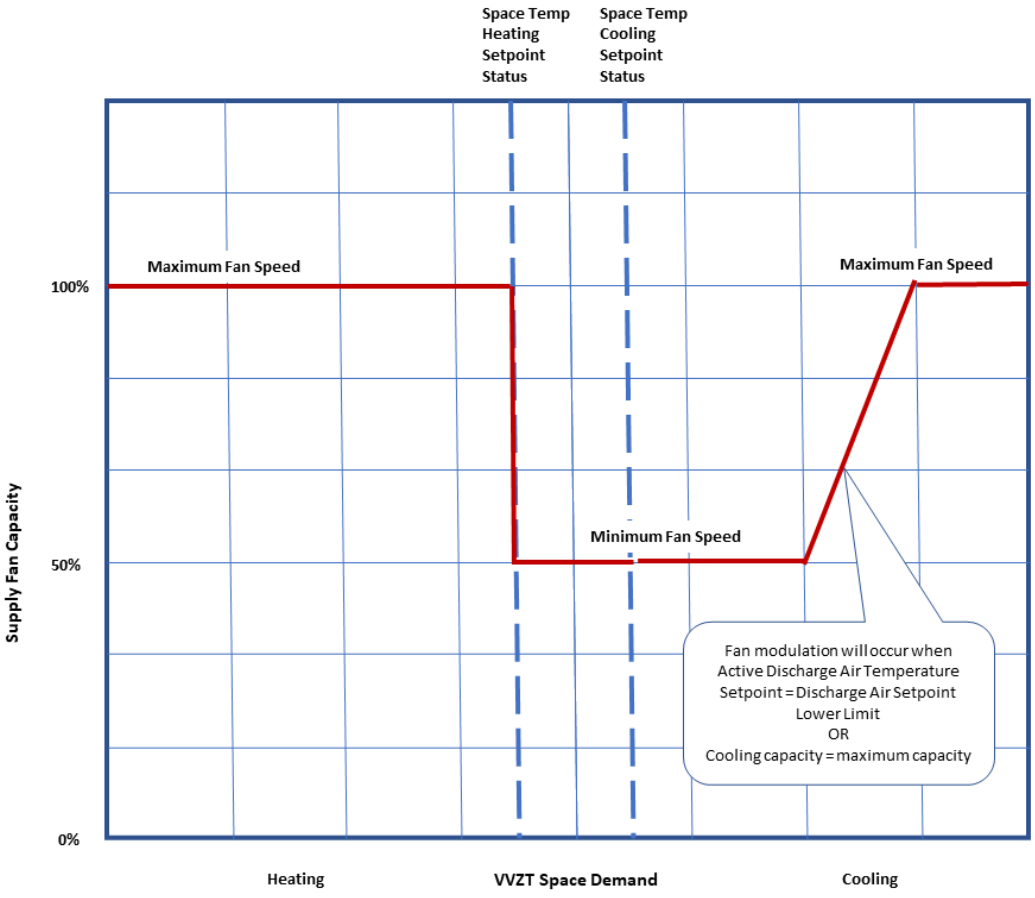
Single Zone Variable Air Volume

Single Zone VAV operates in DX and economizer cooling modes of operation when configured for System Type: VVZT (variable speed indoor fan type) or CVZT (multi speed indoor fan type). A valid space temperature and discharge air temperature sensor are required input. If the discharge air temperature input becomes invalid, the control automatically reverts to Single Loop Space Temperature Control. Symbio™ 700 operates in Single Loop Space Temperature Control in heating modes when staged gas heat and staged electric heat are configured.

Single Zone VAV is a control method of space temperature control that operates the supply fan at the lowest allowed for the cooling capacity required to satisfy the load in the space. As cooling capacity increases, the supply fan speed will increase accordingly until it reaches 100% of its allowed range of operation.

When there is no demand for heating or cooling the supply fan operates at the minimum speed setting while providing ventilation according to occupancy setpoints, see the following figure. Alternately, the supply fan can be setup to cycle off when there is no demand for heating or cooling via Supply Fan Configuration Command setting.

Figure 9. Supply fan sequence of operation



Single Zone VAV – Cooling

Cool

When there is a space cooling demand, DX cooling will initialize and increase to satisfy space temperature. The supply fan will continue to operate at a low, fixed speed at low cooling capacities until the cooling capacity requires additional airflow to keep compressor operation optimal. The supply fan speed will increase as the demand for cooling capacity increases. The minimum supply fan speed is calculated by DX cooling capacity. As the space requires additional cooling capacity, Space Temperature Control will calculate a lower discharge air temperature setpoint increasing DX cooling which in turn increases the supply fan speed. Cooling capacity and supply fan speed both can increase to 100%.

If the unit discharge air temperature reduces to the Discharge Air Temperature Minimum Cool Limit setting or DX cooling reaches 100% capacity, cooling capacity increases will hold while the supply fan speed will continue to increase to 100% or modulate to manage space comfort cooling.

Cool – Economizer

If the unit is configured for a modulating outdoor air damper and conditions are suitable for economizer cooling, the supply fan will operate at minimum speed while the economizer damper modulates between the Outdoor Air Damper Minimum Position Setpoint and 100% to satisfy the discharge air temperature setpoint. If the economizer damper reaches 100% open and additional cooling capacity is required, the supply fan will increase toward 100% to provide additional cooling to the space.

Cool – Economizer + DX

Economizer Cooling + DX is a mode of operation when both economizer and DX cooling are active. If actively economizing, outdoor air damper is 100% and supply fan speed reaches 100% then DX cooling will be added if the unit is not satisfying space cooling requirements.

If DX cooling is active and economizer cooling enables, the control will transition to increase economizer damper above minimum position to 100% to satisfy space cooling while decreasing DX cooling. DX cooling will steadily be removed as long as economizer cooling is able to satisfy the cooling load. The supply fan operates to the lowest speed possible during the transition.

Supply Air Tempering – Space Temperature Control

If the Supply Air Tempering function is configured and the Discharge Air Temperature local sensor is valid, this function prevents excessively cold discharge air from being supplied to the space. Supply Air Tempering is an option for VVZT and multi-speed supply fan units when auxiliary heat is installed.

Supply Air Tempering is not supported when a Conventional TStat is configured.

The following requirements must be met to allow Supply Air Tempering on a Staged Heat unit:

- The unit is in any heating mode, including Heat, Emergency Heat, Morning Warmup, Max Heat but is not actively heating OR
- The unit is in any cooling mode except night purge, but not actively cooling and cooling capacity has been OFF for 5 minutes.

If the discharge air temperature drops to the Discharge Air Temperature Minimum Cool Limit - Active and the Space Temperature is less than the Active Space Temp Cooling Setpoint Status – 0.5°F, the Supply Air Tempering function will bring ON one stage of auxiliary heat.

Supply Air Tempering terminates if the Discharge Air Temperature rises to 10°F ABOVE the Discharge Air Temperature Minimum Cool Limit - Active, or the Space Temperature rises to the Space Temp Cooling Setpoint Status. Additionally, if the Space Heat Control function determines that one or more stages of Heat are required to meet the Space Temp Heating Setpoint Status, Supply Air Tempering will terminate and the unit will stage heating to meet the current space demand.

Unoccupied Cooling

Unoccupied mode is used when the building is unoccupied, and the space conditions are exceeding temperature limits. The unit is normally off in unoccupied mode, but unoccupied operation is allowed in any heat cool mode except OFF.

If a valid space temperature input rises above the Unoccupied Cooling Setpoint, unit operation starts unoccupied cooling to manage space temperature. The controller operates in Single Loop Space Temperature Control with ventilation disabled. DX or economizer cooling will steadily increase to 100 percent capacity. The supply fan will follow capacity control until it reaches 100 percent fan speed. Cooling continues until space temperature is 4°F less than the Unoccupied Cooling Setpoint, the unit will then cycle off.

Unoccupied Heating

When the unit is in unoccupied mode and the valid space temperature input falls below the Unoccupied Heating Setpoint, unit operation starts unoccupied heating to manage space temperature. The controller operates in Single Loop Space Temperature Control with ventilation disabled. The supply fan starts increases with heating capacity unit both reach 100 percent. Heating continues until space temperature is 4°F greater than the Unoccupied Heating Setpoint, the unit will then cycle off.

Variable Volume Discharge Air Control Sequence of Operation

Variable Volume Discharge Air (VVDA) control uses available heating and cooling capacity to deliver the required temperature at the discharge of the unit. A discharge air temperature sensor is required for operation.

In cooling modes, the control uses cooling sources to deliver air temperature as required by the Discharge Air Cooling Setpoint. In heating modes, the control uses heat sources to deliver air temperature as required by the Discharge Air Heating Setpoint. Primary heat is never used in conjunction with mechanical cooling.

Symbio™ 700 operates in Discharge Air Temperature Control in all heat, cool, and occupancy modes of operation. The typical operating mode is Cooling delivering conditioned air to multiple zones of a building. These units have duct work to supply VAV terminal units. The VAV units modulate to control space comfort.

VVDA units also change over to provide heat (when installed). A building automation system or the VAV Box Relay Output can be used when the heat source requires full airflow. See Heat Types for more information. A valid space temperature input must be provided to perform heating modes such as Daytime Warm Up and Morning Warm Up (when enabled).

Supply Air Tempering

Supply Air Tempering is disabled by default but can be enabled if modulating heating capacity is configured. Supply air tempering initiates if discharge air temperature falls 1°F below the Discharge Air Cooling Setpoint Status setpoint and cooling capacity is 0% for 5 minutes and heating capacity is required to maintain discharge air cooling control. Heat is controlled with a 2°F dead band to the discharge air cooling setpoint. The control will transition back to normal cooling when heat capacity is 0% for 5 minutes or there is a compressor demand to manage the discharge air cooling control. Supply air tempering will also terminate if dehumidification initiates or if the controller effective occupancy is unoccupied.

Changeover Input

Variable Volume Discharge Air units include a binary Changeover Input. An open input requests discharge air cooling operation. A closed input will request the unit to change to discharge air heating. This local input is arbitrated with building management Heat Cool Mode Request commands to determine the active mode of operation.

VAV Box Relay Output

This binary output is provided by the controller to be optionally wired to VAV units in a Multi Zone-VAV system to coordinate system changeover to full airflow heating. The binary output is de-energized when the controller is operating in cooling modes. The binary output energizes when the heat type installed requires full airflow. See Maximum Heat for more details.

Discharge Air Reset

Discharge Air Reset is a method to save energy by resetting the discharge air temperature setpoint as heating and cooling building loads change. When enabled, the controller can independently reset the Discharge Air Heating Setpoint Active and/or Discharge Air Cooling Setpoint Active based on Zone (space temperature), Outdoor Air temperature or Return Air Temperature. The new target setpoints are reported as Discharge Air Setpoint Heating Status and Discharge Air Cooling Setpoint Status.

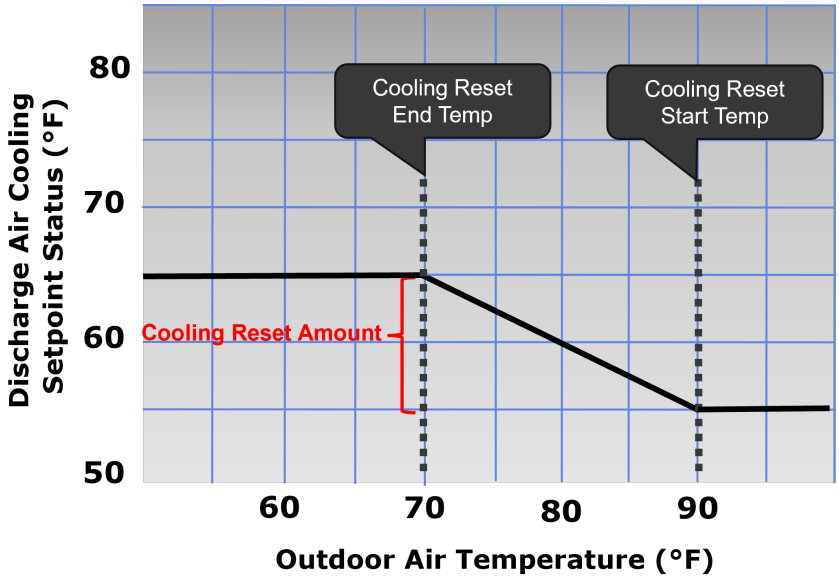
Discharge Air Cooling Reset

The following user adjustable points are used to calculate the Discharge Air Cooling Setpoint reset.

Point Reference	Value
Cooling Reset Type (MV-11111)	Outdoor Air
Cooling Reset Start Temp (AV-11150)	90 °F
Cooling Reset End Temp (AV-11151)	70 °F
Cooling Reset Amount (AV-11152)	10 °F
Discharge Air Cooling Setpoint BAS – Active (AV-11190)	55 °F
Discharge Air Cooling Setpoint Status (AV-11356)	Result of reset calculation

The active Discharge Air Cooling Setpoint BAS (55°F) will be reset 10°F over the range of outdoor air temperature of 90°F (Cooling Reset Start Temperature) to 70°F (Cooling Reset End Temperature). When outdoor air temperature is greater than 90° F, no reset is applied. When outdoor air temperature is 80°F, Discharge Air Cooling Setpoint – Active will be reset 5°F. As a result, discharge air temperature will be controlled to Discharge Air Cooling Setpoint Status of 60°F. When outdoor air temperature is 69°F, discharge air temperature will be controlled to Discharge Air Cooling Setpoint Status of 65°F.

Figure 10. Symbio 700 discharge air reset – cooling



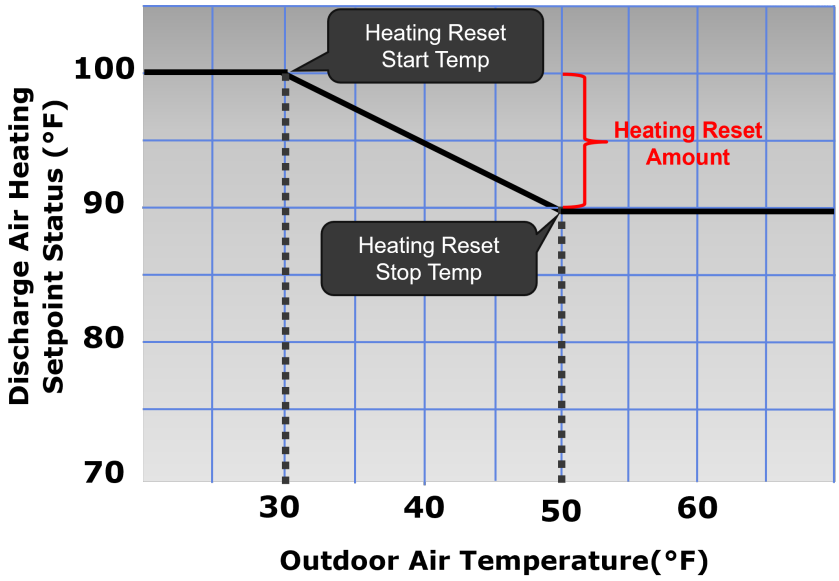
Discharge Air Heating Reset

The following user adjustable points are used to calculate the Discharge Air Heating Setpoint reset.

Point Reference	Value
Heating Reset Type (MV-11136)	Outdoor Air
Heating Reset Start Temp (AV-11241)	30 °F
Heating Reset End Temp (AV-11242)	50 °F
Heating Reset Amount (AV-11343)	10 °F
Discharge Air Heating Setpoint BAS – Active (AV-11191)	100 °F
Discharge Air Heating Setpoint Status (AV-11129)	Result of reset calculation

The active Discharge Air Heating Setpoint BAS (100°F) will be reset 10°F over the range of outdoor air temperature of 30°F (Heating Reset Start Temperature) to 50°F (Heating Reset End Temperature). When outdoor air temperature is less than 30°F, no reset is applied. When outdoor air temperature is 40°F, Discharge Air Heating Setpoint – Active will be reset 5°F. As a result, discharge air temperature will be controlled to Discharge Air Heating Setpoint Status of 95°F. When outdoor air temperature is 55°F, discharge air temperature will be controlled to Discharge Air Heating Setpoint Status of 90°F.

Figure 11. Symbio 700 discharge air reset – heating



Duct Static Pressure Control

Multi Zone-VAV equipment have variable speed supply fans and are often used in Variable Air Volume (VAV) systems consisting of ductwork serving multiple building zones and VAV boxes that control space comfort independent of the rooftop air handler. VAV boxes modulate air volume by a damper that opens and closes to maintain space comfort. In response, duct static pressure increases and decreases. The controller will modulate supply fan speed to maintain Duct Static Pressure relative to the Duct Static Pressure Setpoint.

Duct Static Pressure High Limit

The controller operates the supply fan to maintain duct static pressure below the Duct Static Pressure High Limit setpoint. In the event that duct static pressure approaches the high limit setpoint, the controller will reduce and limit the supply fan speed.

Unoccupied Cooling

Unoccupied Cooling mode is used when the building is unoccupied and the space conditions are exceeding temperature limits. The unit is normally off in unoccupied mode. If a valid space temperature input rises above the Unoccupied Cooling Setpoint, unit operation starts unoccupied cooling to manage space temperature. The controller operates in Discharge Air Control with ventilation disabled while DX or economizer cooling capacity increases to satisfy the Discharge Air Cooling Setpoint. VVDA System Type operate the supply fan under duct static pressure control. Cooling continues until space temperature is 4°F less than the Unoccupied Cooling Setpoint, the unit will then cycle off.

Unoccupied Heating

When the unit is in unoccupied mode and the valid space temperature input falls below the Unoccupied Heating Setpoint, unit operation starts unoccupied heating to manage space temperature. The controller operates in Discharge Air Control with ventilation disabled while heating capacity increases to satisfy the Discharge Air Heating Setpoint. The supply fan operates according to heat type installed, see Heat Types for more information. Heating continues until space temperature is 4°F greater than the Unoccupied Heating Setpoint, the unit will then cycle off.

Heat Cool Modes

Heat Cool Mode Status reports the unit mode of operation. The Symbio™ 700 can receive mode inputs from different external and local input sources that are arbitrated; however, the control active operating mode represents the capacity being delivered to the building and reported via Heat Cool Mode Status. The following modes of operation are supported. All other, unsupported modes are managed as an Auto mode request.

Heat

In this mode, the controls provide heating capacity per heat type installed, heat setpoint, and occupancy. Application requirements such as full or modulating air flow are also considered in heating mode. All forms of cooling capacity are effectively disabled.

If Heat is the requested mode when the unit does not have heat capacity configured or heat is disabled, Heat is reported as Heat Cool Mode Status. If the unit cannot provide heat, the supply fan is enabled to operate (as configured) to provide ventilation during occupied modes of operation.

Cool

Cool mode is reported when the control objective is to provide cooling to maintain building comfort. Direct expansion cooling is the primary cooling source. Cool mode is also reported when Economizer and Dehumidification cooling modes of operation are active.

Fan Only

This mode disables all forms of heating and cooling capacity but operates the fan continuously at minimum speed or modulates to maintain duct static pressure. The outdoor air damper modulates to damper minimum position setpoint to provide ventilation.

Fan Only is also reported in Emergency Override and Ventilation Override Modes. Heat Cool Mode Request can also command the control into Fan Only mode.

Off

Off is the reported mode when unit operation is shutdown due to diagnostics, equipment protections, overrides or normal unit operation when the supply fan is cycled off.

Test

When Service Test is active, Heat Cool Mode Status reports Test.

Maximum Heat

Maximum Heat is a full airflow mode of heating operation with the supply fan operating at maximum speed. The controller does not provide maximum heat; instead, the controller provides heat capacity per the Operating Mode and unit type.

VVDA

Variable Volume Discharge Air control units will transition to Maximum Heat operation in heating modes (for example Occupied Heat, Morning Warm Up, Daytime Warm Up) when the installed heating capacity requires full air flow. Maximum Heat can also be a Heat Cool Mode Request command, which Symbio™ 700 will remain in the mode until the mode is released.

On the transition to Maximum Heat the Symbio 700 will reduce the supply fan to minimum speed, energize the VAV Box Relay and wait the duration of the VAV Box Stroke Time (adjustable). The VAV Box Stroke Time allows the VAV boxes in the system to open. The supply fan speed will then increase to 100 percent or the maximum speed subject to high duct static pressure limit. Once the supply fan reaches its allowed maximum speed, Symbio 700 will enable heating capacity control to Discharge Air Heating Setpoint Active. Ventilation is managed per the operating mode

The transition from Maximum Heat back to modulating air flow control modes is as follows:

1. Heat capacity terminates (if active), a 90 second post heat timer begins.
2. Supply fan speed slows to minimum speed, a 3 minute minimum speed timer begins.
3. VAV Box Relay de-energizes.

4. When the 90 second post heat and 3 minute minimum speed timer expire, modulating air flow control is enabled, operating mode transitions, and capacity control enables.

Space Temperature Control

Space Temperature control units will also accept a Maximum Heat command via Heat Cool Mode Request. On this command the Symbio™ 700 will transition to a heat mode while operating the supply fan at 100 percent capacity. Heat Cool Mode Status will report Maximum Heat to indicate the mode is active.

Morning Warm-up

Morning Warm-up is an optional feature when heat is configured and can be Enabled or Disabled. Morning Warm-up is a mode often used during building unoccupied periods and optimal start applications to rapidly increase the space temperature, as efficiently as possible, before building occupancy. The outdoor air damper minimum position setpoint will be overridden to 0 percent, ventilation air is not provided during Morning Warm-up

For Morning Warm-up to initiate on a transition from Unoccupied to Occupied, Space temperature control units must be in Heat, Max Heat, Emergency Heat or Auto. Discharge air control units can also initiate Morning Warm Up on an Off to Occupied transition.

For all equipment types: On a transition from Unoccupied to Occupied or Occupied-Standby, and with a valid space temperature 1.5° F below the Morning Warm-up Setpoint, morning warm-up operation initiates. Heat Cool Mode Status reports Maximum Heat if the heat type requires full airflow and VAV Box Relay binary output will energize (see Maximum Heat for details). Otherwise, heat types that allow modulating airflow will report Morning Warm Up via Heat Cool Mode Status.

VVDA

The Symbio™ 700 operates in a discharge air control mode delivering heating capacity per Discharge Air Heating Setpoint Active; this provides flexibility when 100 percent heat capacity in not desired. Variable air volume discharge air control units will operate the supply fan per Duct Static Pressure Setpoint if the installed heat type allows; otherwise, the supply fan operates in a full airflow mode and Maximum Heat is reported via Heat Cool Mode Status. When 60 minutes expires or when space temperature equals or exceeds the Morning Warmup Setpoint, Morning Warm Up mode will terminate.

Morning Warm-up mode can also be commanded via Heat Cool Mode Request from a building automation system or external control. In this case, the unit remains in Morning Warm-up mode indefinitely until the commanded mode is removed or changed. When the space temperature exceeds the Morning Warm-up Setpoint, heating capacity is disabled but the supply fan operation continues to circulate air until the Morning Warm-up command is removed. Heat capacity will re-enable if space temperature falls 1.5°F below Morning Warm-up Setpoint.

Space Temperature Control

The Symbio™ 700 operates in a space temperature control mode controlling heating capacity to manage space temperature at the Morning Warm-up Setpoint. The supply fan operates at maximum speed.

When 60 minutes expire or when space temperature equals or exceeds the Morning Warm-up Setpoint, Morning Warm-up mode will terminate.

Morning Warm-up mode can also be commanded via Heat Cool Mode Request from a building automation system or external control. The Symbio 700 controller will continuously control space temperature to the morning warm-up setpoint until the mode is released or changed.

Pre-Cool

Pre-Cool is a mode often used during building unoccupied periods and optimal start applications to rapidly decrease the space temperature, as efficiently as possible, before building occupancy. Pre-Cool mode of operation can be enabled and disabled via Pre-Cool Enable Command. Pre-Cool is reported via Heat Cool Mode Status when active. Ventilation air is not provided during Pre-Cool operation.

VVDA

VVDA equipment types, a transition from Unoccupied to Occupied mode or Off/Shutdown to Occupied, if a valid space temperature input is above the Pre-Cool Setpoint + 1.5°F Pre-Cool Mode will initiate. Symbio™ 700 operates in a discharge air control mode providing discharge air per Discharge Air Cooling Setpoint Active until space temperature drops equal or below the Pre-Cool Setpoint. In this cooling mode the outdoor air damper minimum position is 0%, but the controller will use compressor and/or economizer cooling when suitable. VVDA units will modulate fan speed and manage to the duct static pressure setpoint.

Pre-Cool mode can also be commanded via Heat Cool Mode Request. In this case, the Symbio 700 remains in Pre-Cool indefinitely until the commanded mode is removed. When space temperature is equal or less than Pre-Cool Setpoint, cooling capacity is disabled but the supply fan continues to circulate air until the Pre-Cool command is removed. Cooling capacity control will enable if space temperature rises above Pre-Cool Setpoint + 1.5°F.

Space Temperature Control

Space temperature control units must be in Cool or Auto mode to allow Pre-Cool operation. A transition from Unoccupied to Occupied mode, the controls will initiate Pre-Cool if a valid space temperature input is above the Pre-Cool Setpoint + 1.5°F. The Symbio™ 700 operates in a space temperature control mode controlling space temperature to the Pre-Cool Setpoint. The supply fan operates at maximum speed. When 60 minutes expires or when space temperature is equal or below the Pre-Cool Setpoint, Pre-Cool mode will terminate.

Pre-Cool mode can also be commanded via Heat Cool Mode Request from a building automation system or external control. The Symbio 700 controller will continuously control space temperature to the Pre-Cool setpoint until the mode is released or changed.

Night Purge

Night Purge mode is typically applied in building unoccupied periods when conditions are suitable for economizer cooling; all other forms of cooling capacity are disabled. This mode is only supported via a commanded Heat Cool Mode Request. The controller will not use local inputs and assume outdoor air is suitable for economizing. The outdoor air damper minimum position setpoint will be overridden to
0 percent.

Night Purge operation is terminated when the Night Purge, Heat Cool Mode Request is removed.

VVDA

When active, the Symbio™ 700 operates in a discharge air control mode, modulating the outdoor air damper to provide cooling capacity per Discharge Air Cooling Setpoint Active.

Space Temperature Control

Space temperature control equipment modulates the outdoor air damper to control space temperature to the Space Temp Cooling Setpoint Status setpoint. If space temperature is greater than the space cooling setpoint + 1.5°F the outdoor air damper will be at 100 percent with supply fan at maximum speed. If space temperature is less than the space cooling setpoint -1.5°F the outdoor air damper will be closed with supply fan at minimum speed. Constant volume units, the supply fan operates at 100 percent while Night Purge is active.

Daytime Warm-up

Daytime Warm-up is a feature of VVDA units that can be Enabled, or Disabled. Daytime Warm-up is available with any heat type installed. Daytime Warm-up is a mode used during building occupied periods to recover a critical zone that is too cold. Ventilation air is provided while in Daytime Warm-up mode.

When the Heat Cool Mode Request is Cool (the unit is providing cooling), if a valid space temperature is 3°F below the adjustable Daytime Warm-up Setpoint, daytime Warm-up operation initiates. The unit will operate in a discharge air control mode delivering heating capacity per Discharge Air Heating Setpoint Active. The supply fan operates at max speed on Constant Volume Units. Variable air volume discharge air control units will operate the supply fan to the duct static pressure setpoint and report Heat Cool Mode Status as Heat, if the installed heat type allows modulating air flow. Else if the heat type requires full air flow, Maximum Heat is reported. (See Heat Types and Maximum Heat section for more information).

When the space temperature is equal to or above the Daytime Warm-up Setpoint, daytime Warm-up operation terminates and the Symbio™ 700 will transition back to Cool mode. See Maximum Heat for details. If the requested operating mode is no longer Cool, Daytime Warm-up terminates.

Dehumidification – Hot Gas Reheat

Dehumidification control is a mode of Cooling when hot gas reheat is configured. Symbio™ 700 supports three configurable dehumidification control options: Dew Point (default), Relative Humidity, or Humidistat control for occupied and unoccupied operation. Each has required sensors to be installed for the configurated system type as outlined in the following table.

Table 6. Dehumidification control options per system type

Dehumidification Control	Conventional Thermostat	CVZT/VVZT	VVDA	Required Sensors
Dew Point	—	X	X	Discharge Air Temperature Space Relative Humidity Space Temperature Outdoor Relative Humidity Outdoor Air Temperature Evaporator Entering Refrigerant Temperature
Relative Humidity	X	X	X	Discharge Air Temperature Space Relative Humidity Evaporator Entering Refrigerant Temperature
Humidistat	X	X	X	Discharge Air Temperature Evaporator Entering Refrigerant Temperature

The following Symbio 700 points are provided for dehumidification control.

Dehumidification Enable Command (BV-10106), the default setting is Auto for occupied dehumidification to operate based on the configured dehumidification control option. This setting is editable and when set to Disabled will disable dehumidification during occupied modes of operation.

Unoccupied Dehumidification Enable BAS (BV-11261), the default setting is Enabled for unoccupied dehumidification to operate based on the configured dehumidification control option. This setting is editable and when set to Disabled will disable dehumidification during unoccupied mode.

Dehumidification Status (BV-11255), binary status value that reports dehumidification control is Active or Inactive.

Dehumidification Control Status (AV-11281) reports the hot gas reheat valve status.

Dew Point

Symbio 700 references Space Temperature Active and Space Humidity Active to calculate Space Dew Point. Likewise, Outdoor Air Temperature Active and Outdoor Air Humidity Active are referenced to calculate Outdoor Air Dew Point.

Table 7. Dew point dehumidification points

Instance	Name
AV-11178	Space Temperature Active
AI-10116	Space Humidity Active
AI-10226	Space Dew Point
AV-11338	Space Dew Point Setpoint Active
AV-10228	Space Dew Point Unoccupied Setpoint Active
AV-11113	Space Dehumidification Setpoint Offset BAS - Active
AI-10118	Outdoor Air Temperature Active
AI-10120	Outdoor Air Humidity Active
AI-10117	Outdoor Air Dew Point
AV-11341	Outdoor Air Dew Point Setpoint BAS – Active
AV-11321	Dew Point Setpoint Offset
AV-11342	Unoccupied Dehumidification Timer

Occupied dew point dehumidification initiates:

- Space Dew Point is greater than the Space Dew Point Setpoint Active and
- Outdoor Air Dew Point is greater than the Outdoor Air Dew Point Setpoint BAS - Active

Occupied dew point dehumidification terminates:

- Space Dew Point is less than the Space Dew Point Setpoint Active minus the Dew Point Setpoint Offset or
- Outdoor Air Dew Point is less than the Outdoor Air Dew Point Setpoint BAS - Active minus the Dew Point Setpoint Offset

Unoccupied dew point dehumidification initiates:

- Space Dew Point is greater than the Space Dew Point Unoccupied Setpoint Active and
- Outdoor Air Dew Point is greater than the Outdoor Air Dew Point Setpoint BAS - Active and
- Space has been unoccupied for less than Unoccupied Dehumidification Timer and Space Humidity Active is greater than 65% or
- Space has been unoccupied for greater than or equal to Unoccupied Dehumidification Timer

The Unoccupied Dehumidification Timer is default to 12 hours. Setting the timer to 0 effectively disables the timer and, unoccupied dehumidification initiates only according to Space and Outdoor Dew Point.

Unoccupied dew point dehumidification terminates:

- Space Dew Point is less than the Space Dew Point Unoccupied Setpoint Active minus Dew Point Setpoint Offset or
- Outdoor Air Dew Point is less than the Outdoor Air Dew Point Setpoint BAS – Active minus Dew Point Setpoint Offset or
- Space has been unoccupied for less than Unoccupied Dehumidification Timer and Space Humidity Active is less than 65% minus Space Dehumidification Setpoint Offset BAS - Active

Relative Humidity

Occupied dehumidification control initiates when the Space Humidity Active is greater than the Space Dehumidification Setpoint BAS - Active. Occupied dehumidification terminates when Space Humidity Active is less than Space Dehumidification Setpoint BAS - Active minus Space Dehumidification Setpoint Offset BAS - Active.

Unoccupied dehumidification control initiates when the Space Humidity Active is greater than the Space Dehumidification Unoccupied Setpoint Active. Unoccupied dehumidification control terminates when Space Humidity Active is less than the Space Dehumidification Unoccupied Setpoint Active minus the Space Dehumidification Setpoint Offset BAS - Active.

Table 8. Relative humidity dehumidification points

Instance	Name
AV-11178	Space Temperature Active
AI-10116	Space Humidity Active
AV-11136	Space Dehumidification Setpoint BAS - Active
AV-10214	Space Dehumidification Unoccupied Setpoint Active
AV-11113	Space Dehumidification Setpoint Offset BAS - Active

Humidistat

Dehumidification operation initiates when Customer Options Module, Humidistat binary input is active. When Humidistat is used for Dehumidification control, compressors stage to 100% cooling capacity. Dehumidification operation terminates when Customer Options Module, Humidistat binary input is inactive. Reheat Humidistat Input (BI-11143) reports the status of the hardware input. If a Space Relative Humidity sensor is installed, the Humidistat input will be ignored.

Sequence of Operation

There are two independent processes in dehumidification control with hot gas reheat.

Latent Load Control – The latent load on a unit is the energy required to remove moisture from the air being delivered to space. Additional cooling capacity is added to ensure the evaporator coil is operating below dew point.

Sensible Load Control – The latent load control is adding cooling capacity to remove more moisture from the air; in response, hot gas reheat is utilized to temper the unit discharge air temperature to maintain space comfort; the sensible load.

If space relative humidity is rising, the unit cooling is not operating at a high enough capacity to reach evaporator coil dewpoint; therefore, additional cooling capacity is required. The following examples are for a Voyager 3 unit initially in a normal cooling mode.

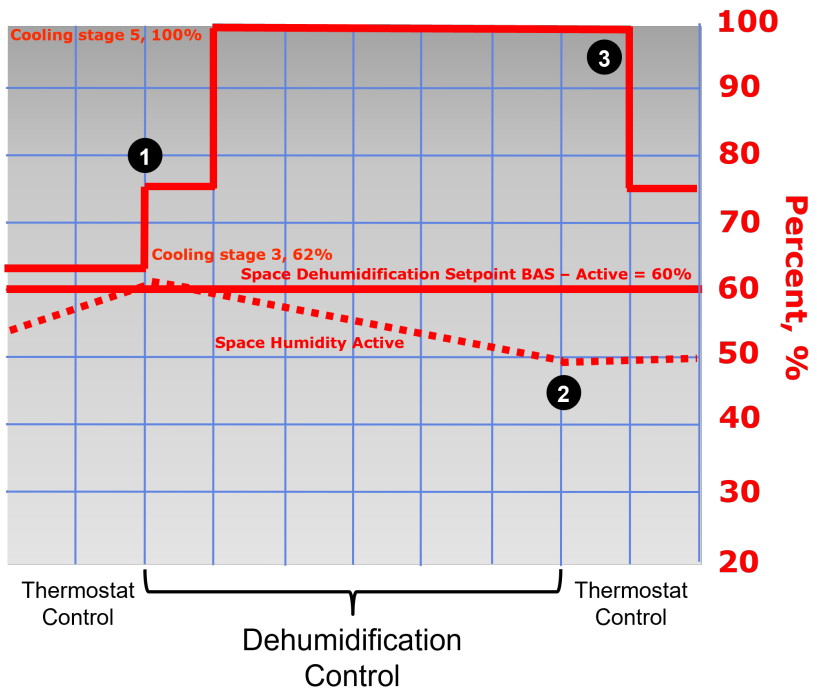
Conventional Thermostat Controlled Units

Dehumidification control can be activated utilizing a space relative humidity. These options are available via Symbio™ 700 configuration, see Appendix B for details. If Space Humidity is selected for dehumidification control, dehumidification activates when Space Humidity Active exceeds the Space Dehumidification Setpoint BAS - Active. If a Humidistat input is used, dehumidification initiates when the Symbio 700 Humidistat input becomes active.

When dehumidification activates, compressors stage to full capacity, economizer cooling is disabled. The reheat valve will open and modulate to an adjustable Discharge Air Reheat Setpoint BAS (AV-11314), 70°F default.

Conventional Thermostat Dehumidification – Latent Load Control

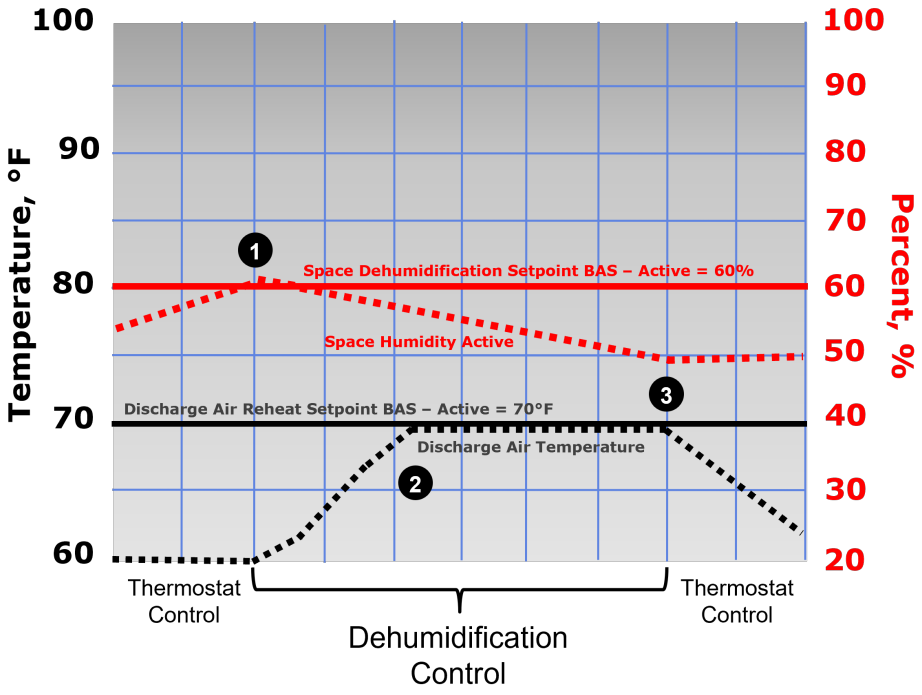
Figure 12. Convention thermostat dehumidification - latent load control



1. When Space Humidity Active exceeds the Space Dehumidification Setpoint BAS - Active or the Humidistat input becomes active, and Dehumidification Status becomes active. Compressor cooling capacity increases to full cooling capacity. The supply fan operates at the minimum defined speed for the active cooling capacity.
2. When Space Humidity Active is reduced below the defined dehumidification low limit or the Humidistat input becomes inactive, dehumidification disables, Dehumidification Status changes to inactive. In this example dehumidification terminates when Space Humidity Active drops below 50% (Space Dehumidification Setpoint BAS - Active minus Space Dehumidification Offset BAS - Active (10%) = 50%).
3. Compressor latent load control reverts to normal thermostat cooling control.

Conventional Thermostat Dehumidification – Sensible Load Control

Figure 13. Conventional thermostat dehumidification - sensible load control



1. When Space Humidity Active exceeds the Space Dehumidification Setpoint BAS - Active Dehumidification Status becomes active. Compressor cooling capacity increases to full cooling capacity, 100%. The supply fan operates at the minimum defined speed for the active cooling capacity. The hot gas reheat valve opens and initializes to 100% capacity. The reheat valve status is reported by Dehumidification Control Status (AV-11281).
2. The hot gas reheat valve modulates to control discharge air temperature to the Discharge Air Reheat Setpoint BAS (default 70°F), editable. The setpoint value being used for control is reported by Discharge Air Reheat Setpoint BAS – Active. Discharge air temperature will increase as the hot gas reheat valve opens.
3. When Space Humidity Active is reduced below the defined dehumidification low limit, dehumidification disables and Dehumidification Status changes to inactive. The discharge air temperature decreases as the hot gas reheat valve closes. In this example dehumidification terminates when Space Humidity Active drops below 50% (Space Dehumidification Setpoint BAS - Active minus Space Dehumidification Offset BAS - Active (10%) = 50%). Control reverts to normal thermostat cooling control.

Conventional Thermostat Dehumidification Termination

Dehumidification with hot gas reheat will terminate or disable in any of the following conditions.

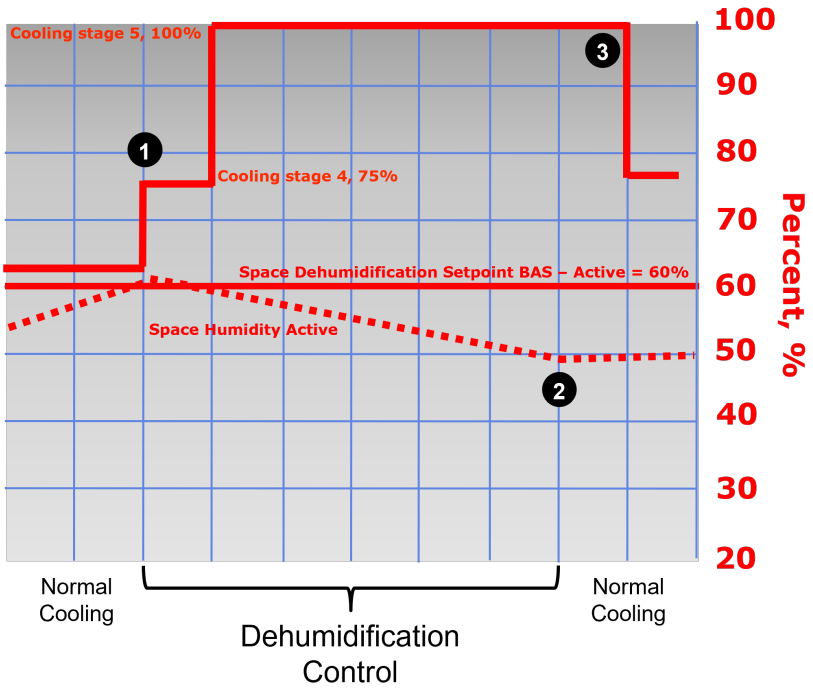
- Thermostat: Y1+Y2 call
- Thermostat: Heat call
- Evaporator Entering Refrigerant Temperature < 35°F, for 5 minutes all compressors will be commanded off. Evaporator Entering Refrigerant Temperature > 45°F, re-enables compressor operation.
- The outdoor air temperature is less than 40°F or greater than 100°F.
- Required sensor for the configured dehumidification control fails or is not installed.

CVZT/VVZT Units

Symbio™ 700 adds compressor cooling stages when dehumidification is requested. Hot gas reheat is added to offset the additional cooling and modulates to maintain space comfort.

CVZT/VVZT Dehumidification – Latent Load Control

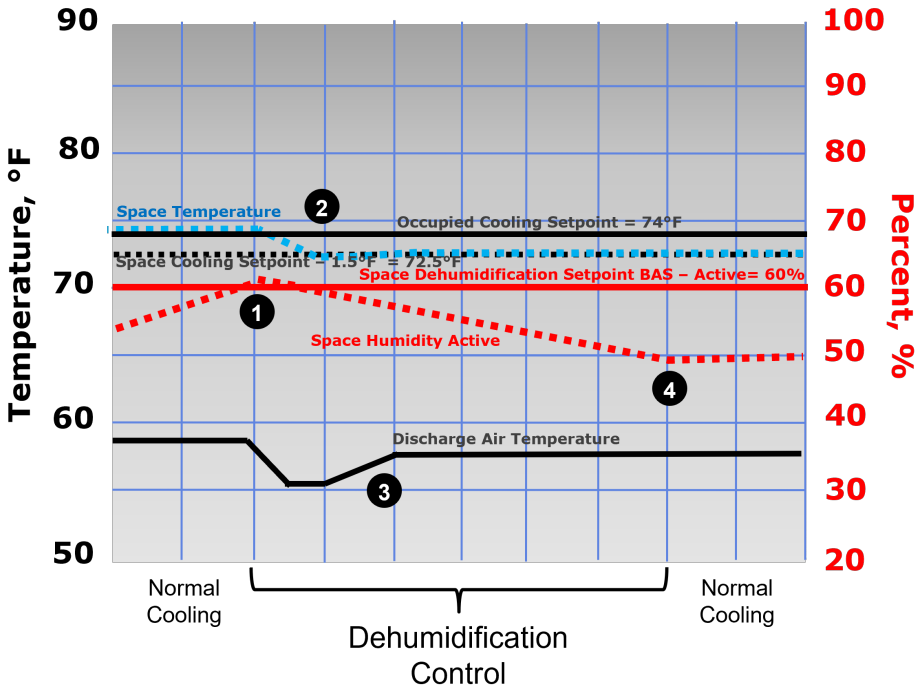
Figure 14. CVZT/VVZT dehumidification - latent load control



1. When Space Humidity Active exceeds the Space Dehumidification Setpoint BAS - Active (60% in this example), Dehumidification Status becomes active, and cooling stages are added to 100% capacity. Economizer cooling disables. In the example above, compressor cooling capacity increases from stage 3 (62%) to stage 4 (75%) to stage 5 (100%). This control response lowers the evaporator coil temperature below dewpoint, removing more moisture and delivering drier air to space. The supply fan speed is reduced if operating above the minimum speed defined for the compressor stage.
2. When Space Humidity Active is reduced below the defined dehumidification low limit, dehumidification disables, Dehumidification Status changes to inactive. In this example dehumidification terminates when Space Humidity Active drops below 50% (Space Dehumidification Setpoint BAS - Active minus Space Dehumidification Offset BAS - Active (10%) = 50%). Compressor latent load control reverts to normal cooling control.
3. Compressor stages unload as necessary to manage the space sensible load.

CVZT/VVZT Dehumidification – Sensible Load Control

Figure 15. CVZT/VVZT dehumidification - sensible load control



1. When Space Humidity Active exceeds the space dehumidification setpoint, Dehumidification Status becomes active, and cooling capacity increases. This will cause discharge air temperature to decrease and potentially space temperature to decrease.
2. If space temperature falls below the Occupied Cooling Setpoint – 1.5°F, the hot gas reheat valve will open and initialize to 100% capacity. Dehumidification Control Status reports the hot gas reheat valve capacity. The reheat valve modulates to control space temperature at Occupied Cooling Setpoint – 1.5°F (in this example 72.5°F). The hot gas reheat valve is controlled to deliver discharge air temperature to the Symbio 700 calculated Discharge Air Temperature Setpoint Active (AV-10281). As less hot gas reheat is required to maintain space temperature, the calculated Discharge Air Temperature Setpoint Active decreases.
3. Discharge Air Temperature will decrease with added compressor cooling stages, but Discharge Air Temperature will increase as the hot gas reheat valve opens.
4. When Space Humidity Active is reduced below the defined dehumidification low limit, dehumidification disables, Dehumidification Status changes to inactive, and the hot gas reheat valve will close. In this example dehumidification terminates when Space Humidity Active drops below 50% (Space Dehumidification Setpoint BAS - Active minus Space Dehumidification Offset BAS - Active (10%) = 50%). Control reverts to normal cooling.

Unoccupied Dehumidification

If the space temperature control unit is Off in unoccupied mode and there is a dehumidification demand due to dew point, relative humidity or humidistat; the Symbio 700 will start the unit to dehumidify the space. Heat Cool Mode Status will report Cool mode, Dehumidification Status will be active, and compressors will stage as in Occupied mode. In Unoccupied, hot gas reheat will control discharge air temperature to a space neutral 72.5°F until the dehumidification demand terminates. If the Heat Cool Mode Request is Off, the unit will remain off.

CVZT/VVZT Dehumidification Termination

Dehumidification with hot gas reheat will terminate or disable in any of the following conditions.

- Space temperature is less than 60°F
- Dehumidification Enable Command (point) is Disabled
- Unoccupied Dehumidification Enable BAS (point) is Disabled

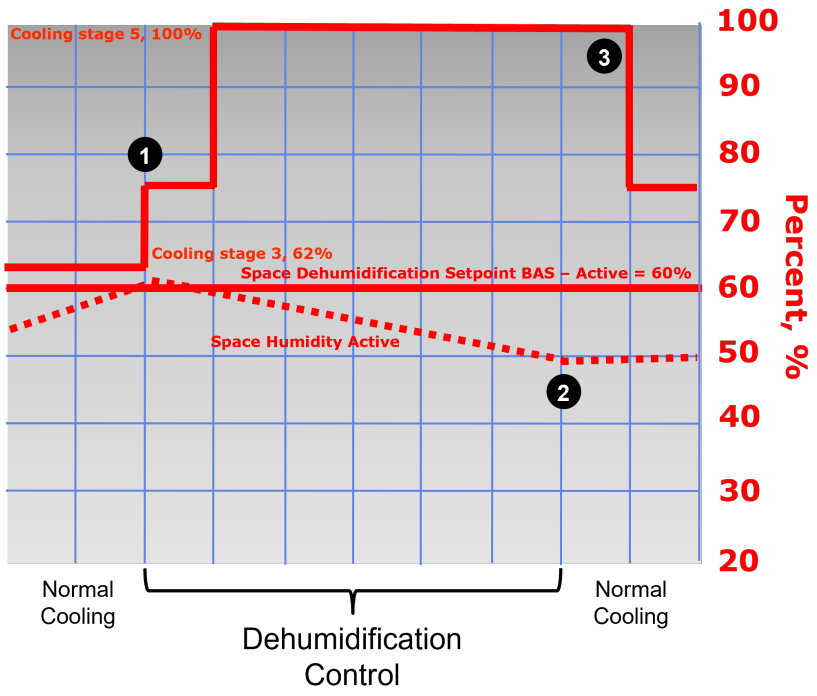
- Evaporator Entering Refrigerant Temperature < 35°F, for 5 minutes all compressors will be commanded off. Evaporator Entering Refrigerant Temperature > 45°F, re-enables compressor operation.
- The outdoor air temperature is less than 40°F or greater than 100°F.
- Required sensor for the configured dehumidification control fails or is not installed.

VVDA Units

Variable Volume Discharge Air units continue to modulate the supply fan to maintain duct static pressure control. Compressors will stage to achieve a calculated evaporator dew point temperature. The reheat valve modulates to maintain discharge air temperature at the Discharge Air Cooling Setpoint Status setpoint.

VVDA Dehumidification – Latent Load Control

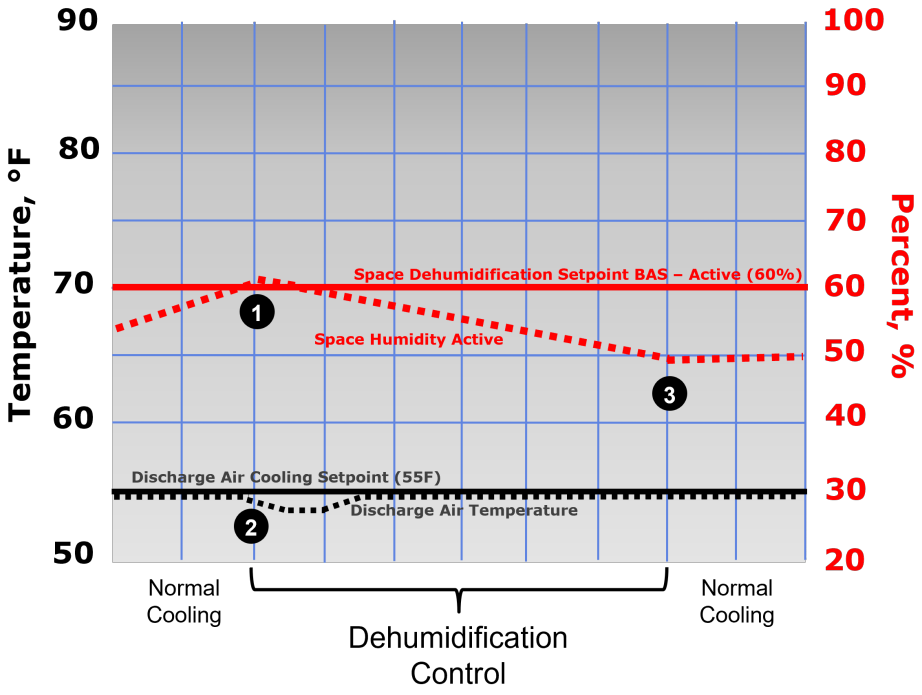
Figure 16. VVDA dehumidification - latent load control



1. When Space Humidity Active exceeds the Space Dehumidification Setpoint BAS - Active (60% in this example), Dehumidification Status becomes active, and a cooling stage is added. Economizer cooling disables. In the example above, compressor cooling capacity increases from stage 3 (62%) to stage 4 (75%) to stage 5 (100%). This control response attempts to lower the evaporator coil temperature below dewpoint, remove more moisture and deliver drier air to space. The supply fan continues duct static pressure control.
2. When Space Humidity Active is reduced below the defined dehumidification low limit, dehumidification disables, Dehumidification Status changes to inactive. In this example dehumidification terminates when Space Humidity Active drops below 50% (Space Dehumidification Setpoint BAS - Active minus Space Dehumidification Offset BAS - Active (10%) = 50%). Compressor latent load control reverts to normal cooling control.
3. Compressor stages unload as necessary, controlling to the Discharge Air Cooling Setpoint.

VVDA Dehumidification – Sensible Load Control

Figure 17. VVDA dehumidification - sensible load control



1. The unit is in normal discharge air cooling and controlling to a Discharge Air Cooling Setpoint Status (55°F). When Space Humidity Active exceeds the space dehumidification setpoint, Dehumidification Status becomes active, and cooling capacity increases.
2. Discharge Air Temperature will decrease with added the compressor cooling stages, and the hot gas reheat valve will open and modulate to maintain discharge air temperature to the Discharge Air Cooling Setpoint Status (55°F).
3. When Space Humidity Active is reduced below the defined dehumidification low limit, dehumidification disables, Dehumidification Status changes to inactive, and the hot gas reheat valve will close. In this example dehumidification terminates when Space Humidity Active drops below 50% (Space Dehumidification Setpoint BAS - Active minus Space Dehumidification Offset BAS - Active (10%) = 50%). Unit reverts to normal discharge air cooling.

Unoccupied Dehumidification

A space temperature sensor is required for a VVDA (multi-zone VAV) unit to operate in an unoccupied mode (standalone). Unoccupied dehumidification is only supported when the unit is operating to satisfy a space cooling demand. Once the space cooling demand is satisfied, if there is a space dehumidification demand, dehumidification will become active. Heat Cool Mode Status will report Cool mode, Dehumidification Status will be active, and compressors will stage as in Occupied mode. In Unoccupied, hot gas reheat will control discharge air temperature to a space neutral 72.5°F until the dehumidification demand terminates. If the Heat Cool Mode Request is Off, the unit will remain off.

VVDA Dehumidification Termination

Dehumidification with hot gas reheat will terminate or disable in any of the following conditions.

- Space temperature, when installed, is less than 60°F and the reheat valve is at full capacity for 10 minutes.
- Dehumidification Enable Command (point) is Disabled
- Unoccupied Dehumidification Enable BAS (point) is Disabled
- Evaporator Entering Refrigerant Temperature < 30°F, for 5 minutes, all compressors will be commanded off. Evaporator Entering Refrigerant Temperature > 45°F, re-enables compressor operation.
- The outdoor air temperature is less than 40°F or greater than 100°F.
- Required sensor for the configured dehumidification control fails or is not installed.

Dehumidification Purge Cycle

During dehumidification, a 3-minute compressor cooling purge cycle initiates when there is no change in the reheat pumpout solenoid according to the Reheat Purge Interval (AV-11315), 120 minutes adjustable. If the state of the reheat pumpout solenoid changes, the Reheat Purge Internal timer is reset. During the dehumidification purge cycle, the modulating hot gas reheat valves move to 50% position and compressors go to full capacity.

Comfort Purge Cycle

During normal compressor cooling, a 3-minute compressor cooling purge cycle initiates when the reheat circuit accumulates operation according to the Comfort Purge Interval (AV-11314), 90 minutes adjustable. If the state of the reheat pumpout solenoid changes, the Reheat Purge Internal timer is reset. The modulating hot gas reheat valves move to a 50% position and compressors go to full capacity.

Reheat Pumpout Solenoid

During cooling modes of operation (reheat valve closed), a reheat pumpout solenoid will energize to remove refrigerant from the reheat coil. For heat pump configurations the Reheat Pumpout Relay will be energized during compressor heating operation and demand defrost. In all other modes, the reheat pumpout solenoid is de-energized. Reheat Pumpout Solenoid Status (BI-11187) reports the Off/On status of the solenoid.

Heat Types

Three heating types are supported:

- Staged gas
- Modulating gas heat
- Staged electric

Single zone VAV and multi-speed constant volume units will increase supply fan speed with increased heating capacity. Multi-Zone VAV units will operate the supply fan at full airflow when staged heating is configured, see Maximum Heat for details. The supply operates in duct static pressure control when modulating heat is configured.

All heat types, when the control terminates heating capacity or exists a heating mode, a post heat timer is enforced. This keeps supply fan On to remove heat from the unit before transitioning to a cooling mode or cycling the supply fan off.

Staged Gas Heat

Symbio™ 700 supports two stages of gas heat as primary heat capacity with a Modbus interface to a VB1287 staged ignition controller. The Modbus interface provides operating details and diagnostics to the Symbio 700 controller.

- **Gas heat capacity (350, 400 MBh) —** has one manifold with one burner with a 2-stage gas valve.
- **Gas heat capacity (600 MBh) —** has a split manifold and two, 2-stage gas valves.
- **Gas heat capacity with two 400 MBh furnaces in series (800 MBh) —** the first furnace has a VB1287 staged ignition controller with a 2-stage gas valve and the second furnace has a second VB1287 staged ignition controller with a two-stage gas valve.

All 2-stage valves in the system are controlled the same: all driven off, low-fire, or high-fire. The first stage, low-fire, is approximately 75% of total heat capacity. Except 600 MBh, low-fire is approximately 66% of total heat capacity.

Sequence of Operations

1. An initial call for heat the Symbio 700 creates a heat demand via Modbus to the gas heat ignition controller(s)
2. A draft inducer is energized, an air pressure switch closes, and a pre-purge initiates for 15 seconds.
3. At end of pre-purge period, the ignition source is activated with two-stage gas valve(s) at minimum fire, and the main gas valve opens for trial for ignition
4. Burners ignite (and cross light) and flame is sensed.
5. After a warm-up period of 10 seconds, gas heat is managed under Symbio 700 control at stage (low fire) on all burners (Running state).
6. With increasing heating demand, Symbio 700 will command the ignition controller(s) to high fire, stage 2 full capacity.
7. Symbio 700 enforces a three-minute period between increasing or decreasing gas heat stages. The three-minute interstage timer is not applicable to conventional thermostat-controlled units.

As heat demand decreases:

1. Gas heat will be commanded from high fire to low fire.
2. If there is no heat demand, gas heat is staged off.
3. The draft inducer will remain on for a post purge time of 30 seconds.

The 800 MBh gas heat, two furnace configuration has added redundancy. If the first staged furnace is not available or locked out, the Symbio 700 will control the second staged furnace to satisfy the heating demand. Likewise, if the second staged furnace is not available or locked out, first staged furnace is controlled to satisfy heating demand.

Modulating Gas Heat

Symbio™ 700 supports modulating gas heat for primary heat with a Modbus interface to a VB1285 modulating ignition controller. The Modbus interface provides operating details and diagnostics to the Symbio 700 controller.

- **Gas heat capacity (350, 400, 600 MBh) with a single furnace —** has a main gas valve, modulating gas valve (burner 1) and a single-stage gas valve (burner 2) on a single split manifold.
- **Gas heat capacity with two 400 MBh furnaces in series (800 MBh) —** the first furnace has a VB1285 modulating ignition controller with a modulating gas valve and the second furnace has a VB1287 staged ignition controller with a two-stage gas valve.

Sequence of Operation

1. An initial call for heat the Symbio 700 creates a heat demand via Modbus to the gas heat ignition controller(s).
2. A draft inducer is energized, an air pressure switch closes for the initial pre-purge for 30 seconds.
3. The modulating gas valve opens to approximately 35%, depending on gas heat size.
4. At end of pre-purge period, the ignition source is activated, and the main gas valve opens for the trial for ignition period (up to 8 seconds).
5. The modulating burner ignites, and flame is sensed.
6. After a warm-up period of 10 seconds, “run” is displayed on the VB1285 ignition controller and gas heat is managed under Symbio 700 control

Single furnace (350, 400, 600 MBh)

1. The Symbio 700 commands the modulating burner open with increasing heat demand until Heat Capacity Primary Status = 50%.
2. At this capacity, if there is a continued demand for heat, the staged gas valve opens on burner 2. Symbio 700 enforces a three-minute period between adding or subtracting a stage on the split manifold (burner 2).
3. When flame is sensed on burner 2, the modulating gas valve reverts to minimum fire and Heat Capacity Primary Status will report 60% (natural gas) or 67% (liquid propane). If flame is not sensed on the split manifold burner 2, gas heat will continue on the primary burner and continue to call for heat on the split burner as required.
4. The modulating gas valve will again open with increasing heat demand.

As heat demand decreases:

1. The modulating burner will decrease capacity.
2. At Heat Capacity Primary Status = 60% (natural gas) or 67% (liquid propane), the second staged burner is turned off.
3. The modulating burner will increase to full capacity before decreasing with decreasing heat demand.
4. If there is no heat demand, gas heat is staged off.
5. The draft inducer will remain on for a post purge time of 30 seconds.

Dual furnace (800 MBh)

1. The Symbio 700 commands the VB1285 modulating burner open with increasing heat demand until Heat Capacity Primary Status = 47%.
2. At this capacity, if there is a continued demand for heat, the VB1287 staged gas valve is commanded to low fire on furnace 2.
3. When flame is sensed on burner 2, the VB1285 modulating gas valve reduces its firing rate to a minimum position.
4. The modulating gas valve will resume opening with increasing heat demand until 74% (natural gas) or 76% (liquid propane).
5. If there is a continued demand for heat, the VB1287 staged gas valve opens to high fire on furnace 2.
6. The VB1285 modulating gas valve position reduces the firing rate, but the valve will continue to open with increasing heat demand to 100% heat capacity.
7. Symbio 700 enforces a three-minute period between adding or subtracting stages of heating on the VB1287 staged ignition controller.

As heat demand decreases:

1. The modulating burner will decrease capacity and the staging process is reversed.
2. When Primary Heating Capacity Status = 75% (natural gas) or 77% (liquid propane), if heat demand continues to decrease, the VB1287 staged burner is commanded from high fire to low fire.
3. In response, the modulating burner will increase in capacity.
4. The modulating burner will then decrease with reduced heat demand.
5. At Primary Heating Capacity Status = 48% (natural gas) or 55% (liquid propane), if heat demand continues to decrease, the VB1287 staged burner is commanded Off.
6. The modulating burner will increase in capacity before decreasing with heat demand.
7. If there is no heat demand, the modulating gas valve and main gas valve is command off.

8. The draft inducer will remain on for a post purge time of 30 seconds.

The 800 MBh gas heat, two furnace configuration has added redundancy. If the first modulating furnace is not available or locked out, the Symbio 700 will control the second staged furnace to satisfy the heating demand. Likewise, if the staged furnace is not available or locked out, modulating furnace is controlled to satisfy heating demand.

Supply Fan Operation

VVZT/CVZT

The supply fan operates at a minimum speed of 58% when gas heat is less than 15% capacity. When gas heat capacity is 15% or greater the supply fan speed will increase with capacity on a defined supply fan curve. If Discharge air temperature reaches the heating maximum setpoint before the supply fan is at 100%, the supply fan speed will increase to provide additional heating capacity.

VVDA

The supply fan operates under duct static pressure control.

Electric Heat – Staged

When staged electric heat is configured in the unit, the controller manages two binary outputs to provide 2-stages of heat control; depending on size of the electric heat installed. When 2 stages of electric heat are installed each stage of electric heat is 50% of available heating capacity.

Outdoor Air Damper Control

Economizer Cooling

Symbio™ 700 supports a 0 to 100 percent economizer damper which requires a discharge air temperature sensor to be installed for economizer cooling. There are four configurable economizer types: fixed dry bulb, differential dry bulb, reference enthalpy, and comparative enthalpy. Each type, enable, and the high limit disable criteria are defined as follows.

Table 9. Economizer types

Type	Economizer Status	Calculation
Fixed Dry Bulb	Enable	Outdoor Air Temperature < Economizer Outdoor Air Enable Setpoint – Economizer Dry Bulb Enable Offset
	Disable	Outdoor Air Temperature > Economizer Outdoor Air Enable Setpoint
Differential Dry Bulb	Enable	Outdoor Air Temperature < (Return Air Temperature - Economizer Dry Bulb Enable Offset - Economizer Dry Bulb Disable Return Air Offset)
	Disable	Outdoor Air Temperature < (Return Air Temperature - Economizer Dry Bulb Disable Return Air Offset)
Reference Enthalpy	Enable	Outdoor Air Enthalpy < (Economizer Outdoor Air Enthalpy Enable Setpoint – Economizer Cooling Reference Enthalpy Offset) AND Outdoor Air Temperature < (Economizer Outdoor Air Enable Setpoint – Economizer Dry Bulb Enable Offset)
	Disable	Outdoor Air Enthalpy > Economizer Outdoor Air Enthalpy Enable Setpoint OR Outdoor Air Temperature > Economizer Outdoor Air Enable Setpoint
Comparative Enthalpy	Enable	Outdoor Air Enthalpy < (Return Air Enthalpy – Economizer Cooling Reference Enthalpy Hysteresis Offset) AND Outdoor Air Temperature < (Economizer Outdoor Air Enable Setpoint - Economizer Dry Bulb Enable Offset)
	Disable	Outdoor Air Enthalpy > Return Air Enthalpy OR Outdoor Air Temperature > Economizer Outdoor Air Enable Setpoint

While the configuration parameters will be used to determine which method of economizer control will be utilized, the table below describes the needed sensor data for each control method.

Table 10. Sensor data

Economizer Enable Method	Required Sensor Data
Comparative Enthalpy	Outdoor Air Temperature
	Outdoor Air Humidity
	Return Air Temperature
	Return Air Humidity
Reference Enthalpy	Outdoor Air Temperature
	Outdoor Air Humidity
Dry Bulb	Outdoor Air Temperature
Differential Dry Bulb	Outdoor Air Temperature
	Return Air Temperature

Outdoor Air Damper Control

When conditions are suitable for economizer operation, the outdoor air damper modulates between a calculated outdoor air damper minimum position (based on Supply Fan Compensation and Demand Controlled Ventilation) and 100 percent open. Economizing will not allow additional mechanical cooling until the damper position is 100 percent and supply fan has reached 100 percent capacity for 5 minutes. If economizer cooling becomes disabled, the damper will revert to minimum position control, and transition to mechanical cooling.

A building automation system can directly command economizer operation via Economizer Airside Enable (auto, enable, disable). If commanded Enable, the controller will start economizer cooling, regardless of outdoor air conditions. If commanded Disable, economizer cooling will be disabled (except if a mode of Night Purge is commanded to the controller). If commanded Auto, the controller will use the configured Economizer high limit method and input values to determine if economizer cooling is available.

Ventilation Control

On equipment installed with a 0 to 100% Economizer or 0 to 100% Economizer w/Traq, the Symbio™ 700 will control the outdoor air damper to provide minimum ventilation requirements based on the specific options installed, enabled features and mode of operation. During normal occupied periods of heating and cooling modes of operation, the outdoor air damper maintains ventilation requirements. However, the following modes of operation will override the damper minimum position setpoint to 0%.

- Morning Warm Up
- PreCool
- Night Purge
- Unoccupied Heat
- Unoccupied Cool
- Off

0 to 100% Economizer

The following sections describe all the options and functions that can be installed and enabled with a 0 to 100% Economizer damper. Tables are provided at the bottom of this section to summarize the setpoints and settings used for all combinations.

Supply Fan Compensation

The outdoor air damper minimum position is modulated to provide outdoor air based on supply fan speed. When the supply fan increases speed, the outdoor air damper minimum position is reduced to prevent over ventilation. When the supply fan decreases speed the outdoor air damper minimum position is increased to maintain design outdoor air requirements. There are three user editable design minimum settings to linearize damper position with the fan curve during Occupied and Occupied Bypass modes of operation.

- Design Minimum OA Damper Position at Min Fan Capacity (25 percent default)
- Design Minimum OA Damper Position at Mid Fan Capacity (15 percent default)
- Design Minimum OA Damper Position at Full Fan Capacity (10 percent default)

There are three additional user editable design minimum settings for Occupied-Standby mode of operation.

- Standby Minimum OA Damper Position at Min Fan Capacity (25 percent default)
- Standby Minimum OA Damper Position at Mid Fan Capacity (15 percent default)
- Standby Minimum OA Damper Position at Full Fan Capacity (10 percent default)

Outdoor Air Minimum Position Control

Available only when supply fan compensation is disabled, this function provides a fixed damper position for minimum outdoor air requirements. The damper is controlled to a position determined by Economizer Minimum Position Setpoint.

Remote Minimum Position Control

With an installed Customer Connection Module and Remote Minimum Position is configured in the Symbio™ 700, a wired potentiometer can be used to adjust the outdoor air damper minimum position setpoint in the range of 0 to 50%. The setting is reported via Remote Minimum Position. If Remote Minimum Position input and Demand Controlled Ventilation is installed, the Remote Minimum Position provides the minimum damper position setpoint at full fan capacity. Tables below provide details of when the Remote Minimum Position input is utilized in minimum ventilation control.

When Remote Minimum Position is installed with Demand Controlled Ventilation, the Remote Minimum Position input is used in place of Design Minimum OA Damper Position at Full Fan Capacity setpoint. See tables below for more details.

Demand Controlled Ventilation (DCV)

Demand controlled ventilation reduces energy consumption by reducing the outdoor air damper below design minimum ventilation based on space CO_2. When Demand Controlled Ventilation is configured and Supply Fan Compensation is enabled, DCV resets the outdoor air damper minimum position based on space CO_2 and supply fan speed. Decreasing CO_2 levels will decrease damper position below the Design Minimum toward the DCV minimum damper position setpoint. Increasing CO_2 level will increase damper position toward design minimum setpoint. DCV requires a valid space CO_2 value from a building

management system or wired sensor. If Space CO_2 value is invalid or Supply Fan Compensation is disabled, the Symbio™ 700 will revert to Outdoor Air Minimum Position Control.

Demand controlled ventilation setpoints used in all methods.

- Space CO_2 High Limit
- Space CO_2 Low Limit

Occupied and Occupied-Bypass mode. If the supply fan is at 100 percent and CO_2 is at the Space CO_2 High Limit (1500 ppm default), the outdoor air damper will be positioned at Design Minimum OA Damper Position at Full Fan Capacity (10 percent default). As CO_2 in the space reduces below the high limit, the outdoor air damper will close. If CO_2 falls below the Space CO_2 Low Limit, the damper position will be at DCV Minimum OA Damper Position at Full Fan Capacity (5 percent default).

Occupied-Standby mode. The damper will reset, based on CO_2, between the Standby Minimum OA Damper Position at Full/Mid/Min Fan Capacity and DCV Minimum OA Damper Position at Full/Mid/Min Fan Capacity.

The following tables define the Symbio 700 setpoints for Demand Controlled Ventilation control, which depend on the features enabled and configuration options installed. Each table is based on the supply fan and damper type installed. Supply Fan Compensation is a feature that can be enabled or disabled. Demand Controlled Ventilation and Remote Minimum Position are Symbio 700 configurations.

Table 11. 0 to 100 percent economizer – variable speed supply fan

Supply Fan Compensation	Demand Controlled Ventilation	Remote Minimum Position	Occupancy Status	Outdoor Air Damper Controlling Setpoints
Enabled	Installed	Installed	Occupied, Occupied Bypass	Remote Minimum Position (Full Fan Capacity) Design Minimum OA Damper Position at Mid Fan Capacity Design Minimum OA Damper Position at Min Fan Capacity DCV Minimum OA Damper Position at Full Fan Capacity DCV Minimum OA Damper Position at Mid Fan Capacity DCV Minimum OA Damper Position at Min Fan Capacity
Enabled	Installed	Not Installed	Occupied, Occupied Bypass	Design Minimum OA Damper Position at Full Fan Capacity Design Minimum OA Damper Position at Mid Fan Capacity Design Minimum OA Damper Position at Min Fan Capacity DCV Minimum OA Damper Position at Full Fan Capacity DCV Minimum OA Damper Position at Mid Fan Capacity DCV Minimum OA Damper Position at Min Fan Capacity
Enabled	Not Installed or Disabled	Installed	Occupied, Occupied Bypass	Remote Minimum Position (Full Fan Capacity) Design Minimum OA Damper Position at Mid Fan Capacity Design Minimum OA Damper Position at Min Fan Capacity
Enabled	Not Installed or Disabled	Not Installed	Occupied, Occupied Bypass	Design Minimum OA Damper Position at Full Fan Capacity Design Minimum OA Damper Position at Mid Fan Capacity Design Minimum OA Damper Position at Min Fan Capacity
Enabled	Installed	Installed or Not Installed	Occupied Standby	Design Minimum OA Damper Position at Full Fan Capacity Design Minimum OA Damper Position at Mid Fan Capacity Design Minimum OA Damper Position at Min Fan Capacity Standby Minimum OA Damper Position at Full Fan Capacity Standby Minimum OA Damper Position at Mid Fan Capacity Standby Minimum OA Damper Position at Min Fan Capacity

Table 11. 0 to 100 percent economizer – variable speed supply fan (continued)

Supply Fan Compensation	Demand Controlled Ventilation	Remote Minimum Position	Occupancy Status	Outdoor Air Damper Controlling Setpoints
Disabled	Installed or Not Installed	Installed or Not Installed	Occupied, Occupied Bypass, Occupied Standby	Economizer Minimum Position Setpoint

Table 12. 0 to 100 percent economizer – multi-speed supply fan (2–speed)

Supply Fan Compensation	Demand Controlled Ventilation	Remote Minimum Position	Occupancy Status	Outdoor Air Damper Controlling Setpoints
Enabled	Installed	Installed	Occupied, Occupied Bypass	**Supply Fan at 100%:** Remote Minimum Position DCV Minimum OA Damper Position at Full Fan Capacity **Supply Fan at minimum speed:** Design Minimum OA Damper Position at Min Fan Capacity DCV Minimum OA Damper Position at Min Fan Capacity
Enabled	Installed	Not Installed	Occupied, Occupied Bypass	**Supply Fan at 100%:** Design Minimum OA Damper Position at Full Fan Capacity DCV Minimum OA Damper Position at Full Fan Capacity **Supply Fan at minimum speed:** Design Minimum OA Damper Position at Min Fan Capacity DCV Minimum OA Damper Position at Min Fan Capacity
Enabled	Not Installed or Disabled	Installed	Occupied, Occupied Bypass	**Supply Fan at 100%:** Remote Minimum Position **Supply Fan at minimum speed:** Design Minimum OA Damper Position at Min Fan Capacity
Enabled	Not Installed or Disabled	Not Installed	Occupied, Occupied Bypass	**Supply Fan at 100%:** Design Minimum OA Damper Position at Full Fan Capacity **Supply Fan at minimum speed:** Design Minimum OA Damper Position at Min Fan Capacity
Enabled	Installed	Installed or Not Installed	Occupied Standby	**Supply Fan at 100%:** Standby Minimum OA Damper Position at Full Fan Capacity DCV Minimum OA Damper Position at Full Fan Capacity **Supply Fan at minimum speed:** Standby Minimum OA Damper Position at Min Fan Capacity DCV Minimum OA Damper Position at Min Fan Capacity
Enabled	Not Installed or Disabled	Installed or Not Installed	Occupied Standby	**Supply Fan at 100%:** Standby Minimum OA Damper Position at Full Fan Capacity **Supply Fan at minimum speed:** Standby Minimum OA Damper Position at Min Fan Capacity

Table 12. 0 to 100 percent economizer – multi-speed supply fan (2–speed) (continued)

Supply Fan Compensation	Demand Controlled Ventilation	Remote Minimum Position	Occupancy Status	Outdoor Air Damper Controlling Setpoints
Disabled	Installed or Not Installed	Installed or Not Installed	Occupied, Occupied Bypass, Occupied Standby	Economizer Minimum Position Setpoint BAS

0 to 100% Economizer with TRAQ

The outdoor air damper is modulated to provide outdoor air flow (cfm) required by the Outdoor Air Minimum Flow setpoint. See Table 13, p. 46 for addition information and settings.

The provided air flow measurement capabilities include periodic, automatic calibration; however, air flow gain and offset settings are also provided for field adjustment.

Demand Controlled Ventilation with TRAQ

When Demand Controlled Ventilation (DCV) and 0 to 100% Economizer with Traq are installed, DCV calculates the Minimum OA Flow Target (setpoint) based on space CO_2. When Space CO_2 is at or above the Space CO_2 High Limit, the Minimum OA Flow Target will be at Design Min Outdoor Air Flow setpoint. When Space CO_2 reduces and is below the high limit, Minimum OA Flow Target is calculated to a lower setpoint. When Space CO_2 is at or below the Space CO_2 Low Limit, the Minimum OA Flow Target will be at DCV Min Outdoor Air Flow setpoint. The following table summarizes the setpoints and settings when Demand Control Ventilation is installed or not installed.

If Demand Controlled Ventilation is installed but is disabled because a Space CO_2 Sensor failure (for example), Traq damper control will revert to Outdoor Air Minimum Flow Setpoint BAS.

Table 13. 0 to 100% Economizer with TRAQ

Demand Control Ventilation	Description	Outdoor Air Damper Controlling Setpoints
Not installed or disabled	Damper controls to outdoor air minimum flow setpoint	**Flow Setpoints:** Outdoor Air Minimum Flow Setpoint BAS Outdoor Air Minimum Flow Setpoint BAS - Active
Installed	Outdoor Air Flow setpoint reset based on space CO_2.	**Flow Setpoint:** Minimum OA Flow Target **Settings:** Design Min Outdoor Air Flow Setpoint DCV Min Outdoor Air Flow Setpoint

0 to 50 percent Motorized Damper

When a 0 to 50 percent motorized damper is installed, with any supply fan type, the outdoor air damper is controlled to a setpoint from a Remote Minimum Position (wired input), Motorized Damper Position Setpoint, or Economizer Minimum Position Setpoint BAS. The following table summarizes which setpoint is in control of the damper position based on options installed and enabled.

Table 14. 0 to 50 percent motorized damper

Supply Fan Compensation	Remote Minimum Position	Occupancy Status	Outdoor Air Damper Controlling Setpoints
Enabled	Installed	Occupied, Occupied Standby	Remote Minimum Position
Enabled	Installed	Unoccupied	0% (damper closed)
Enabled	Not Installed	Any	Motorized Damper Position Setpoint
Disabled	Installed or Not Installed	Any	Economizer Minimum Position Setpoint BAS

Demand Controlled Ventilation (DCV) – Thermostat Control

When the Symbio™ 700 is configured for Conventional Thermostat Control, 0 to 100 percent Economizer Damper and Demand Controlled Ventilation (Installed), the controller provides an Occupancy binary input that can be used to control Occupied and Unoccupied modes of operation. In Unoccupied mode, DCV is disabled, and the outdoor air damper minimum position is effective
0 percent. In Occupied mode, DCV will control the outdoor air damper based on Space CO_2 while the supply fan is ON, as described in the previous sections.

In Occupied mode, if the supply fan cycles OFF, the controller will continue to monitor Space CO_2. If Space CO_2 exceeds the Space CO_2 High Limit setpoint for 15 minutes, the supply fan will turn ON and operate at minimum speed and outdoor air damper at the Design Minimum OA Damper Position. If Space CO_2 falls below the Space CO_2 High Limit – 200 ppm, the supply fan will cycle OFF. The supply fan will also cycle off if occupancy changes to Unoccupied.

Relief Damper Control – Statitrac

Symbio™ 700 Statitrac control is a highly accurate and efficient method of maintaining building pressure control with a large rooftop HVAC unit. To maintain a desired building pressure, Statitrac uses a pressure transducer to measure Space Static Pressure and compare that value to two user adjustable setpoints.

Setpoint Name	Description	Range and Default
Space Static Pressure Setpoint	Desired building pressure setpoint	-0.2 to 0.36 iwc; 0.08 iwc
Space Static Pressure Setpoint Deadband	Deadband around the Space Static Pressure Setpoint used by the controls for proper Relief Damper operation	0.02 to 0.2 iwc; 0.04 iwc

Relief Fan control turns the relief fan on based on the Relief Enable Position Setpoint (user selected setpoint based on Outdoor Air Damper position) and the relief dampers are then modulated in response to Space Static Pressure as shown in the below table.

Condition	Statitrac Response
Space Static Pressure < Space Static Pressure Setpoint – Space Static Pressure Setpoint Deadband	Relief Dampers modulates close
Space Static Pressure Setpoint – Space Static Pressure Setpoint Deadband ≤ Space Static Pressure ≤ Space Static Pressure Setpoint + Space Static Pressure Setpoint Deadband	Relief Damper holds current position
Space Static Pressure ≥ Space Static Pressure Setpoint + Space Static Pressure Setpoint Deadband	Relief Damper modulates open

Economizer and return air dampers are modulated independent relief dampers based on ventilation control and economizer cooling requests.

Statitrac function will provide and output signal from the Indoor Options Module every 60 seconds for the purpose of calibrating the transducer.

If a ventilation override mode is active with Statitrac, the relief fan will turn on and the relief damper will open to 100%.

If the Space Pressure Sensor fails, a diagnostic will be annunciated in the “Active Alarm” and the unit will revert to using Fresh Air Damper Tracking. If the failure on the Space Pressure Sensor clears, Statitrac Space Pressure Control will become active again.

Relief Damper Control – Fresh Air Damper Tracking

The Symbio™ 700 controller offers Relief Fan with Fresh Air Damper Tracking as an option for controlling building pressure with Voyager 3 large commercial rooftop HVAC unit. To keep the space pressure from getting high, the relief damper will track the position of the outdoor air damper anytime the relief fan is energized.

Relief Fan control turns the relief fan on based on the Outdoor Air Damper ≥ Relief Enable Position Setpoint (user selected setpoint based on Outdoor Air Damper position). The relief damper is then modulated to track the position of the outdoor air damper.

When the Outdoor Air Damper < Relief Enable Position Setpoint, the relief fan will de-energize and the relief damper will close.

If a ventilation override mode is active with Fresh Air Damper Tracking, the relief fan will turn on and the relief damper will open to 100%.

General Support Sequences

Supply Fan Speed Setpoint Adjustment

The Symbio™ 700 provides a Supply Fan Maximum Speed Setpoint and a Supply Fan Minimum Speed Setpoint that adjust the maximum and minimum settings provided from the VFD. The details of how the setpoints affect supply fan operation are provided as follows:

- Supply Fan Maximum Speed Setpoint
  - Range: 50 to 100%
  - Operation: This setpoint “trims” the maximum fan speed, based on the configured maximum frequency within the VFD parameter.
  - Example: maximum frequency = 60Hz
    - Supply Fan Maximum Speed Setpoint @ 75% yields a maximum allowed frequency of 45Hz VFD output.
- Supply Fan Minimum Speed Setpoint
  - Range: 0-100%
  - Operation: This setpoint adjusts the minimum allowed frequency based on the maximum allowed frequency. This cannot drive the fan lower than the minimum frequency set within the VFD parameters.
  - Example: VFD Maximum Frequency = 60Hz; VFD Minimum Frequency = 22Hz
    - Supply Fan Maximum Speed Setpoint @ 75% yields a maximum allowed frequency of 45Hz VFD output.
    - Supply Fan Minimum Speed Setpoint @ 75% yields a minimum allowed frequency of 34Hz VFD output (75% of 45Hz).
- Minimum and Maximum Speed Setpoints interact to ensure that the minimum defined fan speed at a given equipment operating condition is maintained.

Supply Fan VFD Bypass

All units have an option to provide Supply Fan VFD Bypass control. This option allows for emergency fan operation if the VFD drive were to fail or become inoperative. The VFD bypass option consists of Contactor VFD Run and Bypass Contactor, which are controlled from the Indoor Options Module. During normal operation, Contactor VFD Run (VRC) is energized for VFD control of the supply fan motor. When VFD Bypass is enabled, the Contactor VFD Run will de-energize, disconnecting the VFD drive output to the supply fan motor. The Bypass Contactor will then be energized or de-energized to control the supply fan On (at line frequency) and Off.

Enabling VFD Bypass Mode

Supply Fan VFD Bypass Mode can be enabled at any time. The unit will remain in VFD Bypass mode until manually terminated. VFD bypass mode will persist over a power failure. No supply fan proving is provided when VFD Bypass is enabled. For VVDA units, Duct Static Pressure High Limit remains active and will cause an immediate supply fan shutdown if the duct static pressure exceeds the maximum limit.

Supply Fan VFD Bypass is enabled via the Symbio Service and Installation Mobile App or the onboard user interface by selecting **Utilities** | **Start VFD Bypass Sequences**.

1. The supply fan and all unit operation is commanded Off. A 40 second timer starts to ensure the supply fan is off and has coasted down to a complete stop.
2. Contactor VFD Run (VRC) de-energizes, disconnecting the VFD output to the supply fan motor.
3. Diagnostic — Supply Fan Bypass Enable diagnostic is active
4. VVDA units — The VAV Box Relay is commanded ON (energized) to indicate full airflow mode to the system, Heat Cool Mode Request will be report Max Heat for all modes of operation, and the VAV Box Stroke Time will count down. All unit operation is disabled until the VAV Box Stroke Time has timed out.
5. VFD Bypass is now active, and the supply fan control logic will turn the Supply Fan Bypass Contactor (BC) On and Off to run and stop the supply fan. All other equipment cooling, heating, ventilation operation and safeties remain active during VFD Bypass.

Disabling VFD Bypass Mode

Supply Fan VFD Bypass Mode can be disabled at any time. Supply Fan VFD Bypass is disabled via the Symbio Service and Installation Mobile App or the onboard user interface by selecting **Utilities | Stop VFD Bypass Sequences**.

1. The supply fan is commanded Off by de-energizing the Bypass Contactor (BC). A 40 second timer starts to ensure the supply fan is off and has coasted to a complete stop.
2. VVDA units — The VAV Box Relay is commanded Off (de-energized) and Heat Cool Mode Status will report Off.
3. All unit operation remains Off for 40 seconds.
4. Diagnostic — Supply Fan Bypass Enable diagnostic is inactive
5. Contactor VFD Run (VRC) will energize, connecting the VFD output to the supply fan motor.
6. VFD Bypass is now inactive and the Symbio 700 will resume supply fan control by communicating commands to the VFD.

Compressor Minimum Runtime

Under all normal running conditions, a 3–minute minimum ON and OFF timer is maintained for each compressor. Once a compressor is turned ON, it remains on for a minimum of 3 minutes. Once a compressor is turned OFF, it remains off for a minimum of 3 minutes. System overrides that require immediate shutdown of the equipment, test modes, and compressor diagnostics/protection functions can override these 3–minute timers. However for normal temperature and thermostatic-based control, these minimum ON/OFF timers are maintained.

Compressor Proof of Operation

For each compressor. a Compressor Proving binary input is used to monitor the state of an auxiliary switch that is used to indicate compressor motor contactor status. Under normal operation, detected operation indicates that all safety devices within the compressor safety circuit are in their normal state. The switch operates as OPEN when the compressor motor is OFF and CLOSED when the compressor motor is ON.

Refer to the Diagnostics section for specific diagnostics generated based on the Compressor Proving signals.

Compressor Low Pressure Cutout Control

For each compressor/circuit, a normally CLOSED low pressure cutout input is monitored for equipment protection on the Symbio™ 700. When a low pressure event is active, the input becomes OPEN and diagnostics are generated as described below. Refer to the Diagnostics section below for specific diagnostics that are generated based on the circuit Low Pressure Cutout inputs.

Evaporator Defrost Control

To prevent frost build-up on the indoor coil during low ambient conditions, compressor operation is monitored and controlled accordingly, relative to outdoor air temperature.

Evaporator Defrost Control can be initiated through two means, based on the FroStat input.

FroStat Installed (default):

- A FroStat input can also be used to directly request the Evaporator Defrost Control function
- When the unit is running in an effective **Cool** mode, the FroStat input will directly control the FroStat diagnostic. If the FroStat input CLOSES, the diagnostic will be annunciated.
- When the unit is running in an effective **Heat** mode, and the Refrigeration System = Heat Pump, the FroStat diagnostic will be controlled **Inactive** until the following are true:
  - FroStat input is CLOSED
  - One or More Compressors have been active for Heat Pump Heating for more than 30 seconds.

Once the above two conditions are met, the FroStat Diagnostic will become Active.

- The FroStat diagnostic is an Auto-Reset diagnostic such that it will be reset when the FroStat input is OPEN in either effecting unit mode.
- If the FroStat diagnostic becomes active, the Compressor Output(s) will de-energize until the FroStat diagnostic is cleared. The supply fan continues to operate during the FroStat diagnostic, so long as it is still requested by a heating or cooling function.

Refrigerant Leak Detection System

Equipment with R-454B refrigerant may require a refrigerant leak detection system (LDS) based on the refrigerant charge. The refrigerant detection system consists of a refrigerant detection sensor that communicates information to the Symbio 700 controller. When the refrigerant detection sensor is in a normal operating state, the equipment provides normal heating, cooling, and ventilation. The refrigerant sensor creates an alarm signal when the sensed refrigerant concentration has exceeded a predetermined threshold.

When the refrigerant detection system is in an alarm state, the Symbio 700 mitigates the refrigerant leak alarm condition for a minimum of 5 minutes after the refrigerant detection system has reset to a normal state.

Mitigation actions:

- All compressor operation is disabled.
- The supply fan operates continuously at or above a defined minimum speed.
- After the supply fan operation is proven; electric heat, gas heat, economizer cooling and ventilation operate normally.
- Alarm output energizes on the Customer Options Module (if installed).

The following Symbio 700 points are available on BACnet or Modbus. The points are not available on LonTalk.

Table 15. Refrigerant Detection System points

Point Reference	Description
Refrigerant Leak Detection System Input	The Active state maintained for the duration of the refrigerant detection system alarm state.
Refrigerant Mitigation Active	Active when the Symbio controller is in a refrigerant mitigation state. This point is also for coordination of building multi-zone VAV systems.
Diagnostic: Refrigerant Concentration Sensor A	Active when Refrigerant Leak Sensor is in the alarm state. Resets, inactive when the Refrigerant Leak Sensor alarm has reset; refrigerant concentration has reduced below the lower concentration threshold.
Diagnostic: Refrigerant Leak Sensor Failure Sensor A	Diagnostic point that shall be active when the refrigerant sensor reports a fault.
Refrigerant LFL Concentration Sensor A	Concentration reported by the refrigerant leak sensor, expressed in percent of LFL (Lower Flammability Limit).
Refrigerant LFL Concentration Alarm Threshold Sensor A	0-100% of LFL (Lower Flammability Limit), value represents the percent of refrigeration concentration level the refrigerant detection sensor will alarm.
Refrigerant Leak Sensor Communication Status Sensor A	Communication status of the Modbus communicating refrigerant detection system. Loss of communication shall initiate refrigerant mitigation actions.

A Multi-Zone VAV application with the supply fan off, the supply fan operation is delayed for the duration of the VAV Box Stroke Time (default 6 minutes, editable) to allow time for VAV boxes to open. The Symbio controller energizes the VAV Box Relay Output and the Refrigerant Mitigation Active point will be active to signal building systems to drive VAV boxes open. When using LON communications, the VAV Box Relay Output must be used to coordinate with building multi-zone VAV system.

In the event the control system detects the refrigerant sensor has failed, become disconnected or unpowered; the controls will respond with a Diagnostic: Refrigerant Leak Sensor Failure, Refrigerant Mitigation Active, and the Alarm Output are active. The Symbio controller provides all mitigation actions (outlined above) for the duration of the sensor failure mode.

Building fire and smoke systems utilizing Ventilation Override or Emergency Override will override the refrigerant detection system. Service Test mode and the Emergency Stop input will also override the refrigerant detection system.

After mitigation actions are completed and the condition triggering mitigation actions are no longer active, the unit will return back to normal operation.

Building Automation System Support Sequences

Occupancy Mode

During expected occupied periods, the system will control to the user selected cooling and heating setpoints.

- The unoccupied setpoint temperatures are often adjusted higher for cooling (setup) and lower for heating (setback) to reduce building operating cost.
- Regardless of how the (occupied) fan mode is set, the supply fan mode is forced to AUTO during unoccupied periods to reduce supply fan operating costs.

There are two mechanisms available to control when units should switch between occupied and unoccupied modes:

- Building controllers provide signals to the unit to request occupied or unoccupied operation. Time-of-day scheduling within building controllers typically determine when the switching should occur.
- Stand Alone Unoccupied control is initiated by a contact closure that causes the unit to begin unoccupied control. In this mode, the controller will use the Unoccupied Cooling and Heating setpoints to determine capacity control needs. Stand Alone Unoccupied control is only applicable when the unit is not being controlled by a conventional thermostat interface.

Timed Override

Exceptions to the time-of-day scheduling are required when unusual or difficult to schedule events cause a space to become occupied during a scheduled unoccupied period. The Timed Override function provides a mechanism for an occupant to signal the system that the space is actually occupied and override the time-of-day schedule to provide occupied control for some limited time period. It also provides a mechanism to return the system to unoccupied mode when the space is no longer occupied.

There are two methods of requesting or terminating timed override on a Symbio™ 700 control system:

- BAS
  - Timed Override Request value can be set to three discrete values:
    - **Idle [1]**— Controller will not override Unoccupied Mode.
    - **On [2]**— When the value is changed from Idle to On, the controller overrides Unoccupied Mode and the unit operates in Occupied Mode. The Occupied Bypass Timer and the Timed Override Timer is Active is set to ‘Active’. The controller remains in Timed Override for the duration of the Occupied Bypass Time.
    - **Cancel [3]**— When value is changed from On to Cancel, the unit’s unoccupied status is no longer overridden and the Occupied Bypass Timer will be set to 0.
- Zone Sensor Initiate/Terminate buttons – Some zone sensors have a Timed Override Initiate and Terminate buttons.

When Timed Override is initiated by any of the above methods, the controller starts a timer using the user selected Occupied Bypass Time to control the duration of the Timed Override event. The range for Occupied Bypass Time is 0 to 240 minutes with a default of 120 minutes.

When the unit is operating in Occupied Bypass mode, the Occupied Bypass Timer duration can be extended by initiating a Timed Override Request again. This extends the Occupied Bypass Timer duration by the value of the user selected Occupied Bypass Time.

When Timed Override is terminated by any of the above methods, the controller will exit Occupied Bypass and will again begin to use unoccupied control setpoints.

Unit Stop

The Unit Stop feature allows for immediate shutdown of all devices in the equipment when initiated. When a Unit Stop request is received, the following actions are taken:

- All equipment control binary outputs are de-energized
  - Indoor fan
  - Compressors
  - Condenser fans
  - Unloader solenoids
  - Heat stages

- All equipment control analog outputs are set to their minimum/off command values.
- All communicating devices, such as supply fans, are commanded to their off state.
- All control algorithms are initialized to their normal startup values and held until the stop request is released.

The Unit Stop request can be initiated from the following sources:

- Unit Stop Command
- Equipment Shutdown Input
- Emergency Override BAS
- Phase Monitor

If a Unit Stop is initiated, the source of the Unit Stop can be determined by the Unit Stop Source Point and other status/diagnostic points.

Capacity Limit Control

The Symbio™ 700 provides the following capabilities to lockout or limit heat and cool capacity installed in the equipment (these capabilities interact):

- Capacity Lockout points have highest priority.
- When Heat Lockout Command and Cooling Lockout BAS are false (not locked out); the control will limit capacity based on Cooling Capacity Enable and Primary Heat Enable.
- Last in priority is Demand Limit Request BAS which enables independent limits on cooling and heating capacity.

See the following sections for more details of each.

Priority	Cooling Limits	Heating Limits
1	Cooling Lockout BAS	Heat Lockout Command
2	Cooling Capacity Enable	Primary Heat Enable BAS
3	• Demand Limit Request BAS • Demand Limit Input (binary input) • Cooling Demand Limit Capacity Enable Setpoint	• Demand Limit Request BAS • Demand Limit Input (binary input) • Heating Demand Limit Capacity Enable Setpoint

Capacity Lockouts

Capacity Lockout points are available to the building automation to provide a method to override or lockout DX Cooling, Gas Heating, and Electric Heating. Cooling Lockout BAS will disable all DX cooling capacity while economizer operation is allowed. Heat Lockout Command disables all forms of installed primary heating capacity.

Cooling Capacity Enable

Cooling Capacity Enable is a building automation interface point used to limit the DX cooling capacity of the equipment. It will not limit economizer cooling. The 0-100% value limits the amount of cooling capacity; default is 100%. The cooling stages allowed = (Limit % * number of stages), round down to the nearest integer.

Heat Primary Enable

Primary Heat Enable BAS is a building automation interface point used to limit all forms of primary and secondary (gas, electric, compressor) heat installed in the equipment. The 0-100% value limits the amount of heating capacity; default is 100%. Staged heating stages allowed = (Heat Primary Enable * number of stages), round down to the nearest integer. Modulating heat allowed = Heat Primary Enable (%).

Demand Limit

When Demand Management is configured for Demand Limit, demand limits can be applied via building management points or the Demand Limit Input (Symbio 700 J16-1 and J16-2). The primary purpose of this function is to limit power consumption of heating and cooling capacities installed in the equipment. Demand Limit does not apply to economizer cooling nor hot gas reheat.

If no building management is writing to the Demand Limit Request BAS point, demand limit can be enabled or disabled by the Demand Limit Input (hardware binary input). Otherwise, if Demand Limit Request BAS is being commanded, it will have priority over the hardware binary input. The result of the arbitration between Demand Limit Request BAS and the Demand Limit Input is reported via Demand Limit Request BAS – Active point.

When Demand Limit Request BAS – Active point is Limit (true), Cooling Demand Limit Capacity Enable Setpoint (0-100%) and Heating Demand Limit Capacity Enable Setpoint (0-100%) apply limits to cooling and heating capacity, respectively. The power consumption result will depend on number of heating and cooling stages installed and how each map to the capacity calculation. Cooling Demand Limit Capacity Enable Setpoint limits compressor stages of operation however will not limit economizer cooling. Heating Demand Limit Capacity Enable setpoint limits the stages of electric heat; however, Heating Demand Limit Capacity Enable will not limit gas heat.

Calculation: Number of heating or cooling stages allowed = (Limit % * number of stages), round down to the nearest integer.

Example: 5-stage Cooling Only unit with 2-stage primary heat

When Demand Limit Request BAS – Active is Limited and Heating Demand Limit Capacity Enable setpoint is 60%, 2-stage primary heat installed, limits operation to 1-stage of heat.

(60% * 2) = 1.2, round down to nearest integer = 1.

When Demand Limit Request BAS – Active is Limited and Cooling Demand Limit Capacity Enable setpoint is 90%, 5-stages cooling installed, limits operation to 4-stages of cooling.

(90% * 5) = 4.5, round down to nearest integer = 4.

Remote Capacity Control

The Symbio™ 700 controls support the Remote Capacity Control function for all system types except VVDA. This function allows a user to directly control the unit capacity rather than allowing the internal algorithm to provide control.

The following features/functions can be requested directly via points, rather than relying on internal temperature or thermostat control sequences:

- Supply Fan Speed
- Cooling Capacity
- Heating Capacity

For Thermostat Controlled equipment, the Thermostat Inputs must be **OPEN** i.e. requesting **OFF** mode before the Cooling or Heating Capacity requests will be honored.

For Space Temperature controlled equipment, the Heat Cool Mode Request must be set to **Fan Only** before the Cooling or Heating Capacity requests will be honored.

For each entity, there is an **Enable** point to enable or disable the remote capacity control, and an analog value point to allow the user to request an analog capacity value. The analog value is translated to the appropriate value per function.

Supply Fan Speed Command overrides will be honored in all non-Heat or Off modes. When Supply Fan Speed Command is enabled, and the Supply Fan Speed Command value is 0 percent, the system will interpret as an **OFF** mode request, and all active capacity will be set to **OFF**; fan, heating, and cooling.

All equipment safeties and limitations will be in-place while the Remote Capacity Control functions are being leveraged:

- Minimum fan speeds as defined per active capacity will be maintained
- Compressor Minimum ON/OFF times will be maintained
- All system diagnostics will be maintained.

Emergency and Ventilation Override

This feature has two options for initiating an override request, either through the optional hardwired Ventilation Override terminals via Customer Connection Module, or by initiating a request through the Emergency Override point. Ventilation Override via the hard-wired binary inputs has priority over Emergency Override.

Ventilation Override

When configured for the Ventilation Override option, applying 24 volts to one of the three Ventilation Override inputs manually activates Ventilation Override. Three inputs are provided on the Customer Option Module supporting Ventilation Override functionality:

- Pressurize mode
- Purge mode
- Exhaust mode

If more than one mode is requested at the same time, the Pressurize request will have priority followed by Purge, and then Exhaust. When Any Ventilation Override mode is active, all heating and cooling is turned off. For the case where the unit is required to turn Off via hardwired interface, the Equipment Shutdown binary input is used.

Emergency Override

All units with Symbio™ 700 support the remote Emergency Override command functionality. Within this point, there are addition enumerations versus the hardware Ventilation Override interface:

- 1 = EMERG_NORMAL
- 2 = EMERG_PRESSURIZE
- 3 = EMERG_DEPRESSURIZE
- 4 = EMERG_PURGE
- 5 = EMERG_SHUTDOWN
- 6 = EMERG_FIRE

Pressurize, Depressurize, and Purge map to their respective Ventilation Override Modes directly. , Emerg_Shutdown and Emerg_Fire are unique to Emergency Override. Both of those are treated as a remote shutdown request for the equipment.

Equipment Operation

Emergency and Ventilation Override requests/actions will take priority over normal equipment timing events, such as compressor minimum ON/OFF/Inter-stage timers.

During an Emergency or Ventilation Override sequence, all temperature control algorithms are initialized to an inactive state until the Emergency or Ventilation Override request is cleared. For each override request, the unit will operate in a pre-determined state until override requests are cleared.

For detailed unit operation during Emergency or Ventilation Override, refer to the Appendix section of this document.

Service Test Mode

Service Test Mode can be used to initiate certain operating modes of the equipment. See the following sections for more details associated with this feature.

Service Test Timeout

Service Test Timeout (Minute) is a user selected time value. Once Service Test Mode has been initiated, and this timer expires, the controls are forced to leave Service Test Mode and return to normal unit operation.

- **Minimum value** - 1 minute
- **Maximum value** - 120 minutes
- **Default value** - 60 minutes

Timer Initiate: When any value for Service Test State Request is chosen other than Inactive, the controller sets the Service Test Timeout to the user selected value and the unit begins to operate as described in the tables below. It continues in operation until the Service Test Timeout Timer reaches 0 **OR** until the user chooses a different Service Test State Request.

Timer Terminate: if the Service Test Timeout timer has reached 0, the controls sets the Service Test Stage Request to Inactive and the unit returns to normal unit operation. If the Service Test Timeout Timer has not reached 0, the user can set the Service Test Stage Request to Inactive to exit the active Service Test State Request and return to normal unit operation.

Timer Reset: if the Service Test Timeout timer has not reached 0, the user can select the Service Test Stage Request to any value other than Inactive. The controller resets the Service Test Timeout Timer to the user selected value and the unit operates as describe in the tables below for the new request.

Leaving Service Test Mode

There are three ways to leave Service Test Mode:

- When the Service Test Timeout timer expires, the unit will leave Service Test Mode.
- Service Test State Request is set to **Inactive**.
- The controller goes through a power cycle or reset.

Service Test Mode – CVZT, VVZT, and VVDA

The table below provides unit operation for each stage of service test depending on the unit configuration. The table describes the service test mode states and expected unit response. For all service test mode operations, “IN CONTROL” refers to Symbio™ 700 algorithms controlling the unit. For instance, in all service test mode states, the condenser fan will be controlled as needed to ensure safe unit operation.

Table 16. Service test mode states — cooling only

Service Test Step	Supply fan On/Off	Supply fan speed (VVZT, CVZT)	Supply fan speed (VVDA)[a]	VAV box relay request (VVDA)	Outdoor air damper position	Compr cool stage	Aux heat stage	Relief fan On/Off	Reheat pumpout solenoid On/Off[b]	Reheat valve 1	Heat cool mode status (VVZT, CVZT)	Heat cool mode status (VVDA)[c]
Inactive	In Control	In Control	In Control	In Control	In Control	In Control	In Control	In Control	In Control	In Control	In Control	In Control
Fan On	ON	Min	Min	Open	Min	0	0	In Control	OFF	0	Test	Maximum Heat
Fan On Econ Open	ON	Min	Min	Open	100	0	0	In Control	OFF	0	Test	Maximum Heat
High Fan Speed Econ Open	ON	100	100	Open	100	0	0	In Control	OFF	0	Test	Maximum Heat
Ventilation Low Fan Speed	ON	Min	Min	Open	Design Min	0	0	In Control	OFF	0	Test	Maximum Heat
Ventilation Mid Fan Speed	ON	Effective Mid[d]	Effective Mid[d]	Open	Design Mid	0	0	In Control	OFF	0	Test	Maximum Heat
Ventilation High Fan Speed	ON	100	100	Open	Design Max	0	0	In Control	OFF	0	Test	Maximum Heat
Cool 1	ON	Min for Capacity	100	Open	Min	1	0	In Control	ON	0	Test	Maximum Heat
Cool 2	ON	Min for Capacity	100	Open	Min	2	0	In Control	ON	0	Test	Maximum Heat
Cool 3	ON	Min for Capacity	100	Open	Min	3	0	In Control	ON	0	Test	Maximum Heat
Cool 4	ON	Min for Capacity	100	Open	Min	4	0	In Control	ON	0	Test	Maximum Heat
Cool 5	ON	Min for Capacity	100	Open	Min	5	0	In Control	ON	0	Test	Maximum Heat
Reheat	ON	Min for Capacity	100	Open	Min	5	0	In Control	OFF	50	Test	Maximum Heat
Heat 1	ON	Min for Capacity[e]	100	Open	Min	0	1	In Control	OFF	0	Test	Maximum Heat
Heat 2	ON	100	100	Open	Min	0	2	In Control	OFF	0	Test	Maximum Heat

Table 16. Service test mode states — cooling only (continued)

Service Test Step	Supply fan On/Off	Supply fan speed (VVZT, CVZT)	Supply fan speed (VVDA)[a]	VAV box relay request (VVDA)	Outdoor air damper position	Compr cool stage	Aux heat stage	Relief fan On/Off	Reheat pumpout solenoid On/Off[b]	Reheat valve 1	Heat cool mode status (VVZT, CVZT)	Heat cool mode status (VVDA)[c]
Open Reheat Vavle	OFF	0	0	Open	0	0	0	In Control	OFF	85[f]	Test	Maximum Heat
Close Reheat Valve	OFF	0	0	Open	0	0	0	In Control	OFF	0	Test	Maximum Heat

[a] For VVDA units, Supply Fan Speed Request shall not be 100 before VAV boxes full open. If VAV box stroke timer has not expired, set fan speed to Min to avoid high duct static pressure trip.
[b] Reheat Pumpout Solenoid On/Off Request shall be ON whenever compressor cool or heat stage request is greater than 0, except during the "Reheat" step.
[c] For VVDA units, Heat Cool Mode Status shall be set to Maximum Heat to coordinate VAV Box Relay, when any service test step activated
[d] Supply Fan Speed Request "Effective Mid" equals to *(Effective Maximum Fan Speed + Effective Minimum Fan Speed) / 2*.
[e] For "Heat 1" , the supply fan speed shall run at min speed due to Staged Gas Heat supply fan requirements.
[f] Reheat Valve 1 Request is set to 85% in "Open Reheat Valve 1" step, because 85% is the max operating range during normal operation. Refer to the step motor reference docs for more detail information.

Troubleshooting

The Symbio™ 700 controller provides system shutdown, operational default operation, and communication error handling of the unit. The list of fault conditions below will stop normal operation or change the operation of the unit to a default condition. Diagnostics are indicated in the Active Alarm menu of the onboard user interface and the Symbio Service and Installation mobile app. The following sections provide diagnostic details valuable to troubleshooting.

Data Logs

Data logs are useful when servicing a unit. Many data logs for all core processes are created by default. The exact number depends on unit configuration. The Symbio™ Service and Installation App as well as Tracer TU can be used to view, create, and manage Data Logs. Data logs have 50,000 points trending at 10 second intervals. This gives over 5 days of data. Data logs can be exported to a USB memory stick via a .CSV format. Select property data, that has been configured for recording data over time, can be viewed graphically. Up to 4 sets of data can be viewed at one time using the Symbio Service and Installation App by using View Data Logs in the Tools menu.

Unit Stop Source

Symbio™ 700 provides a Unit Stop Source point that is valuable to understanding the source of a unit shutdown or what is preventing normal unit operation. Unit Stop Source is viewable from the local user interface, Symbio 700 Service and Installation App status screen, or Tracer TU. One of the following values is reported:

- None
- Emergency Stop
- Drain Pan Overflow
- Local HI
- Remote HI
- External Auto Stop
- Phase Monitor
- Emergency Override
- Supply Fan Fault
- Equipment Shutdown Input
- Smoke Detector
- Equipment Limit
- Sensor Failure

Diagnostic Reset

The following section describes how Diagnostics can be resets.

Power-Up Reset or Exception/Override Mode Transition

At power-up or after the unit exits an override mode, such as Service Test, the controller restarts and clears all diagnostics.

Reset Diagnostic Point

A Diagnostic Reset point is available to reset diagnostics by command a Reset. Diagnostic Reset default state is Normal but writing the Reset value to the point will command the controller to reset all diagnostics. The point value reverts to Normal after approximately 5 seconds.

Local User Interface Display

Using the Symbio™ 700 local user interface display, navigate to the Service menu, Diagnostics, then Reset Diagnostic. This operation will cause all diagnostics to reset.

Heat Cool Mode Transition Reset

When the unit is controlled by a zone sensor, a transition from System Mode Switch Local = OFF to System Mode Switch Local = Heat, Cool, Auto, or Emergency Heat triggers a Reset Diagnostic request.

Diagnostics

Diagnostic descriptions

Column	Description
Diagnostics	The actual text displayed on the user interface display, Symbio Service and Installation App or Tracer TU alarm log.
Target	The system or component directly affected by the diagnostic. Either none, partial or total unit functionality is impaired.
Severity	Warnings may or may not affect unit operation. Normal Shutdown will provide an orderly termination of component control. Immediate Shutdown overrides all normal timers, components and outputs off. Components may have different responses based on the unit's mode of operation.
Persistence	Latching diagnostics require a manual reset at a user interface, commanded reset or cycling power to the controller. See Diagnostic Reset section for all reset options. Non-Latching diagnostics automatically reset and can clear by normal unit operation.
Alarm Indicator Status	Symbio™ 700 supports an Alarm Indicator Status point that, if configured as Installed, drives a relay output on the Customer Connection Module (J10). This point is set to active when a failure occurs that functionally stops a critical component within the unit.
Condition / Response	This describes the conditions the system or component was experiencing at the time the alarm is generated. The response indicates what the unit, system or component will do during the alarm event.

Communication

If 15 seconds passes with no valid communications received from any of the Options modules, the Symbio™ 700 takes the following actions:

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Unit Communications Failure	Unit	Warning	Non-Latching	Y	Active if any diagnostic listed in this table have a "not communicating" status.
Symbio 700 Onboard I/O Communication Status	Unit	Immediate Shutdown	Non-Latching	Y	All Symbio 700 controller inputs and outputs are disabled.
Customer Options Module Communication Status	Partial	Warning	Non-Latching	Y	Functions requiring a Customer Options Module are disabled. If Ventilation Override is inactive during communication loss event, unit will continue operation with component level disables.
	Unit	Immediate Shutdown	Non-Latching	Y	If Ventilation Override is active during communication loss event, unit perform an Immediate Shutdown.
Indoor Options Module Communication Status	Partial	Warning	Non-Latching	Y	Single Zone VAV operation disables. Electric heat operation disables. Economizer cooling disables.
Fresh Air Options Module Communication Status	Partial	Warning	Non-Latching	Y	All economizer outdoor air damper and power exhaust operation are disabled.
Stepper Motor Module Communication Status	—	Warning	Non-Latching	Y	Compressor operation disables.

Sensor and other diagnostics

The following diagnostics are generated from sensor values out of range, or the diagnostic is generated by the Symbio™ 700 controller.

Troubleshooting

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Maintenance Required	None	Warning	Non-Latching	No	The supply fan run hours has exceeded Filter Runtime Hours Setpoint. To reset the diagnostic, set Filter Timer Reset. To disable this diagnostic, set Filter Runtime Hours Setpoint to zero.
Space Temperature Active	Unit	Immediate Shutdown	Non-Latching	YES	CVZT and VVZT space temperature control units will perform an immediate shutdown.
	Component	Warning	Non-Latching	No	VVDA (mulit-zone VAV) control units will disable the following modes of operation. Dehumidification Pre Cool Morning Warmup Daytime Warmup
Outdoor Air Temperature Active	Unit	Warning	Non-Latching	YES	Heat pump Demand Defrost disables, time based outdoor coil defrost will be utilized. Heat Pump Heating Lockout disables.
	Damper	Warning	Non-Latching	YES	Economizer cooling disabled. See Economizer failure modes and system response table.
	Reheat	Warning	Non-Latching	YES	Disable hot gas reheat.
Discharge Air Temperature Local	Component	Warning	Non-Latching	YES	Wired discharge air temperature input is out of range. The following capabilities are disabled: Dehumidification with hot gas reheat Modulating gas heat Single zone vav Economizer cooling
Outdoor Air Humidity Active	Damper	Warning	Non-Latching	YES	Comparative Enthalpy disables, Reference enthalpy disables, control reverts to dry bulb control. See Economizer failure modes and system response table.
Space Humidity Active	Dehumidification	Warning	Non-Latching	No	Dew Point and Space Relative Humidity dehumidification control disables.
Return Air Humidity Active	Damper	Warning	Non-Latching	No	Return air humidity greater than 90%, return air enthalpy calculations with be based on 90% return air humidity. Return air humidity less than 10%, return air enthalpy calculations with be based on 10% return air humidity. See Economizer failure modes and system response table.
Return Air Temperature Input	Damper	Warning	Non-Latching	No	Comparative Enthalpy disables, Reference enthalpy disables, control reverts to dry bulb control. See Economizer failure modes and system response table.
Space CO_2 Concentration Active	Damper	Warning	Non-Latching	No	Demand controlled ventilation disables. Control will revert to supply compensation, if enabled, or the outdoor air damper will revert to Economizer Minimum Position Setpoint.
Phase Monitor Status	Unit	Immediate Shutdown	Latching	YES	Phase Monitor binary input is in an active state.
Diagnostic: High Condensate Level Detected	None	Normal Shutdown	Non-Latching	No	Condensate overflow binary input is closed. If the condensate overflow input opens, the diagnostic will reset.
Diagnostic: Morning Warmup Mode Exceeded 120 Minutes	Heat	Warning	Non-Latching	No	Space conditions have not been satisfied to exit morning warmup mode for more than 120 minutes.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Pre Cool Mode Exceeded 120 Minutes	Cool	Warning	Non-Latching	No	Space conditions have not been satisfied to exit pre cool mode for more than 120 minutes.
Diagnostic: Night Purge Mode Exceeded 120 Minutes	Cool	Warning	Non-Latching	No	Space conditions have not been satisfied to exit night purge mode for more than 120 minutes.
Diagnostic: Condensate Overflow Lockout	Unit	Immediate Shutdown	Latching	YES	The Condensate Overflow Trip occurred three times within 72 hours. Inspect condensate pan and drain.
Diagnostic: Filter Change Required	None	Warning	Non-Latching	No	Clogged Filter Input (BI) is Closed for two continuous minutes while the Supply Fan is On.
Diagnostic: Duct Static Pressure Local Trip	Unit	Warning	Non-Latching	No	Duct static pressure input has gone out of range. Inspect tubing, replace sensor as required.
Diagnostic: Duct Static Pressure Local Lockout	Unit	Immediate Shutdown	Latching	YES	Duct static pressure input has failed three times within one hour. Inspect tubing to the sensor, replace sensor as required.
Diagnostic: Duct Static Pressure Limit Trip	Unit	Immediate Shutdown	Non-Latching	No	Duct static pressure exceeded the Duct Air Static Pressure High Limit Setpoint. The unit will shutdown for 3 minutes before restart.
Diagnostic: Duct Static Pressure Limit Lockout	Unit	Immediate Shutdown	Latching	YES	The supply air pressure has exceeded the Duct Static Pressure High Limit Setpoint for the 3rd consecutive time without the supply fan running for 3 continuous minutes.
Diagnostic: Supply Fan Bypass Enable	None	Warning	Non-Latching	No	Supply fan VFD bypass mode enabled. VFD diagnostics are suppressed.
Space Static Pressure Input	Space Pressure Control	Warning	Non-Latching	No	Sensor input out of range. Control will revert to outdoor air damper tracking. If Supply Fan Compensation is enabled, Supply Fan Compensation will be used for space pressure management. If Supply Fan Compensation is disabled, TRAQ OA Minimum Position Request is disabled and an outdoor air damper setpoint of 10% will be utilized for ventilation.
Diagnostic: Space Pressure Calibration Limit	Space Pressure Control	Warning	Non-Latching	YES	Zeroing, null voltage is out of range for five minutes. Control will revert to outdoor air damper tracking.
Outdoor Air Flow Active	Outdoor Airflow Control	Warning	Non-Latching	YES	Pressure sensor input out of range. If Supply Fan Compensation is enabled, Supply Fan Compensation will be used to determine the minimum position for minimum ventilation. If Supply Fan Compensation is disabled, TRAQ OA Minimum Position Request is disabled and an outdoor air damper setpoint of 10% will be utilized for ventilation.
Diagnostic: Outdoor Airflow Pressure Calibration Limit	Outdoor Airflow Control	Warning	Non-Latching	YES	Zeroing, null voltage is out of range for five minutes. Control will revert to outdoor air damper tracking.

Refrigerant Detection System

Equipment with R–454B refrigerant may require a refrigerant detection system based on the refrigerant charge. The refrigerant detection system consists of a refrigerant detection sensor that communicates the following diagnostics to the Symbio™ 700 controller via Modbus.

Diagnostic	Target	Severity	Persistence	Alarm Output	Condition / Response
Diagnostic: Refrigerant Concentration	Compressor	Immediate Shutdown	Non-Latching	YES	Refrigerant detection sensor detected a concentration above a limit, the following mitigation actions are taken for 5 minutes after the refrigerant sensor resets. Compressor operation disabled, supply fan operation at or above minimum speed, ventilation enabled, economizer cooling and non-compressor heating allowed.
Diagnostic: Refrigerant Sensor Failure	Compressor	Immediate Shutdown	Non-Latching	YES	Refrigerant detection sensor detected a failure, the following mitigation actions are taken until the refrigerant sensors is replaced or the failure is resolved. Compressor operation disabled, supply fan operation at or above minimum speed, ventilation enabled, economizer cooling and non-compressor heating allowed.
Refrigerant Detection System Communication Status	Compressor	Immediate Shutdown	Non-Latching	YES	Modbus communication has been lost with the refrigerant detection sensor, check wiring, ensure the sensor is receiving power, replace sensor as required. Compressor operation disabled, supply fan operation at or above minimum speed, ventilation enabled, economizer cooling and non-compressor heating allowed.

The refrigerant detection sensor has a green and red LED to indicate operating status as described in the following table.

Operating status	Green LED	Red LED	Symbio 700 Response
No power to the sensor or sensor failure.	OFF	OFF	Refrigerant Detection System Communication Status = 2, Not Communicating. Refrigerant Mitigation Active point in active state.
Power up initiate	ON	OFF	None
Power-up self-test failure	ON	ON	Diagnostic: Refrigerant Sensor Failure active. Refrigerant Mitigation Active point in active state.
Warm up delay	ON	OFF	None
Normal runtime mode	Heartbeat	OFF	None
LFL (Lower Flammability Limit) alarm, refrigerant concentration detected above setpoint limit.	OFF	ON	Diagnostic: Refrigerant Concentration active state. Refrigerant Mitigation Active point in active state.
Failure resulting in loss of function, sensor may need to be replaced.	OFF	Blinking	Diagnostic: Refrigerant Sensor Failure active state. Refrigerant Mitigation Active point in active state.
Sensor near end of life	Blinking	Heartbeat	None
Warning sensor is operating outside of its expected/normal range due to excessive temperature or humidity. The sensor is not damaged and should return to a normal operating state when under normal conditions.	ON	Heartbeat	None

Supply Fan VFD (Voyager 3, Foundation, Odyssey)

The following diagnostics are available through direct data monitoring from the Modbus controlled VFD or data monitoring.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Supply Fan Failure	Unit	Immediate Shutdown	Latching	YES	If the supply fan is requested to run, but the VFD Running Status is False (read via Modbus communication) for 40 continuous seconds.
Diagnostic: Supply Fan Proving Failure	Unit	Immediate Shutdown	Non-Latching	YES	VFD drive is reporting via Modbus that the supply fan is not running.
Diagnostic: VFD Fault Supply Fan - 1	Unit	Immediate Shutdown	Non-Latching	YES	VFD drive is reporting via Modbus that there is a fault.
Diagnostic: VFD Supply Fan Ground Fault - 1	Unit	Immediate Shutdown	Non-Latching	YES	VFD drive is reporting via Modbus that there is a ground fault. There is a discharge from the output phases to ground, either in the cable between the frequency converter and the motor or in the motor itself. Troubleshooting: Turn off the frequency converter and remove the ground fault. Measure the resistance to ground the motor cables and the motor with a megohmmeter to check for ground faults in the motor.
Diagnostic: VFD Supply Fan Motor Current Overload - 1	Unit	Immediate Shutdown	Non-Latching	YES	VFD drive is reporting via Modbus that there is a current overload condition. The fault is that the frequency converter is overloaded by more than 100% for too long.
Diagnostic: VFD Supply Fan Short Circuit - 1	Unit	Immediate Shutdown	Non-Latching	YES	VFD drive is reporting via Modbus that there is short circuit condition. There is short-circuiting in the motor or on the motor terminals.
Diagnostic: VFD Supply Fan Broken Belt - 1	Unit	Immediate Shutdown	Non-Latching	YES	VFD drive motor torque is below the broken belt torque (current) value. And the converter output frequency is above or equal to 15Hz.
Supply Fan VFD Communication Status	Unit	Immediate Shutdown	Non-Latching	YES	Modbus communication to the supply fan VFD has exceeded 30 seconds. Check Modbus wiring.

Compressor

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: FroStat Trip	Circuit	Immediate Shutdown	Non-Latching	No	Compressor circuit operating in the cooling mode when the FroStat input became active (closed). Supply fan continues to operate until the FroStat input is inactive (open).
Diagnostic: Comp 1 Proving Trip Diagnostic: Comp 2 Proving Trip Diagnostic: Comp 3 Proving Trip	Circuit	Immediate Shutdown	Non-Latching	No	When a compressor output is commanded ON and it has been running for more than 5 seconds, if the associated proving input opens. Circuit is disabled for 15 minutes. NOTICE: Diagnostic: Compressor X Proving Trip is an indication of a HPC (high pressure cutout) or DTL (discharge temperature limit) opening the compressor contactor circuit.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Compressor 1 Proving Lockout Diagnostic: Compressor 2 Proving Lockout Diagnostic: Compressor3 Proving Lockout	Circuit	Immediate Shutdown	Latching	YES	Two scenarios can cause this diagnostic to be active: Scenario 1: If a refrigeration circuit accumulates 4 consecutive Diagnostic: Comp X Proving Trips during the same compressor call for heating or cooling, a Diagnostic: Compressor X Proving Lockout is generated. Note: If the call for the compressor operation terminates, the counter is set to zero. Scenario 2: If a compressor associated proving input does not CLOSE within 5 seconds of the compressor startup.
Diagnostic: Circuit 1 LPC Trip Diagnostic: Circuit 2 LPC Trip	Circuit	Immediate Shutdown	Non-Latching	No	Low pressure cutout binary input is open, circuit is disabled for 3 minutes.
Diagnostic: Circuit 1 LPC Lockout Diagnostic: Circuit 2 LPC Lockout	Circuit	Immediate Shutdown	Latching	YES	If four LPC trip events occur within 3 minutes (R-410A) or 10 minutes (R-454B) of initial circuit operation. Requires a manual reset.
Diagnostic: Compressor 1 Contactor Failure Diagnostic: Compressor 2 Contactor Failure Diagnostic: Compressor 3 Contactor Failure	Circuit	Immediate Shutdown	Latching	YES	If a compressor proving input becomes Active for 5 continuous seconds when the associated compressor output is Inactive.
Diagnostic: Reheat Valve 1 Fault	Compressor	Immediate Shutdown	Latching	YES	Hot gas reheat valve 1 motor fault associated with undervoltage, overcurrent, for thermal shutdown. All compressor operation is disabled.
Diagnostic: Reheat Valve 2 Fault	Compressor	Immediate Shutdown	Latching	YES	Hot gas reheat valve 2 motor fault associated with undervoltage, overcurrent, for thermal shutdown. All compressor operation is disabled.

Compressor Diagnostic Descriptions

Diagnostic: Compressor X Proving Trip

NOTICE: Diagnostic: Compressor X Proving Trip is an indication of a HPC (high pressure cutout) or DTL (discharge temperature limit) opening the compressor contactor circuit.

When a compressor output is commanded ON and it has been running for more than 5 seconds, if the associated proving input opens:

- Diagnostic: Comp X Proving Trip is active.
- Command the associated compressor output OFF immediately.
- Command any compressor output OFF that is on the same refrigeration circuit as the compressor which had the proving input trip.
- The Circuit is disabled for 15 minutes.

After a 15-minute compressor proving timeout has expired and Diagnostic: Compressor 1 Proving Lockout is not active:

- Diagnostic: Comp X Proving Trip diagnostic is reset.
- If the cooling stage is still requested ON, the circuit is allowed to stage again.

Diagnostic: Compressor X Proving Lockout

There are two cases that can cause a Diagnostic: Compressor X Proving Lockout:

- If a refrigeration circuit accumulates 4 consecutive Diagnostic: Comp X Proving Trips during the same compressor operating cycle, a Diagnostic: Compressor X Proving Lockout is generated.

Note: *If the call for the compressor operation terminates, the counter is set to zero.*

- A compressor proving input does not close within 5 seconds of the compressor output commanded On.

If a Diagnostic: Compressor X Proving Lockout is active the following will occur:

- All compressors on the associated circuit are shutdown immediately and locked out, a diagnostic reset action is required to clear the diagnostic.
- Diagnostic: Compressor X Proving Lockout diagnostic point is active.
- The alarm output is active.

Diagnostic: Compressor X Contactor Failure

If a compressor proving input becomes active for 5 continuous seconds when the associated compressor command output is commanded Off, Diagnostic: Compressor X Contactor Failure is active:

- All compressors on the associated circuit are de-energized immediately and they are locked out until a Reset Diagnostic is initiated.
- The Compressor X Contactor Failure diagnostic point is active
- The alarm output is active.

Diagnostics – Low Pressure Cutout

The following operation is enforced based on the state of the circuit LPC input:

Prior to Compressor Start-up:

- If a compressor output is Off and its circuit's LPC input is open, compressor operation is not inhibited, and the Diagnostic: Circuit X LPC Trip point is not active.

After Compressor Start-up:

An LPC Bypass Delay function delays low-pressure cutout after compressor start-up until a pre-determined amount of time passes in low ambient conditions. The length of the delay is determined based on ambient temperature:

- If the Outdoor Air Temperature Active is less than 40°F, the LPC Bypass Delay is set to 60 seconds.
- If the Outdoor Air Temperature Active is between 40°F and 50°F, the LPC Bypass Delay is set to 30 seconds.
- If the Outdoor Air Temperature Active is greater than 50°F, the LPC Bypass Delay is 0 seconds.

Diagnostic: Circuit X LPC Trip

- All compressor outputs on the effected circuit are commanded OFF.
- The Diagnostic: Circuit X LPC Trip point is active.
- The circuit is disabled for 3 minutes.
- The circuit LPC trip counter is incremented.

After the 3-minute circuit disable has expired, if the unit is not under a lockout event:

- The Diagnostic: Circuit X LPC Trip point is reset.
- If the compressor stage is still requested ON, the circuit is allowed to stage again.

LPC Trip Count reset:

- R-410A units, if the Circuit runs for 3 minutes without a low-pressure trip, its LPC Trip Count is reset to 0.
- R-454B units, If the circuit runs for 10 minutes without low pressure, the LPC timer resets to 0.

Heat pumps, if the Outdoor Air Temperature is less than 0°F or if the unit is in active defrost, the low-pressure cutout input state is ignored.

Diagnostic: Circuit X LPC Lockout

If a circuit LPC trip counter accumulates 4 low pressure events without the circuit running for the 3-minute or 10-minute reset time (counter is not reset), Diagnostic: Circuit X LPC Lockout is active.

- All compressors on the associated circuit are de-energized immediately and they are locked out until a Reset Diagnostic is initiated.
- Diagnostic: Circuit X LPC Lockout point is active.
- The alarm output is active.

Outdoor Air Damper

Symbio™ 700 provides Economizer fault detection and diagnostics which can impact or alter the controllers ability to provide economizer cooling.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
FDD: Excessive Outdoor Air	Economizer cooling	Warning	Non-Latching	YES	Economizer cooling disabled (Economizer Airside Status = Inactive), and outdoor air damper command is less than the outdoor air damper position feedback – 10% for five minutes. Economizer operation is disabled.
FDD: Outdoor Air Damper Not Modulating	Economizer cooling	Warning	Non-Latching	YES	Economizer cooling disabled (Economizer Airside Status = Inactive), and outdoor air damper command is less than the outdoor air damper position feedback + 10% for five minutes. Economizer operation is disabled.
FDD: Unit Economizing When It Should Not	Economizer cooling	Warning	Non-Latching	YES	Economizer cooling enabled (Economizer Airside Status = Active), and outdoor air damper command is less than the outdoor air damper position feedback – 10% for five minutes. Economizer operation is disabled.
FDD: Unit Not Economizing When It Should	Economizer cooling	Warning	Non-Latching	YES	Economizer cooling enabled (Economizer Airside Status = Active), and outdoor air damper command is less than the outdoor air damper position feedback + 10% for five minutes. Economizer operation is disabled.

Economizer failure modes and system response

Economizer Control	Failure Mode	Economizer Cooling Response
Comparative Enthalpy	Return Air Humidity Sensor	Absolute (Reference) Enthalpy
	Return Air Temperature Sensor	Absolute (Reference) Enthalpy
	Outdoor Air Temperature Sensor	Economizer Cooling Disabled- OA Damper Closes to Economizer Minimum Position Setpoint Active
	Outdoor Air Humidity Sensor	Absolute Temperature (Reference Dry Bulb)
Reference Enthalpy	Outdoor Air Temperature Sensor	Economizer Cooling Disabled- OA Damper Closes to Economizer Minimum Position Setpoint Active
	Outdoor Air Humidity Sensor	Absolute Temperature (Reference Dry Bulb)
Reference Dry Bulb	Outdoor Air Temperature Sensor	Economizer Cooling Disabled- OA Damper Closes to Economizer Minimum Position Setpoint Active
Differential Dry Bulb	Outdoor Air Temperature Sensor	Economizer Cooling Disabled- OA Damper Closes to Economizer Minimum Position Setpoint Active
	Return Air Temperature Sensor	Absolute Temperature (Reference Dry Bulb)
Fresh Air Options Module	Communications Failure	All economizing operation is discontinued

Modulating Gas Heat

The following diagnostics are available through direct data monitoring from the Modbus controlled gas heat ignition module (VB1285) or through data monitoring.

Diagnostic	Target	Severity	Persistence	Alarm Output	Condition / Response
Diagnostic: Gas Heat Lockout Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1285 ignition controller. This diagnostic is active anytime a gas heat ignition module lockout condition exists (if the diagnostic is or is not listed in this table).
Diagnostic: Gas Heat Unexpected Flame Manif 1 Burner 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1285 ignition controller and supply fan operation stays On continuous. Unexpected flame on manifold 1, burner 1. VB1285 ignition controller lockout duration will remain in effect for 10 to 300 seconds after error condition is cleared or the power is cycled. VB1287 LED Display [E08] Possible causes: Gas valve may be stuck open or malfunctioning. Flame rod maybe malfunctioning.
Diagnostic: Gas Heat Unexpected Flame Manif 1 Burner 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1285 ignition controller and supply fan operation stays On continuous. Unexpected flame on manifold 1, burner 2. VB1285 ignition controller lockout duration will remain in effect for 10 to 300 seconds after error condition is cleared or the power is cycled. VB1287 LED Display [E18] Possible causes: Gas valve may be stuck open or malfunctioning. Flame rod maybe malfunctioning.
Diagnostic: Modulating Gas Invalid ID Plug Manifold 1	Heat	Immediate Shutdown	Non-Latching	No	Call for heat is removed to the VB1285 ignition controller. VB1285 ignition controller ID Plug contains invalid data. Possible cause: Corrupt ID Plug. Plug not connected properly.
Diagnostic: Modulating Gas Heat Configuration Invalid	Heat	Immediate Shutdown	Non-Latching	No	The Symbio™ 700 configuration does not match the VB1285 ignition controller ID Plug heat configuration. All burners on the manifold are unavailable until the condition is corrected. Diagnostic also occurs when communication fails to the gas heat ignition module; Symbio 700 is unable to read the ignition module ID Plug heat configuration. VB1285 ignition controller functioning properly. LED Display [normal activity] Possible causes: Wrong ID Plug installed on VB1287 ignition controller. Symbio 700 mis-configured.
Diagnostic: Gas Heat Weak Flame Manif 1 Burner 1	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat. VB1285 ignition controller has detected a weak flame on manifold 1, burner 1. LED Display [A05] This diagnostic is an indication that the signal from the flame sensor is weak. Possible causes: Dirty flame rod. Improper flame rod installation/position. Improper gas valve pressure.
Diagnostic: Gas Heat Weak Flame Manif 1 Burner 2	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat. VB1285 ignition controller has detected a weak flame on manifold 1, burner 2. This diagnostic is an indication that the signal from the flame sensor is weak. LED Display [A15] Possible causes: Dirty or weak flame rod. Improper flame rod installation/position. Improper gas valve pressure.

Diagnostic	Target	Severity	Persistence	Alarm Output	Condition / Response
Diagnostic: Gas Heat Insufficient Combustion Air	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat. VB1285 ignition controller has detected insufficient combustion air and will reduce heat capacity while the condition exists. LED Display [A03] Possible causes: Blocked vent, high vent temperatures, high elevation or the inducer fan is not able to run at the required speed. Note: At extreme elevations, an A03 code may be normal operation.
Diagnostic: Modulating Gas Primary Limit Open Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1285 ignition controller. VB1285 ignition controller has detected the roll out switch open, an open fuse, or the thermal cut out. All burners on the manifold are temporarily unavailable. Lockout duration will remain in effect for 10 to 300 seconds after error condition is cleared or the power is cycled. LED Display [E02]. Voyager 3, 600 MBH gas furnaces, low gas pressure or high gas pressure will cause a Primary Limit Open, [E02] diagnostic. Possible causes: Roll out switch, open fuse, or the TCO (thermal cut out).
Diagnostic: Modulating Gas Heat Open Fuse Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1285 ignition controller. LED Display [E13] VB1285 ignition controller has detected a fuse is open. All burners on the manifold are unavailable.
Diagnostic: Gas Heat Failed Ignition Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1285 ignition controller. VB1285 ignition controller has four failed ignition attempts on the primary burner and retries have been exhausted. Heat is locked out for 1-hour. The ignition controller will attempt to restart after the 1-hour lockout period. Power may be cycled to reset. LED Display [E01] Possible causes: Plugged vent on gas pressure regulator. Spark ignitor is bad. The high voltage wire is bad.
Diagnostic: Gas Heat Failed Ignition Manifold 1 Burner 2	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat on the primary burner and operate at limited 50% capacity. VB1285 ignition controller, manifold 1 split burner has four failed ignition attempts and retries have been exhausted. Split burner is locked out for 1-hour. The ignition controller will attempt to restart the split burner after the 1-hour lockout period. Power may be cycled to reset. LED Display [A11] Possible causes: Plugged vent on gas pressure regulator, failed flame rod. The high voltage wire is bad.
Diagnostic: Modulating Gas Valve Failure Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1285 ignition controller. The VB1285 modulating gas valve actuator did not reach a Park or Full-On position. Lockout duration will remain in effect for 10 to 300 seconds after error condition is cleared or the power is cycled. LED Display [E03] Possible cause: Bad gas valve.

Diagnostic	Target	Severity	Persistence	Alarm Output	Condition / Response
Diagnostic: Modulating Gas Control Board Failure Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1285 ignition controller. VB1285 Board Failure. Lockout duration will remain in effect for 10 to 300 seconds after error condition is cleared. LED Display [888] Possible cause: Bad wiring. Board failure.
Diagnostic: Gas Heat Air Pressure Sensor Reading Low	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1285 ignition controller. Air pressure switch failed to open during pre-purge calibration process. The inducer fan increases until the air pressure switch closes, then the inducer fan decreases until the air pressure switch opens (the switch failed to open). Display Code [E04] Possible cause: Incorrect ID Plug or incorrect air pressure switch installed. Insufficient air due to blocked vent, pneumatic tube between inducer fan static tap and the air pressure switch has an obstruction.
Diagnostic: Gas Heat Air Pressure Sensor Reading High	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1285 ignition controller. Air pressure switch failed to close during pre-purge calibration process. VB1285 Display Code [E05] Possible cause: Pneumatic tubing connections or tubing is damaged. Verify pneumatic tubing is connected to VB1285 negative connection. Defective air pressure switch.
Diagnostic: Gas Heat Loss of Inducer Motor Control	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat. VB1285 ignition controller, the air pressure is not modulating down at minimum inducer drive. Display Code [A07] Possible cause: Pneumatic tubing. Connect a manometer and verify the pressure on the VB1285 display matches the manometer. If pressures do not match cycle control power on the VB1285. If problem still exists, the VB1285 ignition module and/or the inducer fan may need replacing.
Diagnostic: Gas Heat Air Sensor Null Pressure Check	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat. VB1285 ignition controller air sensor pressure check out of tolerance. Display Code [A08] Possible cause: The VB1285 air pressure on-board sensor is out of tolerance. Cycle control power to the VB1285. If problem still exists, the VB1285 ignition module may need replacing.
Diagnostic: Gas Heat Limited Low Fire	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat. Flame loss at low fire results in an auto-adjustment that limits the burner turn down during the rest of the current call for heat. VB1285 Display Code [A04] Possible cause: Wind caused loss of flame at low fire.
Gas Heat Ignition Module 1 Communication Status	Heat	Warning	Non-Latching	YES	Gas heat operation on ignition module 1 disabled. Check Modbus wiring.

VB1285 Ignition Module Display Code Callout

On power up:

H-A	Heatco Model Series
600	Gas Heating Input (600 MBh)
-10, -5	10:1 modulation turndown, 5:1 modulation turndown
nat, LP	Natural Gas, Propane Gas
001	Version of ID Chip
OFF	Standby Mode – awaiting call for heat or increase in firing rate demand
Pur	Purge – 30 second purge period
IGn	Ignition sequence – 6 second trial for ignition and flame signal proving
HEA	Warm up – 10 second warm up period (ignores firing rate demand)
Run	Normal run operation

During normal run operation the display will cycle through multiple codes:

Run	Normal run operation
Fr=	Firing rate percentage
100	Numerical value of firing rate percentage (numerical values will change)
Alr	Inducer fan pressure
1.80	Numerical value of inducer pressure (numerical values will change)

During abnormal operation the display will cycle through multiple codes:

A01	Alert Code (There are nine different alert functions)
Fr=	Firing rate percentage
100	Numerical value of firing rate percentage (numerical values will change)
Alr	Inducer fan pressure
1.80	Numerical value of inducer pressure (numerical values will change)

Staged Gas Heat

The following diagnostics are available through direct data monitoring from the Modbus controlled gas heat ignition module (VB1287) or through data monitoring.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Gas Heat Lockout Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Symbio 700 call for heat is removed to the VB1287 ignition controller. This diagnostic is active anytime a gas heat ignition module lockout condition exists (if the diagnostic is or is not listed in this table). VB1287 on manifold 1 has a gas heat lock out diagnostic.
Diagnostic: Gas Heat Flame Rod Aged Manifold 1	Heat	Warning	Non-Latching	No	The Symbio 700 will continue call for heat. Ignition module LED code: Yellow [1,1][a] Warning, VB1287 detected the flame rod on manifold 1 is aged and may need service.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Gas Heat Unexpected Flame Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller and supply fan operation stays On. Ignition module LED code: RedVB1287 Ignition Controller has two multi-color LEDs [LED1, LED2]. Each LED has a designated flash pattern that indicates an operating status or diagnostic condition. For example: When LED code [1,2] is active, LED1 will flash Red 1-time, LED2 will flash Red 2-times. VB1287 controlled detected an unexpected flame on manifold 1. Requires a power reset or the gas ignition controller will auto reset after 1-hour. Possible causes: Gas valve may be stuck open or malfunctioning. Flame rod maybe malfunctioning. Will continue to retry, lockout after 3-trials.
Diagnostic: Staged Gas Heat Invalid ID Plug Manifold 1	Heat	Immediate Shutdown	Non-Latching	No	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,4][a] ID Plug contains invalid data. Ignition controller disables communication and goes into a hard lockout. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Staged Gas Heat Configuration Invalid Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	The Symbio 700 configuration does not match the VB1285 ignition controller ID Plug heat configuration. All burners on the manifold are unavailable until the condition is corrected. Diagnostic also occurs when communication fails to the gas heat ignition module; Symbio 700 is unable to read the ignition module ID Plug heat configuration. Ignition module LED code: Green [normal operation] VB1285 ignition controller functioning properly. Possible causes: Wrong ID Plug installed. Symbio 700 mis-configured.
Diagnostic: Staged Gas Heat Primary Limit Open Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,1][a] VB1287 ignition controller detected the primary limit is open. Requires a power reset or the gas ignition controller will auto reset after 1-hour. Possible causes: This diagnostic is triggered when the cool down timer has expired, therefore indicates longer term failures, likely the roll out switch. Verify Primary Limit input.
Diagnostic: Staged Gas Heat Open Fuse Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [4,5][a] VB1287 ignition controller detected the fuse is open. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Heat Failed Ignition Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [4,5][a] VB1287 ignition controller detected the fuse is open. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Staged Gas Heat Valve Failure Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [1,2][a] Manifold 1 failed ignition 4 times. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Gas Heat Inducer High Pressure Manifold 1	Heat	Warning	Non-Latching	YES	The Symbio 700 will continue call for heat. Ignition module LED code: Red [1,5][a] VB1287 has detected a manifold 1 valve failure. The failure indicates that gas valve state is low when it must be high, or high when it must be low. Requires a power reset or the gas ignition controller will auto reset after 1-hour.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Gas Heat Valve Short Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [1,4][a] VB1287 ignition controller detected the burner gas valve is shorted to 24VAC. Requires a power reset or the gas ignition controller will auto reset after 1-hour. Possible causes: Control needs to be replaced. Faulty wiring.
Diagnostic: Gas Heat Flame Loss Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [1,3][a] The burner on manifold 1, flame loss count is limited to 3, after which the VB1287 ignition controller will lockout. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Staged Gas Heat Control Board Failure Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,5][a] VB1287 has detect an ignition board failure. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Gas Heat Start Failure Manifold 1	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,2][a] VB1287 safety checks at start-up failed or flame circuitry idle state is incorrect.
Gas Heat Ignition Module 1 Communication Status	Heat	Warning	Non-Latching	YES	Gas heat operation on ignition module 1 disabled. Check Modbus wiring.

[a] VB1287 Ignition Controller has two multi-color LEDs [LED1, LED2]. Each LED has a designated flash pattern that indicates an operating status or diagnostic condition. For example: When LED code [1,2] is active, LED1 will flash Red 1-time, LED2 will flash Red 2-times.

VB1287 LED Display Code Operation

The VB287 ignition controller board has two multi-color LEDs, LED1 and LED2. Each LED has a designated flash and color indicating the VB1287 operating state:

- Normal operation (green color)
- Warning codes (yellow color)
- Failure codes (red color)

Normal Operation:

Display Code [LED1, LED2]	Color	Description
1,1	Green	Ignition controller is idle.
1,2	Green	Call for heat. Burner state is PURGE.
1,3	Green	Call for heat. VB1287 prime burner is running. Burner states Gas-On, Warmup, Run.
3,3	Green	Call for heat. VB1287 prime and split burners are running.

Staged Gas Heat – 800 MBh

The following diagnostics are available through direct data monitoring from the Modbus controlled gas heat ignition module or through data monitoring. These diagnostics are associated with an 800 MBh gas furnace where Gas Heat Ignition Module 2 (VB1287) is controlling a two stage 400 MBh module.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Gas Heat Lockout Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. This diagnostic is active anytime a gas heat ignition module lockout condition exists (if the diagnostic is or is not listed in this table). VB1287 on manifold 2 has a gas heat lock out diagnostic.
Diagnostic: Gas Heat Flame Rod Aged Manifold 2	Heat	Warning	Non-Latching	No	The Symbio™ 700 will continue call for heat. Ignition module LED code: Yellow [1,1][1] Warning, VB1287 detected the flame rod on manifold 1 is aged and may need service.
Diagnostic: Gas Heat Unexpected Flame Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller and supply fan operation stays On. Ignition module LED code: Red [1,1][1] VB1287 controlled detected an unexpected flame on manifold 1. Requires a power reset or the gas ignition controller will auto reset after 1-hour. Possible causes: Gas valve may be stuck open or malfunctioning. Flame rod maybe malfunctioning. Will continue to retry, lockout after 3-trials.
Diagnostic: Staged Gas Heat Invalid ID Plug Manifold 2	Heat	Immediate Shutdown	Non-Latching	No	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,4][1] ID Plug contains invalid data. Ignition controller disables communication and goes into a hard lockout. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Staged Gas Heat Configuration Invalid Manifold 2	Heat	Immediate Shutdown	Non-Latching	No	The Symbio 700 configuration does not match the VB1285 ignition controller ID Plug heat configuration. All burners on the manifold are unavailable until the condition is corrected. Diagnostic also occurs when communication fails to the gas heat ignition module; Symbio 700 is unable to read the ignition module ID Plug heat configuration. Ignition module LED code: Green [normal operation] VB1285 ignition controller functioning properly. Possible causes: Wrong ID Plug installed. Symbio 700 mis-configured.
Diagnostic: Staged Gas Heat Primary Limit Open Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,1][1] VB1287 ignition controller detected the primary limit is open. Requires a power reset or the gas ignition controller will auto reset after 1-hour. Possible causes: This diagnostic is triggered when the cool down timer has expired, therefore indicates longer term failures, likely the roll out switch. Verify Primary Limit input.
Diagnostic: Staged Gas Heat Open Fuse Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [4,5][1] VB1287 ignition controller detected the fuse is open. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Heat Failed Ignition Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [4,5][1] VB1287 ignition controller detected the fuse is open. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Staged Gas Heat Valve Failure Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [1,2][1] Manifold 1 failed ignition 4 times. Requires a power reset or the gas ignition controller will auto reset after 1-hour.

Diagnostic	Target	Severity	Persistence	Alarm Indicator	Condition / Response
Diagnostic: Gas Heat Inducer High Pressure Manifold 2	Heat	Warning	Non-Latching	YES	The Symbio 700 will continue call for heat. Ignition module LED code: Red [1,5][1] VB1287 has detected a manifold 1 valve failure. The failure indicates that gas valve state is low when it must be high, or high when it must be low. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Gas Heat Valve Short Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [1,4][1] VB1287 ignition controller detected the burner gas valve is shorted to 24VAC. Requires a power reset or the gas ignition controller will auto reset after 1-hour. Possible causes: Control needs to be replaced. Faulty wiring.
Diagnostic: Gas Heat Flame Loss Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [1,3][1] The burner on manifold 1, flame loss count is limited to 3, after which the VB1287 ignition controller will lockout. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Staged Gas Heat Control Board Failure Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,5][1] VB1287 has detected an ignition board failure. Requires a power reset or the gas ignition controller will auto reset after 1-hour.
Diagnostic: Gas Heat Start Failure Manifold 2	Heat	Immediate Shutdown	Non-Latching	YES	Call for heat is removed to the VB1287 ignition controller. Ignition module LED code: Red [5,2][1] VB1287 safety checks at start-up failed or flame circuitry idle state is incorrect.
Gas Heat Ignition Module 2 Communication Status	Heat	Warning	Non-Latching	YES	Gas heat operation on ignition module 2 disabled. Gas heat will continue via ignition module 1. Check Modbus wiring.

Device Tracker

Symbio™ 700 will keep statistical data of the unit for component starts and component run times (in hours) for the following unit components:

- Compressor 1
- Compressor 2
- Compressor 3
- Circuit 1 Condenser Coil Defrost
- Circuit 2 Condenser Coil Defrost
- Condenser Fan 1
- Condenser Fan 2
- Condenser Fan 3
- Supply Fan
- Electric Heat Stage 1
- Electric Heat Stage 2
- Filter (Runtime only)
- Gas Heat Stage 1
- Gas Heat Stage 2
- Modulating Gas Heat

- Relief Fan
- Filter

If the Filter Runtime hours exceed the value set by the user for the Filter Runtime Hours Setpoint, the Symbio 700 controller activates the Diagnostic: Maintenance Required point. If there is a requirement to reset the component statistical data, the Run Time Reset or Starts Reset points can be accessed through the Symbio Service and Installation mobile app. If the reset points are set to Reset, the Component Run Time and Starts are reset to 0 and the associated reset points are set back to inactive. In the case of the Diagnostic: Maintenance Required point; it will also be reset to Inactive if Filter Timer Reset point is set.

Hardware

The following tables provide troubleshooting information for common sensors. The terminal voltage is measured at the Symbio™ 700 input while the sensor is connected.

Table 17. Temperature sensor (10K thermistor type)

Temp (°F)	Temp (°C)	Resistance (ohms)	Terminal Voltage	Temp (°F)	Temp (°C)	Resistance (ohms)	Terminal Voltage
-10	-23.33	118070	2.30	60	15.56	15314	1.51
-8	-22.22	110558	2.29	62	16.67	14547	1.48
-6	-21.11	103574	2.28	64	17.78	13823	1.45
-4	-20.00	97078	2.27	66	18.89	13139	1.42
-2	-18.89	91032	2.25	68	20.00	12494	1.39
0	-17.78	85403	2.24	70	21.11	11884	1.36
2	-16.67	80160	2.22	72	22.22	11307	1.33
4	-15.56	75272	2.21	74	23.33	10762	1.30
6	-14.44	70715	2.19	76	24.44	10247	1.27
8	-13.33	66464	2.17	78	25.56	9759	1.23
10	-12.22	62496	2.16	80	26.67	9298	1.20
12	-11.11	58791	2.14	82	27.78	8861	1.17
14	-10.00	55329	2.12	84	28.89	8448	1.14
16	-8.89	52094	2.10	86	30.00	8056	1.12
18	-7.78	49069	2.08	88	31.11	7684	1.09
20	-6.67	46240	2.06	90	32.22	7332	1.06
22	-5.56	43592	2.03	92	33.33	6999	1.03
24	-4.44	41112	2.01	94	34.44	6682	1.00
26	-3.33	38790	1.99	96	35.56	6382	0.97
28	-2.22	36613	1.96	98	36.67	6097	0.95
30	-1.11	34573	1.94	100	37.78	5826	0.92
32	0.00	32659	1.91	102	38.89	5569	0.89
34	1.11	30864	1.89	104	40.00	5325	0.87
36	2.22	29178	1.86	106	41.11	5093	0.84
38	3.33	27596	1.84	108	42.22	4872	0.82
40	4.44	26109	1.81	110	43.33	4662	0.79
42	5.56	24712	1.78	112	44.44	4463	0.77
44	6.67	23398	1.75	114	45.56	4273	0.75
46	7.78	22162	1.72	116	46.67	4092	0.73
48	8.89	21000	1.69	118	47.78	3921	0.70

Table 17. Temperature sensor (10K thermistor type) (continued)

Temp (°F)	Temp (°C)	Resistance (ohms)	Terminal Voltage
50	10.00	19905	1.66
52	11.11	18875	1.63
54	12.22	17904	1.60
56	13.33	16989	1.57
58	14.44	16127	1.54

Temp (°F)	Temp (°C)	Resistance (ohms)	Terminal Voltage
120	48.89	3757	0.68
122	50.00	3601	0.66
124	51.11	3452	0.64
126	52.22	3311	0.62
128	53.33	3176	0.60

Table 18. Zone sensor – Setpoint input

Temperature (°F)	Resistance (ohms)	Terminal Voltage (V)
50	889.4	0.204
51	869.9	0.200
52	850.5	0.196
53	831.0	0.192
54	811.5	0.188
55	792.0	0.183
56	772.6	0.179
57	753.1	0.175
58	733.6	0.171
59	714.2	0.167
60	694.7	0.162
61	675.2	0.158
62	655.7	0.154
63	636.3	0.150
64	616.8	0.145
65	597.3	0.141
66	577.8	0.137
67	558.4	0.132
68	538.9	0.128
69	519.4	0.123
70	500.0	0.119
71	480.5	0.115
72	461.0	0.110
73	441.5	0.106
74	422.1	0.101
75	402.6	0.097
76	383.1	0.092
77	363.7	0.088
78	344.2	0.083
79	324.7	0.079
80	305.2	0.074
81	285.8	0.069

Table 18. Zone sensor – Setpoint input (continued)

Temperature (°F)	Resistance (ohms)	Terminal Voltage (V)
82	266.3	0.065
83	246.8	0.060
84	227.3	0.056
85	207.9	0.051

Table 19. Zone sensor – Mode input

System Mode	Fan Mode	Resistance (ohms nom.)	Terminal Voltage (V nom.)
Off	Auto	2.32K	0.47
Cool	Auto	4.87K	0.82
Auto	Auto	7.68K	1.09
Off	On	10.77K	1.30
Cool	On	13.32K	1.43
Auto	On	16.13K	1.54
Heat	Auto	19.48K	1.65
Heat	On	27.93K	1.84
Emergency Heat	Auto	35.0K	1.94
Emergency Heat	On	43.45K	2.03

Table 20. Relative humidity sensor

Relative Humidity (%)	Current (mA)
0	4
6	5
13	6
19	7
25	8
31	9
38	10
44	11
50	12
56	13
63	14
69	15
75	16
81	17
88	18
94	19
100	20

Table 21. CO_2 sensor

CO_2 (PPM)	Terminal Voltage (VDC)
200	1
300	1.5
400	2
500	2.5
600	3
700	3.5
800	4
900	4.5
1000	5
1100	5.5
1200	6
1300	6.5
1400	7
1500	7.5
1600	8
1700	8.5
1800	9
1900	9.5
2000	10

Appendix A

Supply Fan

Multi-Speed/VFD

Table 22. Multi-speed supply fan speeds

Unit Operation	Supply Fan Speed (%)
Off	0
Fan Only	37
Cooling Stage 1	67
Cooling Stage 2	67
Cooling Stage 3	67
Cooling Stage 4	67
Cooling Stage 5	100
Electric/Gas Heat 1	100
Electric/Gas Heat 2	100
Modulating Gas Heat	100

Variable Speed/VFD – Space Temperature Control

Table 23. Variable speed space temperature control supply fan minimum speed (%)

Unit Operation	Supply Fan Speed (%)
Off	0
Fan Only	37
Cooling Stage 1	37
Cooling Stage 2	58
Cooling Stage 3	67
Cooling Stage 4	75
Cooling Stage 5	75
Electric/Gas Heat 1	100
Electric/Gas Heat 2	100
Modulating Gas Heat	58

Variable Speed/VFD – Discharge Air Temperature Control

Table 24. Variable speed discharge temperature control supply fan minimum speed (%)

Unit Operation	Supply Fan Speed (%)
Off	0
Fan Only	37
Cooling Stage 1	37
Cooling Stage 2	37
Cooling Stage 3	37

Table 24. Variable speed discharge temperature control supply fan minimum speed (%) (continued)

Unit Operation	Supply Fan Speed (%)
Cooling Stage 4	37
Cooling Stage 5	37
Electric/Gas Heat 1	100
Electric/Gas Heat 2	100
Modulating Gas Heat	37
Daytime Warm-up	100
Morning Warm-up	100

Compressor Staging

27.5 to 35 ton (60 Hz) and 22.9 to 29.2 ton (50 Hz)

Table 25. Standard and high efficiency

Compressor Stage[a]	Conventional Thermostat	Cooling Capacity %	Lead / Lag	Compressor Output			Compressor		
				BO-4 (P8-5)	BO-6 (P9-5)	BO-8 (P10-5)	CPR1	CPR2	CPR3
0	Open	0	—	OFF	OFF	OFF	OFF	OFF	OFF
1[b]	Y1 or Y2	25	—	**ON**	OFF	OFF	**ON**	OFF	OFF
2	—	38	Lead	OFF	ON	OFF	OFF	ON	OFF
2	—	38	Lag	OFF	OFF	ON	OFF	OFF	ON
3[c]	—	62	Lead	ON	**ON**	OFF	ON	**ON**	OFF
3[c]	—	62	Lag	**ON**	OFF	ON	**ON**	OFF	ON
4	—	75	—	OFF	ON	**ON**	OFF	ON	ON
5[d]	Y1+Y2	100	—	ON	ON	**ON**	ON	ON	ON

[a] Oil Management. Oil management is a function that aids oil distribution in the refrigerant system. If compressors operate at a stage continuously for a specified period, an oil management sequence is enforced.
[b] Stage 1 Oil management. If operating at compressor stage-1 for 19 continuous minutes, control will go to stage 3 for 30 seconds.
[c] Stage 3 Oil management. If operating at compressor stage-3 for 30 continuous minutes, control will go to stage 2 for 30 seconds.
[d] Stage 5 Oil management. If operating at compressor stage 5 for 30 continuous minutes, control will go to stage 4 for 30 seconds.

40 to 50 ton (60 Hz) and 33.3 to 41.7 ton (50 Hz)

Below table defines the compressor staging sequence for the following unit configurations:

- Standard efficiency units
- High efficiency units with hot gas reheat installed
- High efficiency units with a 5-row evaporator coil design and hot gas reheat is not installed.

Compressor Stage[a]	Conventional Thermostat	Cooling Capacity %	Lead / Lag	Compressor Output			Compressor		
				BO-4 (P8-5)	BO-6 (P9-5)	BO-8 (P10-5)	CPR1	CPR2	CPR3
0	Open	0	—	OFF	OFF	OFF	OFF	OFF	OFF
1[b]	Y1 or Y2	25	—	**ON**	OFF	OFF	**ON**	OFF	OFF
2[c]	—	38	Lead	OFF	ON	OFF	OFF	ON	OFF
2[d]	—	38	Lag	OFF	OFF	ON	OFF	OFF	ON

Compressor Stage[a]	Conventional Thermostat	Cooling Capacity %	Lead / Lag	Compressor Output			Compressor		
				BO-4 (P8-5)	BO-6 (P9-5)	BO-8 (P10-5)	CPR1	CPR2	CPR3
3[e]	—	62	Lead	ON	**ON**	OFF	ON	**ON**	OFF
3[e]	—	62	Lag	**ON**	OFF	ON	**ON**	OFF	ON
4[f]	—	75	—	OFF	ON	**ON**	OFF	ON	ON
5[g]	Y1+Y2	100	—	ON	ON	**ON**	ON	ON	ON

[a] Extended Oil Management. Oil management is a function that aids oil distribution in the refrigerant system. If compressors operate at a stage continuously for a specified period, an oil management sequence is enforced. In addition, if compressor cooling operation at different stages accumulates 30 minutes without operating an oil management sequence, oil management will be enforced at the active compressor stage.

[b] Stage 1 Oil management. If operating at compressor stage-1 for 30 continuous minutes the following oil management compressor staging sequences alternate:

- Sequence 1: stage-2 (lead) for 1 minute, then stage-3 (lead) for 1 minute.
- Sequence 2: stage-2 (lag) for 1 minute, then stage-3 (lag) for 1 minute.

[c] Stage 2 (lead) Oil management. If operating at compressor stage-2 (lead) for 30 continuous minutes, control will go to stage-1 for 1 minute, then stage-3 (lag) for 1 minute.

[d] Stage 2 (lag) Oil management. If operating at compressor stage-2 (lag) for 30 continuous minutes, control will go to stage-1 for 1 minute, then stage-3 (lead) for 1 minute.

[e] Stage 3 (lead or lag) Oil management. If operating at stage-3 (lead or lag) for 30 continuous minutes, control will go to stage-4 for 1 minute.

[f] Stage 4 Oil management. If operating at compressor stage-4 for 30 continuous minutes, control will go to stage-3 (lag) for 1 minute, then stage-3 (lag) for 1 minute.

[g] Stage 5 Oil management. If operating at compressor stage-5 for 30 continuous minutes, control will go to stage-3 for 1-minute.

Table 26. High efficiency, 6-row evaporator coil design and hot gas reheat is not Installed

Compressor Stage[a]	Conventional Thermostat	Cooling Capacity %	Lead / Lag	Compressor Output			Compressor		
				BO-4 (P8-5)	BO-6 (P9-5)	BO-8 (P10-5)	CPR1	CPR2	CPR3
0	Open	0	NA	OFF	OFF	OFF	OFF	OFF	OFF
1[b]	Y1 or Y2	25	NA	**ON**	OFF	OFF	**ON**	OFF	OFF
2	NA	38	Lead	OFF	ON	OFF	OFF	ON	OFF
2	NA	38	Lag	OFF	OFF	ON	OFF	OFF	ON
3[c]	NA	62	Lead	ON	**ON**	OFF	ON	**ON**	OFF
3[c]	NA	62	Lag	**ON**	OFF	ON	**ON**	OFF	ON
4	NA	75	—	OFF	ON	**ON**	OFF	ON	ON
5[d]	Y1+Y2	100	—	ON	ON	**ON**	ON	ON	ON

[a] Oil Management. Oil management is a function that aids oil distribution in the refrigerant system. If compressors operate at a stage continuously for a specified period, an oil management sequence is enforced.

[b] Stage 1 Oil management. If operating at compressor stage-1 for 19 continuous minutes, control will go to stage-3 for 30 seconds.

[c] tage 3 Oil management. If operating at compressor stage-3 for 30 continuous minutes, control will go to stage-2 for 30 seconds.

[d] Stage 5 Oil management. If operating at compressor stage-5 for 30 continuous minutes, control will go to stage-4 for 30 seconds.

Condenser Fan Operation

Thermostat, CVZT, VVZT, and VVDA

Table 27. 27.5 to 35 ton standard and high efficiency

Active CF Cool Stage	Outdoor Air Temperature Active (°F)	Dehumidification Status	Condenser Fan Output		
			Fan 1	Fan 2	Fan 3
0	X	X	OFF	OFF	OFF
1	< 75	X	ON	OFF	OFF
1	> = 75	X	ON	ON	OFF
1	> = 85	X	ON	ON	ON
2	< 63	X	ON	OFF	OFF

Table 27. 27.5 to 35 ton standard and high efficiency (continued)

Active CF Cool Stage	Outdoor Air Temperature Active (°F)	Dehumidification Status	Condenser Fan Output		
2	> = 63	X	ON	ON	OFF
2	> = 75	X	ON	ON	ON
3	< 65	X	ON	OFF	OFF
3	> = 65	X	ON	ON	OFF
3	> = 75	X	ON	ON	ON
4	< 65	X	ON	OFF	OFF
4	> = 65	X	ON	ON	OFF
4	> = 75	X	ON	ON	ON
5	< 45	INACTIVE	ON	OFF	OFF
5	> = 45	INACTIVE	ON	ON	OFF
5	> = 65	INACTIVE	ON	ON	ON
5	<80	ACTIVE	ON	OFF	OFF
5	>=85	ACTIVE	ON	ON	OFF

Table 28. 40 ton standard and high efficiency

Active CF Cool Stage	Outdoor Air Temperature Active (°F)	Dehumidification Status	Condenser Fan Output		
			Fan 1	Fan 2	Fan 3 and 4
0	X	X	OFF	OFF	OFF
1	< 70	X	ON	OFF	OFF
1	> = 70	X	ON	ON	OFF
1	> = 85	X	ON	ON	ON
2	< 70	X	ON	OFF	OFF
2	> = 70	X	ON	ON	OFF
2	> = 85	X	ON	ON	ON
3	< 45	X	ON	OFF	OFF
3	> = 45	X	ON	ON	OFF
3	> = 80	X	ON	ON	ON
4	< 45	X	ON	OFF	OFF
4	> = 45	X	ON	ON	OFF
4	> = 80	X	ON	ON	ON
5	< 45	INACTIVE	ON	OFF	OFF
5	> = 45	INACTIVE	ON	ON	OFF
5	> = 70	INACTIVE	ON	ON	ON
5	<80	ACTIVE	ON	ON	OFF
5	>=85	ACTIVE	ON	ON	ON

Table 29. 50 ton standard and high efficiency

Active CF Cool Stage	Outdoor Air Temperature Active (°F)	Dehumidification Status	Condenser Fan Output		
			Fan 1	Fan 2	Fan 3 and 4
0	X	X	OFF	OFF	OFF
1	< 65	X	ON	OFF	OFF
1	> = 65	X	ON	ON	OFF
1	> = 80	X	ON	ON	ON
2	< 65	X	ON	OFF	OFF
2	> = 65	X	ON	ON	OFF
2	> = 80	X	ON	ON	ON
3	< 40	X	ON	OFF	OFF
3	> = 40	X	ON	ON	OFF
3	> = 75	X	ON	ON	ON
4	< 40	X	ON	OFF	OFF
4	> = 40	X	ON	ON	OFF
4	> = 75	X	ON	ON	ON
5	< 40	INACTIVE	ON	OFF	OFF
5	> = 40	INACTIVE	ON	ON	OFF
5	> = 65	INACTIVE	ON	ON	ON
5	<80	ACTIVE	ON	ON	OFF
5	>=85	ACTIVE	ON	ON	ON

Electric Heat

Table 30. Electric heat staging

Unit Operation	Unit Response
Electric Heat Stage 1	Electric Heat Stage 1 Output ON
Electric Heat Stage 2	Electric Heat Stage 1 and 2 Outputs ON

Gas Heat

Table 31. Gas heat staging

Unit Operation	Unit Response
Gas Heat Stage 1	Gas Valve Stage 1 Running
Gas Heat Stage 2	Gas Valve Stage 2 Running

Emergency and Ventilation Override

Table 32. Emergency and ventilation override

Inputs	Outputs					
Emergency Override BAS	**Supply Fan On/ Off Request**	**Supply Fan Speed Request**	**Outdoor Air Damper**	**Relief Fan**	**VAV Box Relay [(a)]**	**Heat Cool Mode Status**
Point	**State**	**%**	**State**	**State**	**State**	**Point**
2 = EMERG_PRESSURIZE	ON	100	100%	Off	Energized	Fan Only
3 = EMERG_DEPRESSURIZE	OFF	0	0%	On/100%	De-energized	Fan Only
4 = EMERG_PURGE	ON	100	100%	On/100%	Energized	Fan Only
5 = EMERG_SHUTDOWN	OFF	0	0%	Off	De-energized	OFF
6 = EMERG_FIRE	OFF	0	0%	Off	De-energized	OFF
1 = EMERG_NORMAL	Auto	Auto	Auto	Auto	Auto	Auto

[(a)] Variable Volume Discharge Air units

Space Setpoint Adjustment

Zone sensors with an internal or external setpoint adjustment provide the controller with a local setpoint (50 to 85°F or 10 to 29.4°C). The internal setpoint adjustment is concealed under the zone sensor cover. To access the setpoint adjustment, remove the zone sensor cover. Some external setpoints (when present) are displayed on the digital display zone sensor front cover. When the local setpoint adjustment is used to determine the setpoints, all unit setpoints are calculated based on the local setpoint value, the configured setpoints, and the active mode of the controller. The controller determines the effective space setpoint based on the following:

- Local wired setpoint input
- Occupancy mode
- Heating or cooling mode (space demand)
- Space setpoint high and low limits (configured)

Single Setpoint

Heat mode:

- Occupied mode: Space Temperature Setpoint Active = Space Temperature Setpoint (arbitrated) - Occupied Offset
- Occupied standby mode: Space Temperature Setpoint Active = Space Temperature Setpoint (arbitrated) - Standby Offset
- Unoccupied mode: Space Temperature Setpoint Active = Unoccupied Heating Setpoint Cool mode

Cool mode:

- Occupied mode: Space Temperature Setpoint Active = Space Temperature Setpoint (arbitrated) + Occupied Offset
- Occupied standby mode: Space Temperature Setpoint Active = Space Temperature Setpoint (arbitrated) + Standby Offset
- Unoccupied mode: Space Temperature Setpoint Active = Unoccupied Cooling setpoint

When a building automation system or other controller communicates a setpoint to the controller, the controller ignores the local setpoint input and uses the communicated value (default operation). The exception is when the system is in unoccupied mode and the controller always uses the unoccupied setpoints. After the controller completes all setpoint calculations, the calculated occupied setpoint is validated against the following configured space setpoint limits:

- Heating setpoint high limit
- Heating setpoint low limit
- Cooling setpoint high limit
- Cooling setpoint low limit

These setpoint limits apply only to the occupied and occupied standby, heating, and cooling setpoints. They do not apply to the unoccupied heating and cooling setpoints. When the controller is in the unoccupied mode, it always uses the unoccupied heating and cooling setpoints.

Dual Setpoint

When Symbio™ 700 is configured for system types CVZT or VVZT, the controls can be configured for Dual Setpoint control. Dual Setpoint provides independent space cooling setpoint and space heating setpoint inputs to the controller. It also allows an external source to write to independent the space cooling and heating setpoints.

Appendix B

Voyager 3 Configuration

The following table describes the Symbio™ 700 configuration and options available with reference to the Voyager 3 model number.

Symbio 700 Configuration Item	Symbio 700 Configuration Option	Equipment Model Number	Description
System Type	CVZT	Digit 16 = 8, E, F, G	Constant Volume Zone Temperature control, provides discrete supply fan speeds at heating and cooling capacities. Select CVZT when the equipment is controlled from a conventional thermostat or a zone sensor.
	VVZT	Digit 16 = 6, 7, C, D	Variable Volume Zone Temperature control, also known as single zone VAV, modulates the supply fan with capacity providing space temperature control to a space.
	VVDA	Digit 16 = 4, 5, A, B	Variable Volume Discharge Air control, also known has multi-zone VAV, provides supply fan duct static pressure control and controls to a discharge air temperature setpoint.
Space Controller	Conventional TStat	Digit 16 = 8, E, F, G	Equipment is being controlled from a conventional thermostat in the space.
	Dual Setpoint Zone Sensor	Digit 16 = 6, 7, 8, C, D, E, F, G	Heating and Cooling setpoint zone sensor installed. Applicable to CVZT and VVZT System Type.
	Single Setpoint Zone Sensor	Digit 16 = 6, 7, 8, C, D, E, F, G	Single setpoint zone sensor installed. Applicable to CVZT and VVZT System Type.
Refrigeration System	Cooling Only	Digit 1, 2 = **TC** – DX Cooling No Heat **TE** – DX Cooling, Electric Heat **YC** – DX Cooling, Natural Gas Heat	Direct expansion cooling unit with or without primary heating capacity.
Refrigerant	R-410A	Digit 7 = C	Equipment designed for R-410A.
	R-454B	Digit 7 = D	Equipment designed for R-454B
Tonnage (60 Hz)	27.5	Digit 4, 5, 6 = 330	Designates the equipment capacity in tons of cooling.
	30	Digit 4, 5, 6 = 360	
	35	Digit 4, 5, 6 = 420	
	40	Digit 4, 5, 6 = 480	
	50	Digit 4, 5, 6 = 600	
Tonnage (50 Hz)	22.9	Digit 4, 5, 6 = 275	
	25.4	Digit 4, 5, 6 = 305	
	29.2	Digit 4, 5, 6 = 350	
	33.3	Digit 4, 5, 6 = 400	
	41.7	Digit 4, 5, 6 = 500	
Refrigeration Circuit	Single Manifold	Not applicable	Type of refrigeration circuit

Symbio 700 Configuration Item	Symbio 700 Configuration Option	Equipment Model Number	Description
Voltage	208/230/60	Digit 8 = E, F	208Vac or 230Vac, 60Hz
	380/415/50	Digit 8 = C, D	380Vac or 415Vac, 50Hz
	460/60	Digit 8 = 4	460Vac, 60Hz
	575/60	Digit 8 = 5	575Vac, 60Hz
Efficiency	Standard	Digit 29 = O, J	Standard Efficiency
	High	Digit 29 = K, L	High Efficiency
Hot Gas Reheat	Not Installed, Modulating	Digit 32 = 0	Not Installed
		Digit 32 = T	Modulating Hot Gas Reheat installed
Dehumidification Control	Humidistat	Not applicable	Dehumidification will be initiated based on a Humidistat binary input when modulating hot gas reheat is installed.
	Relative Humidity	Not applicable	Dehumidification will be initiated based on a Space Relative Humidity when modulating hot gas reheat is installed.
	Dew Point	Not applicable (default setting)	Default setting when modulating hot gas reheat is installed. Dehumidification will be initiated based on a Space Dew Point and Outdoor Air Dew Point.
Primary Heating Source	Not Installed	Digit 1, 2 = TC	No primary heating installed.
	Electric	Digit 1, 2 = TE	Cooling only unit with primary electric heat installed.
	Gas	Digit 1,2 = YC	Cooling only unit with primary staged or modulating gas heat installed.
Primary Heating Type	Staged	Digit 1, 2 = TE or YC Digit 9 = L, H, J, K	Staged electric heat or staged gas heat installed.
	Modulating	Digit 1, 2 = YC Digit 9 = M, P, R, T	Modulating gas heat installed.
Primary Heating Capacity	350 MBh Low Heat 400 MBh Low Heat 600 MBh High Heat 800 MBh High Heat	Digit 9 = L, J (staged low heat) Digit 9 = H, K (staged high heat) Digit 9 = M, R (modulating low heat) Digit 9 = P, T (modulating high heat)	Options when gas heat is installed.
Primary Heating Stages	1	Digit 9 = A, B	Electric heat installed.
	2	Digit 9 = C, D, E (electric heat) Digit 9 = L, H, J, K (staged gas heat)	Electric heat or staged gas heat installed.
Outside Air	Not Installed	Digit 15 = A, B	No outdoor air damper installed, or 0-25% Manual Damper
	0 to 100% Economizer	Digit 15 = C, D, E, F, G, H, J, K, L	0 to 100% Economizer installed
	0 to 100% Economizer w/TRAQ	Digit 15 = 1, 2, 3, 4, 5, 6, 7, 8, 9	0 to 100% Economizer with Traq damper installed

Appendix B

Symbio 700 Configuration Item	Symbio 700 Configuration Option	Equipment Model Number	Description
Economizer Type	Dry Bulb	Digit 15 = C, F, 1, 7	Economizer cooling operation enables based on outdoor air temperature (dry bulb).
	Differential Dry Bulb	Not applicable	Economizer cooling operation enables based on outdoor air temperature and space temperature difference.
	Reference Enthalpy	Digit 15 = D, 2, 5, 8	Economizer cooling operation enables based on outdoor air temperature and outdoor air humidity.
	Comparative Enthalpy	Digit 15 = E, 3, 6, 9	Economizer cooling operation enables based on outdoor air temperature and outdoor air humidity (enthalpy) vs. return air temperature and return air humidity (enthalpy).
Demand Controlled Ventilation	Disabled, Enabled	Not applicable	Demand Controlled Ventilation is a 0 to 100% economizer damper option when a Space CO_2 sensor is installed.
Remote Minimum Position	Not Installed	Not applicable	Not installed
	Installed	Not applicable	Remote minimum position potentiometer installed
Space Pressure Control	Not installed	Digit 11 = 0, 1	Not installed, or Barometric Relief
	Relief Fan Only	Digit 11 = 2, 3, 7, 8	Power Exhaust Fan with 0 to 100% Economizer installed.
	Relief Fan with Fresh Air Damper Tracking	Digit 11 = 4, 5	Fresh Air Damper Tracking Power Exhaust Fan with 0 to 100% Economizer installed.
	Relief Fan with StatiTrac	Digit 11 = 6, 9	Power Exhaust with Statitrac (space static pressure control) installed
Clogged Filter	Not Installed	Not applicable	Not installed
	Installed	Digit 31 = R	Clogged Filter switch installed
FroStat	Not Installed	Not applicable	Not installed
	Installed	Not applicable	FroStat installed
Condensate Overflow Switch	Not Installed	Digit 28 = 0, M	Condensate overflow switch not installed
	Installed	Digit 28 = 1, 2	Condensation switch installed
Humidistat	Not Installed	Not applicable	Not installed
	Installed	Not applicable	Humidistat installed.
Humidity Sensor	Not Installed	Not applicable	Not installed
	Installed	Not applicable	Space Humidity Sensor installed.
CO_2 Sensor	Not Installed	Not applicable	Not installed
	Installed	Not applicable	Space CO_2 Sensor installed.
Discharge Temperature Sensor	Not Installed	Not applicable	Not installed
	Installed	Digit 31 = P	Discharge Air Temperature Sensor installed
External Auto/Stop	Not Installed	Not applicable	Not installed
	Installed	Not applicable	External Auto/Stop binary input installed, requires a Customer Options Module.

Symbio 700 Configuration Item	Symbio 700 Configuration Option	Equipment Model Number	Description
Ventilation Override	Not Installed	Not applicable	Not installed
	Installed	Digit 23 = G	Ventilation Override installed, requires a Customer Options Module.
Alarm Indicator	Not Installed	Not applicable	Not installed
	Installed	Not applicable	An installed Alarm Indicator requires a Customer Options Module.
Demand Management	None	Not applicable	Symbio 700 Demand Shed/Demand Limit binary input (J16) is disabled.
	Demand Limit	Not applicable	Symbio 700 binary input (J16) enables and disables energy Demand Limit function. See Demand Limit is this document.
	Demand Shed	Not applicable	Symbio 700 binary input (J16) enables and disables energy Demand Shed function. The Demand Shed is enabled, space heating and cooling setpoints will be offset 4°F (default setting).
Supply Air Tempering	Disabled	Not applicable	Disabled, Supply Air Tempering functional is disabled.
	Enabled	Not applicable	Enabled, Supply Air Tempering function is allowed.
VFD Bypass	Not Installed	Digit 16 = 8, F, 4, A, 6, C	Not installed
	Installed	Digit 16 = E, G, 5, B, 7, D	VFD Bypass is installed.
Airflow Path	Downflow (Up-Flow)	Digit 3 = D, F	Traq dampers are installed with an up-flow return air path.
	Horizontal	Digit 3 = H, R	Traq dampers are installed with a horizontal return air path.
Evaporator Coil Row Count[a]	5 Rows	Not applicable	Unit evaporator coil is a 5-row design.
	6 Rows	Not applicable	Unit evaporator coil is a 6-row design.

[a] Configuration question only applies to 33.3, 40, 41.7, 50-ton high efficiency units without hot gas reheat.

Notes

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